

World Action Models are Zero-shot Policies

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<https://dreamzero0.github.io>



Figure 1: **Overview.** By jointly predicting video and action, World Action Models (WAMs) inherit world physics priors that enable 1) effective learning from diverse, non-repetitive data, 2) open-world generalization, 3) cross-embodiment learning from video-only data, and 4) few-shot adaptation to new robots.

世界动作模型是零样本策略

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图1: **概述.** 通过联合预测视频和动作, 世界动作模型 (WAMs) 继承了世界物理先验, 从而实现了1) 从多样、非重复数据中有效学习, 2) 开放世界泛化, 3) 通过仅视频数据进行跨具身化学习, 以及4) 对新机器人的少样本适应。

Abstract

State-of-the-art Vision-Language-Action (VLA) models excel at semantic generalization but struggle to generalize to unseen physical motions in novel environments. We introduce DREAMZERO, a World Action Model (WAM) built upon a pretrained video diffusion backbone. Unlike VLAs, WAMs learn physical dynamics by predicting future world states and actions, using video as a dense representation of how the world evolves. By jointly modeling video and action, DREAMZERO learns diverse skills effectively from heterogeneous robot data without relying on repetitive demonstrations. This results in over $2\times$ improvement in generalization to new tasks and environments compared to state-of-the-art VLAs in real-robot experiments. Crucially, through model and system optimizations, we enable a 14B autoregressive video diffusion model to perform real-time closed-loop control at 7Hz. Finally, we demonstrate two forms of cross-embodiment transfer: video-only demonstrations from other robots or humans yield a relative improvement of over 42% on unseen task performance with just 10–20 minutes of data. More surprisingly, DREAMZERO enables few-shot embodiment adaptation, transferring to a new embodiment with only 30 minutes of play data while retaining zero-shot generalization.

1. Introduction

Recent robotic foundation models, termed Vision-Language Action models (VLAs), extend pretrained Vision-Language Models (VLMs) to predict motor actions (Bjorck et al., 2025; Black et al., 2024; Brohan et al., 2023; Gemini Robotics Team, 2025; Kim et al., 2024). While VLAs successfully inherit linguistic priors to generalize across diverse language instructions, especially manipulating diverse objects (Brohan et al., 2023), their generalization to novel environments and, more critically, to new motions or skills remains limited (Guruprasad et al., 2025; Zhou et al., 2025). For example, VLAs can successfully execute “move coke can to Taylor Swift” (Brohan et al., 2023) by leveraging the web knowledge acquired during VLM pretraining to identify the target location, and connecting it to the learned move skill from the robot data. However, they fail at a task like “untie the shoelace” if that specific skill was not present in the robot training data. Although VLM priors encode *what* to do at a semantic level, they lack representations of *how* actions should be executed with precise spatial awareness, aligned with geometry, dynamics, and motor control (Chen et al., 2024; Feng et al., 2025). As a result, VLAs often struggle to adapt to new environments or generalize to novel tasks beyond the distribution of expert demonstrations, without explicitly collecting large-scale task- and environment-specific action data.

In this paper, we present DREAMZERO, a 14B robot foundation model built upon a pretrained image-to-video diffusion backbone (Team Wan, 2025). We term this architecture a *World Action Model (WAM)*—a foundation model designed to predict both actions and visual future states in an aligned manner. Initialized from video diffusion models trained on web-scale video data, WAMs leverage rich spatiotemporal priors to jointly generate future frames and actions conditioned on language instructions and observations. This shifts action learning from dense state–action imitation to inverse dynamics—aligning motor commands with predicted visual futures. Consequently, we observe that this enables (1) effective learning from robot data that are heterogeneous trajectories collected during the execution of useful behaviors in real-world settings, rather than relying solely on carefully repeated demonstrations (2) zero-shot generalization to new tasks in new environments, and (3) efficient cross-embodiment transfer.

This approach yields three core advancements that distinguish DREAMZERO from prior work, including other WAMs (Kim et al., 2026; Liang et al., 2025; Pai et al., 2025). First, DREAMZERO unlocks new generalization capabilities beyond traditional VLAs and previous WAMs—across environments, across tasks, and across embodiments (Figure 2 and Figure 3). Compared to the state-of-the-art pretrained VLAs, we observe more than a $2\times$ improvement in average task progress on environment and task generalization benchmarks. Second, DREAMZERO demonstrates that generalist policies can be learned effectively from diverse, heterogeneous data, breaking away from the conventional wisdom that generalist robot policies require multiple repeated

摘要

最先进的视觉-语言-动作模型（VLA）在语义泛化上表现出色，但在新环境中泛化到未见过的物理运动时却面临挑战。我们推出了DreamZero——一种基于预训练视频扩散骨干网络构建的世界动作模型（WAM）。与VLA不同，WAM通过预测未来世界状态和动作来学习物理动力学，将视频作为世界如何演变的密集表征。通过联合建模视频和动作，DreamZero能够从异构机器人数据中有效学习多样技能，而无需依赖重复演示。在真实机器人实验中，与最先进的VLA相比，DreamZero在新任务和新环境上的泛化能力提升了超过 $2\times$ 。关键在于，通过模型与系统优化，我们使一个14B的自回归视频扩散模型能够以7赫兹的频率执行实时闭环控制。最后，我们展示了两种跨具身迁移形式：来自其他机器人或人类的仅视频演示，仅需10-20分钟的数据即可在未见任务性能上实现超过42%的相对提升。更令人惊讶的是，DreamZero实现了少样本实体适应，仅需30分钟的游戏数据即可迁移到新具身，同时保持零样本泛化能力。

1. 引言

近期的机器人基础模型，即视觉-语言-动作模型（VLAs），扩展了预训练的视觉-语言模型（VLMs）以预测运动动作（Bjorck等人，2025；Black等人，2024；Brohan等人，2023；Gemini Robotics Team，2025；Kim等人，2024）。尽管VLAs成功继承了语言先验，从而能够在多样化的语言指令下实现泛化，尤其是在操作多样物体时（Brohan等人，2023），但它们在新环境中的泛化能力，以及更关键地，对新的运动或技能的泛化能力仍然有限（Guruprasad等人，2025；Zhou等人，2025）。例如，VLAs可以通过利用VLM预训练期间获得的网络知识来识别目标位置，并将其与从机器人数据中学习到的移动技能联系起来，成功执行“将可乐罐移到Taylor Swift旁边”（Brohan等人，2023）。然而，如果某项特定技能（如“解开鞋带”）未出现在机器人训练数据中，它们就会失败。尽管VLM先验在语义层面上编码了“做什么”，但它们缺少如何执行动作的表征——即缺乏精确的空间感知、与几何、动力学和运动控制的对齐（Chen等人，2024；Feng等人，2025）。因此，VLAs常常难以适应新环境或泛化到超出专家演示分布的新任务，除非显式收集大规模任务和特定环境下的动作数据。

在本文中，我们提出DREAMZERO，这是一个基于预训练图像到视频扩散骨干网络（Team Wan, 2025）构建的14B机器人基础模型。我们将这种架构称为世界动作模型（WAM）——一种旨在以对齐方式同时预测动作和视觉未来状态的基础模型。WAM从基于网络规模视频数据训练的扩散模型初始化，利用丰富的时空先验，根据语言指令和观测信息联合生成未来帧和动作。这将动作学习从密集的状态-动作模仿转变为逆动力学——使电机指令与预测的视觉未来对齐。因此，我们观察到这实现了：（1）有效从机器人数据中学习，这些数据是在真实世界设置中执行有用行为期间收集的异构轨迹，而非仅依赖精心重复的演示；（2）在新环境中对新任务的零样本泛化；（3）高效的跨具身迁移。

该方法带来了三项核心进展，使DREAMZERO区别于先前工作，包括其他WAMs（Kim等人，2026；Liang等人，2025；Pai等人，2025）。首先，DREAMZERO解锁了超越传统VLAs和先前WAMs的新泛化能力——跨环境、跨任务、跨具身（图2和图3）。与最先进的预训练VLAs相比，我们在环境和任务泛化基准上观察到平均任务进度提升了超过 $2\times$ 。其次，DREAMZERO表明通用策略可以从多样、异构数据中有效学习，打破了通用机器人策略需要多次重复

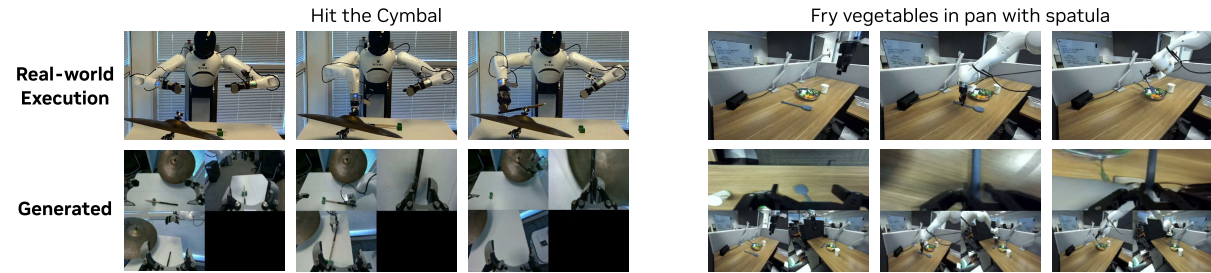


Figure 2: **Joint Video and Action Prediction.** DREAMZERO jointly generates video and action. We observe that the predicted actions closely align with the generated video. The examples are from totally unseen tasks.

demonstrations per task. Although other WAMs show that priors learned from videos prediction improves sample efficiency for action learning compared to VLAs (Liao et al., 2025; Pai et al., 2025), most works still focus on repeated demonstrations. Moreover, the environment generalization of DREAMZERO is retained even after task-specific post-training, outperforming state-of-the-art VLAs by 10% on average task progress. Lastly, we demonstrate two forms of cross-embodiment transfer. First, video-only demonstrations from another robot (YAM) or humans yield a relative improvement of over 42% on unseen task performance for the target robot (AgiBot G1) with just 10–20 minutes of data. Second, and more surprisingly, we show that DREAMZERO enables *few-shot embodiment adaptation*: a model pretrained on AgiBot G1 adapts to an entirely new robot (YAM) with only 30 minutes of play data, retaining zero-shot generalization. To the best of our knowledge, this sets a new benchmark for data-efficient embodiment adaptation.

DREAMZERO is a 14B autoregressive diffusion transformer trained with a teacher-forcing chunk-wise video denoising objective. Our architectural analysis reveals that larger pretrained video diffusion models produce higher-quality video predictions, which directly translates to superior downstream action execution—indicating that policy performance is fundamentally tied to video generation quality. We further find that diverse distribution of the training data is essential for generalization, outperforming multi-task repetitive data with the same amount of hours. Furthermore, we observe that autoregressive architectures lead to smoother robot motions and higher modality alignment between predicted videos and executed actions.

To address the computational overhead inherent to video diffusion models, we introduce a suite of optimizations spanning three categories: (1) algorithmic improvements, including decoupled video and action denoising schedules (DREAMZERO-Flash); (2) system-level parallelism and caching strategies; and (3) low-level optimizations such as quantization, and CUDA kernel tuning. Collectively, these techniques achieve a $38\times$ inference speedup without degrading performance, enabling DREAMZERO to generate action chunks at approximately 7Hz for smooth, real-time robotic control.

Our main contributions are:

- We introduce DREAMZERO, a 14B WAM that jointly predicts video and actions, enabling effective learning from diverse, non-repetitive robot data.
- We demonstrate over $2\times$ improvement in zero-shot generalization to **unseen verbs and motions** compared to state-of-the-art VLAs, while retaining generalization across objects and environments.
- We present model and system optimizations achieving $38\times$ **inference speedup**, enabling real-time closed-loop control at **7Hz**.
- We demonstrate cross-embodiment transfer: video-only data from humans (12 minutes) or other robots (20 minutes) yields a relative improvement of over 42% on unseen tasks, and introduce **few-shot embodiment adaptation**—DREAMZERO pretrained on AgiBot G1 adapts to an entirely new robot (YAM) with only 30 minutes of play data, enabling zero-shot generalization.
- We **open-source** our model weights, inference code, and code to run publicly available real-world (RoboArena)

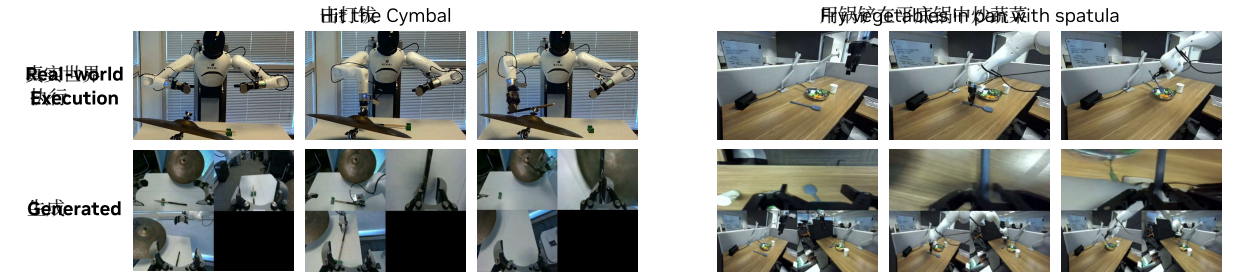


图2: **联合视频与动作预测。** DreamZero联合生成视频和动作。我们观察到预测动作与生成的视频紧密对齐。示例来自完全未见任务。

每次任务演示的传统观念。尽管其他WAMs表明，与VLAs相比，从视频预测中学习的先验能提高动作学习的样本效率（Liao等人，2025；Pai等人，2025），但大多数工作仍聚焦于重复演示。此外，DREAMZERO的环境泛化能力在任务特定后训练后依然保持，在平均任务进度上超越最先进的VLAs 10%。最后，我们展示了两种跨具身迁移形式。首先，来自另一机器人（YAM）或人类仅依靠视频演示，仅需10-20分钟的数据，就能使目标机器人（AgiBot G1）的未见任务性能提升超过42%。其次，更令人惊讶的是，我们表明DREAMZERO实现了少样本实体适应：一个在AgiBot G1上预训练的模型，仅用30分钟的游戏数据就能适应全新的机器人（YAM），并保留零样本泛化能力。据我们所知，这为数据高效的具身适应设立了新基准。

DreamZero是一个14B自回归扩散Transformer，采用教师强制分块视频去噪目标进行训练。我们的架构分析表明，更大的预训练视频扩散模型能产生更高质量的视频预测，这直接转化为更优的下游动作执行——说明策略性能从根本上与视频生成质量挂钩。我们进一步发现，训练数据的多样化分布对泛化至关重要，其表现优于相同小时数的多任务重复数据。此外，我们观察到自回归架构能带来更平滑的机器人运动，以及预测视频与执行动作之间更高的模态对齐。

为了解决视频扩散模型固有的计算开销，我们引入了一套涵盖三个类别的优化方案：（1）算法改进，包括解耦视频与动作去噪调度（DreamZero-Flash）；（2）系统级并行与缓存策略；（3）底层优化，如量化和CUDA内核调优。这些技术共同实现了 $38\times$ 倍的推理加速，且不降低性能，使DreamZero能够以约7赫兹的频率生成动作块，从而实现平滑的实时机器人控制。

我们的主要贡献包括：

- 我们提出了DreamZero，一个14B世界动作模型，能联合预测视频和动作，从而从多样非重复机器人数据中实现高效学习。
- 我们展示了相比最先进的视觉-语言-动作模型，在零样本泛化到**未见动词与动作**上的 $2\times$ 提升，同时保持了跨物体与环境的泛化能力。
- 我们提出了模型与系统优化，实现了 $38\times$ **推理加速**，从而能以**7赫兹**进行实时闭环控制。
- 我们展示了跨具身迁移：来自人类（12分钟）或其他机器人（20分钟）的仅视频数据在未见任务上带来了超过42%的相对提升，并引入了**少样本实体适应**——在AgiBot G1上预训练的DreamZero仅用30分钟的游戏数据即可适应全新的机器人YAM，实现零样本泛化。
- 我们开源了**模型权重**、推理代码以及运行公开真实世界（RoboArena）的代码



Figure 3: **Free-form Evaluation.** DREAMZERO performs a diverse range of tasks when conditioned on natural language instructions, including object manipulation, tool use, and human-robot interaction.

and simulation benchmarks (Polaris and Genie Sim 3.0)¹ at <https://github.com/dreamzero0/dreamzero>.

2. Related Work

2.1. Vision Language Action Models

Utilizing Foundation Models for Robotics. Developing foundation models (Bommasani et al., 2021) for physical artificial intelligence has emerged as a significant research frontier. One line of work involves using existing, pre-trained foundation models as “black-box” reasoners to handle high-level task planning. These works usually involve *modular* systems, where the foundation models generate sequences of instructions, visual traces, or affordances that are subsequently executed by specialized, low-level robotic policies or controllers (Brohan et al., 2023; Driess et al., 2023; Huang et al., 2023; Kumar et al., 2026; Singh et al., 2023). While this modularity simplifies complex planning and enables stronger generalization (Kaelbling and Lozano-Pérez; Lee et al., 2025; Li et al., 2025) and efficiency (Drechkowski et al., 2025), it is contingent upon having a pre-existing library of low-level skills and a robust interface to bridge the gap between abstract reasoning and physical execution. Additionally, these decoupled systems face the risk of compounding errors across modules.

VLAs. On the other hand, end-to-end models such as Vision-Language-Action models (VLAs) (Bjorck et al., 2025; Black et al., 2024; Brohan et al., 2022, 2023; Bu et al., 2025; Gemini Robotics Team, 2025; Kim et al., 2024; Physical Intelligence, 2025; Yang et al., 2025; Ye et al., 2025; Zheng et al., 2025), have gained popularity by moving away from a rigid hierarchy of planning and control, combining language-conditioned semantics and low-level robot actions within the same model. VLAs are often initialized from large vision-language (VLM) models pre-trained on web-scale datasets. While pushing the frontier on visual-semantic knowledge transfer, these models are pre-trained on *static* image-text datasets, which limits their ability to inherit spatiotemporal priors required to transfer knowledge to new physical skills.

Generalization in VLAs. Generalization in VLAs has been mostly demonstrated on object and semantic level (Brohan et al., 2023; Gao et al., 2025) while generalization to completely new skills and environments has

¹Despite only being trained on ~500 hours of real-world data, DREAMZERO shows non-trivial performance on Genie Sim 3.0, which is a simulation benchmark comprised of 100 different tasks *without* being explicitly trained on the 10k hours of simulation training data.



图3: 自由形式评估。DreamZero在自然语言指令条件下执行多样任务, 包括物体操作、工具使用和人机交互。

以及仿真基准 (Polaris和Genie Sim 3.0)¹, 位于<https://github.com/dreamzero0/dreamzero>。

2. 相关工作

2.1. 视觉语言动作模型

将基础模型应用于机器人技术。为物理人工智能开发基础模型 (Bommasani et al., 2021) 已成为一个重要的研究前沿。一条研究路线涉及使用现有的预训练基础模型作为“黑盒”推理器来处理高层任务规划。这些工作通常涉及模块化系统, 其中基础模型生成指令序列、视觉轨迹或可供性, 随后由专门的低级机器人策略或控制器执行 (Brohan et al., 2023; Driess et al., 2023; Huang等人, 2023; Kumar et al., 2026; Singh et al., 2023)。虽然这种模块化简化了复杂规划, 并实现了更强的泛化 (Kaelbling and Lozano-Pérez; Lee et al., 2025; Li et al., 2025) 和效率 (Drechkowski et al., 2025), 但它依赖于预先存在的低级技能库以及一个鲁棒的接口来弥合抽象推理与物理执行之间的差距。此外, 这些解耦系统面临着错误在模块间累积的风险。

VLAs。另一方面, 诸如视觉-语言-动作模型 (VLA) 之类的端到端模型 (Bjorck等人, 2025; Black等人, 2024; Brohan等人, 2022、2023; Bu等人, 2025; Gemini Robotics Team, 2025; Kim等人, 2024; Physical Intelligence, 2025; Yang等人, 2025; Ye等人, 2025; Zheng等人, 2025) 已逐渐流行起来, 它们摆脱了规划和控制的严格层级结构, 将语言条件语义和低级机器人动作结合在同一个模型中。VLA通常从基于网络规模数据集预训练的大型视觉-语言模型 (VLM) 中初始化而来。尽管这些模型在视觉-语义知识迁移方面推动了前沿发展, 但它们是在静态图文数据集上预训练的, 这限制了它们继承时空先验的能力, 而时空先验是将知识迁移到新的物理技能所必需的。

在VLAs中的泛化。 VLAs的泛化主要被证明在物体和语义层面上 (Brohan et al., 2023; Gao et al., 2025) 而泛化到全新的技能和环境已经

¹尽管仅在 ~500 小时的真实世界数据上训练, DreamZero在Genie Sim 3.0上表现不俗, Genie Sim 3.0是一个包含100个不同任务的模拟基准无需在1万小时的模拟训练数据上显式训练。

remained limited. In particular, existing work utilizing VLAs achieves environment generalization by collecting human teleoperation data across hundreds of diverse environments for specific tasks ([Physical Intelligence, 2025](#)). Furthermore, while current VLAs attempt to achieve task generalization by covering a large library of language-conditioned motion primitives ([Gemini Robotics Team, 2025](#)), this approach is fundamentally constrained by the impracticality of capturing the vast amount of possible physical interactions and motions with a fixed set of episode-level language-conditioned tasks. In contrast, video-based world models learn from every consecutive frame pair in the data, while also leveraging large-scale video pretraining to understand physical dynamics.

2.2. Video Model-based Robot Policies

Video Generation in Robotics. Prior works show that video generation models can be used to synthesize robot trajectories and extract executable actions at test-time through various approaches: inverse-dynamics models ([Du et al., 2023](#); [Zhou et al., 2024](#)), optical flow as dense correspondence ([Ko et al., 2024](#)), or trajectory prediction as high-level planning ([Du et al., 2024](#); [Yang et al., 2024](#)). Other works generate human videos—either with 3D tracking ([Liang et al., 2024](#)) or for novel scenes and motions ([Bharadhwaj et al., 2024](#); [Chen et al., 2025](#))—and train policies using point tracking objectives. Most recently, ([Jang et al., 2025](#); [Luo et al., 2025](#)) demonstrated that video generation models can produce synthetic robot data for unseen behaviors in novel environments, leveraging the strong generalization capabilities of these models.

Joint Video and Action Generation. Another line of work couples video and action generation for end-to-end learning. These methods demonstrate that incorporating a world modeling objective alongside action prediction improves multi-task performance, sample efficiency, and generalization to novel scenes and objects. Previous work ([Cheang et al., 2024](#); [Li et al., 2025](#); [Won et al., 2025](#); [Wu et al., 2024](#); [Zhao et al., 2025](#); [Zheng et al., 2025](#); [Zhu et al., 2025](#)) learns to do joint world modeling and action prediction from scratch or from VLAs, while more recent work ([Hu et al., 2024](#); [Kim et al., 2026](#); [Liang et al., 2025](#); [Liao et al., 2025](#); [Pai et al., 2025](#)) leverages pretrained video diffusion models to inherit rich visual dynamics priors. We refer to these models collectively as *World Action Models* (WAMs) since they leverage world modeling capability (predicting the future state) for action prediction. We use the term World Action Models (WAMs) rather than Video Action Models (VAMs) to reflect that video is just one possible world modeling objective—future WAMs may align actions with other predictive modalities such as tactile sensing, force feedback, or learned latent representations. In contrast to prior WAMs, DREAMZERO systematically explores data diversity and scale to expose the full generalization potential of WAMs, adopts an autoregressive architecture better suited for long-horizon world–action modeling, achieves state-of-the-art generalization across both novel tasks and environments, and achieves state-of-the-art cross-embodiment transfer, both learning from different embodiments (video only) and few-shot adaptation to a new embodiment.

Why WAMs. WAMs built upon video diffusion backbones inherit rich spatiotemporal priors from web-scale data, capturing the best of both paradigms: the seamless gradient flow of end-to-end VLAs and dense world modeling supervision for planning. Unlike latent world models ([Assran et al., 2025](#); [Hafner et al., 2019, 2020, 2023](#)), which learn dynamics from scratch in compact latent spaces, WAMs leverage pretrained video representations that already encode physical dynamics from internet-scale data. Central to this approach is learning the joint distribution of video and action—DREAMZERO simultaneously learns both modalities, with video prediction serving as an implicit visual planner that guides action generation. This formulation not only means that improving robotic capabilities reduces to improving video generation, but also enables three capabilities that elude current VLAs: zero-shot generalization to novel tasks, effective learning from heterogeneous robot data, and extremely efficient cross-embodiment transfer from videos. We provide further discussion about the differences between WAMs and alternative world model architectures (e.g., latent-space, 3D point cloud) in Appendix A.

仍然有限。具体来说，现有利用VLAs的工作通过收集数百个多样环境中的特定任务的人类远程操作数据来实现环境泛化 ([Physical Intelligence, 2025](#))。此外，虽然当前VLAs试图通过覆盖大量语言条件化运动原语库来实现任务泛化 ([Gemini Robotics Team, 2025](#))，但这种方法从根本上受到限制，因为用一组固定的片段级别语言条件化任务来捕捉海量可能的物理交互和运动是不现实的。相比之下，基于视频的世界模型从数据中的每一对连续帧中学习，同时利用大规模视频预训练来理解物理动力学。

2.2. 基于视频模型的机器人策略

机器人领域的视频生成。先前的研究表明，视频生成模型可用于合成机器人轨迹，并在测试时通过多种方法提取可执行动作：逆动力学模型（Du等人，2023；Zhou等人，2024）、作为密集对应的光流（Ko等人，2024），或作为高层规划的轨迹预测（Du等人，2024；Yang等人，2024）。其他工作生成人类视频——无论是通过三维追踪（Liang等人，2024），还是针对新场景和动作（Bharadhwaj等人，2024；Chen等人，2025）——并利用点跟踪目标训练策略。最近，（Jang等人，2025；Luo等人，2025）证明视频生成模型能够为未见行为和新环境生成合成机器人数据，充分利用了这些模型强大的泛化能力。

联合视频与动作生成。另一项工作将视频与动作生成相结合，用于端到端学习。这些方法表明，在动作预测中融入世界建模目标，能够提升多任务性能、样本效率以及对新场景和新物体的泛化能力。先前的工作（Cheang等人，2024；Li等人，2025；Won等人，2025；Wu等人，2024；Zhao等人，2025；Zheng等人，2025；Zhu等人，2025）从零开始或基于视觉-语言-动作模型学习联合世界建模与动作预测，而更新的工作（Hu等人，2024；Kim等人，2026；Liang等人，2025；Liao等人，2025；Pai等人，2025）则利用预训练视频扩散模型来继承丰富的视觉动力学先验。我们将这些模型统称为世界动作模型（WAMs），因为它们利用世界建模能力（预测未来状态）来进行动作预测。我们使用术语“世界动作模型（WAMs）”而非“视频动作模型（VAMs）”，以反映视频只是其中一种可能的世界建模目标——未来的WAMs可能将动作与其他预测模态（如触觉感知、力反馈或学习到的潜在表示）对齐。与先前的WAMs相比，DreamZero系统地探索数据多样性和规模以揭示WAMs的全部泛化潜力，采用更适合长时域世界-动作建模的自回归架构，在跨新任务和新环境方面实现了最先进的泛化能力，并且实现了最先进的跨具身化迁移，既能从不同具身（仅视频）中学习，也能通过少样本适应新具身。

为什么WAMs。 基于视频扩散骨干的WAMs继承了来自网络规模数据的丰富时空先验，融合了两种范式的优势：端到端VLA的无缝梯度流，以及用于规划的密集世界建模监督。与在紧凑潜在空间中从头学习动力学的潜在世界模型（Assran等人，2025；Hafner等人，2019，2020，2023）不同，WAMs利用了已从互联网规模数据中编码了物理动力学的预训练视频表示。此方法的核心是学习视频与动作的联合分布——DREAMZERO同时学习这两种模态，其中视频预测作为隐式视觉规划器来指导动作生成。这种公式不仅意味着提升机器人能力等同于改进视频生成，还实现了当前VLA所不具备的三项能力：对新任务的零样本泛化、从异构机器人数据中有效学习，以及基于视频的极其高效的跨具身迁移。我们在附录A中进一步讨论了WAMs与替代世界模型架构（如潜在空间、3D点云）之间的差异。

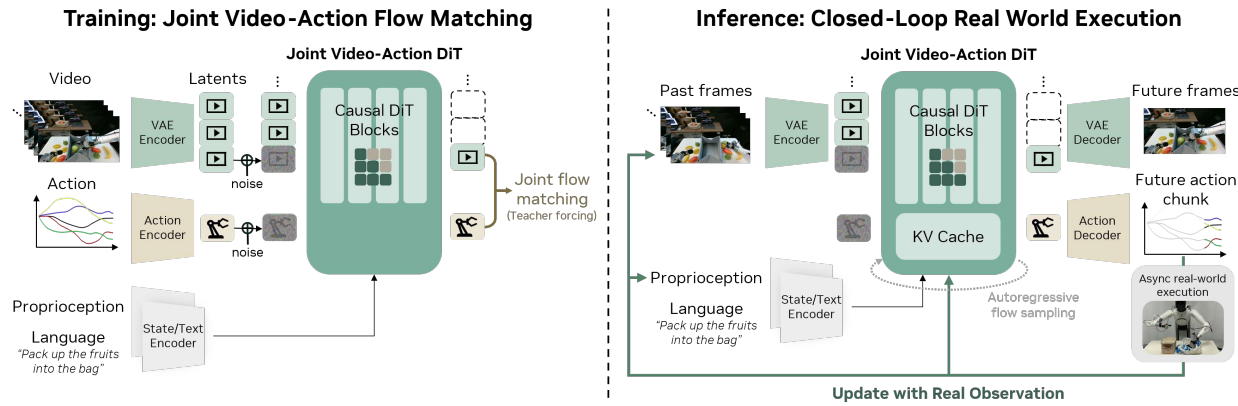


Figure 4: **Model Architecture of DREAMZERO.** The model takes three inputs: visual context (encoded via a VAE), language instructions (via a text encoder), and proprioceptive state (via a state encoder). These are processed by an autoregressive DiT backbone using flow matching, which jointly predicts future video frames and actions through separate decoders. During training (left), for each chunk, the model denoises noisy video and action latents conditioned on clean video context. During inference (right), predictions are executed asynchronously in the real world, and ground-truth observations are fed back into the KV cache to prevent error accumulation.

3. DREAMZERO

Pretrained video diffusion models offer rich spatiotemporal priors from web-scale data, making them attractive backbones for robot policies. However, converting these models into effective World Action Models (WAMs) presents three key challenges: (1) **Video-action alignment**: jointly predicting video and actions requires tight coupling between visual futures and motor commands, yet naively combining separate video and action heads can lead to misalignment; (2) **Architectural design**: it remains unclear whether bidirectional or autoregressive architectures are better suited for WAMs, with implications in modality alignment, error accumulation, and inference efficiency; and (3) **Real-time inference**: video diffusion models require iterative denoising across high-dimensional latent spaces, making them prohibitively slow for closed-loop control.

DREAMZERO addresses these challenges through three design choices. First, we train a single end-to-end model that jointly denoises video and action with a shared objective, ensuring deep integration between modalities. Second, we adopt an autoregressive architecture and exploit the closed-loop setting: after each action chunk is executed, we replace predicted frames with ground-truth observations in the KV cache, eliminating compounding errors while enabling efficient inference via KV caching and preserving native frame rates for precise modality alignment (See right side of Figure 4). Third, we introduce a suite of system-, implementation-, and model-level optimizations that achieve a $38\times$ inference speedup, enabling real-time control at 7Hz. We detail the model architecture in Section 3.1 and real-time execution in Section 3.2.

3.1. Model Architecture

Problem Formulation. DREAMZERO jointly predicts video $\mathbf{o}_{l:l+H}$ and actions $\mathbf{a}_{l:l+H}$ conditioned on language instruction $\mathbf{c}$, proprioceptive state $\mathbf{q}_l$ and visual observation including the current and the past history $\mathbf{o}_{0:l}$ where $H > 0$ is a fixed horizon and l is a random index sampled from a trajectory. Note that joint prediction of video and action is a decomposition of (1) autoregressive video prediction and (2) action prediction from an inverse-dynamics model (IDM):

$$\underbrace{\pi_0(\mathbf{o}_{l:l+H}, \mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{DREAMZERO}} = \underbrace{\pi_0(\mathbf{o}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{video prediction}} \underbrace{\pi_0(\mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l+H}, \mathbf{q}_l)}_{\text{IDM}} \quad (1)$$

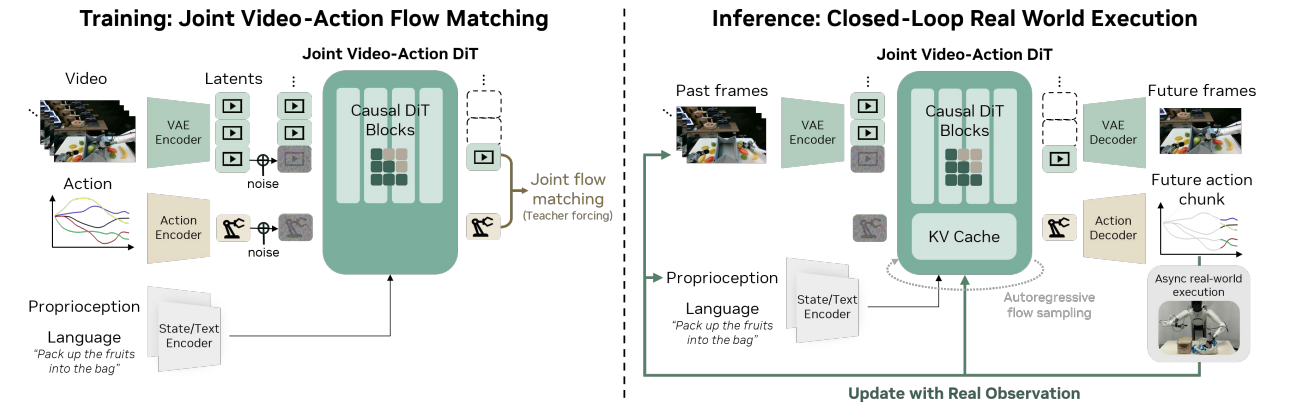


图4: **DreamZero的模型架构。** 模型接受三种输入：视觉上下文（通过VAE编码）、语言指令（通过文本编码器）和本体感知状态（通过状态编码器）。这些输入由自回归DiT骨干网络使用流匹配进行处理，该网络通过独立的解码器联合预测未来的视频帧和动作。在训练期间（左图），对于每个块，模型在干净视频上下文的条件下对噪声视频和动作潜变量进行去噪。在推理期间（右图），预测在真实世界中异步执行，真实观测被反馈到KV缓存中以防止误差累积。

3. DreamZero

预训练视频扩散模型从网络规模数据中提供了丰富的时空先验，使其成为机器人策略的有吸引力的主干网络。然而，将这些模型转化为有效的世界动作模型（WAMs）面临三个关键挑战：(1) **视频-动作对齐**：联合预测视频和动作需要视觉未来与电机指令之间的紧密耦合，然而简单地将独立的视频头和动作头组合可能导致不对齐；(2) **架构设计**：目前仍不清楚双向架构还是自回归架构更适合WAMs，这涉及模态对齐、误差累积和推理效率等方面的影响；(3) **实时推理**：视频扩散模型需要在高维潜空间中进行迭代去噪，使其在闭环控制中过于缓慢。

DREAMZERO 通过三种设计选择解决了这些挑战。首先，我们训练一个单一的端到端模型，通过共享目标联合去噪视频和动作，确保模态之间的深度融合。其次，我们采用自回归架构并利用闭环设定：每个动作块执行后，我们用真实观测替换KV缓存中的预测帧，消除复合误差，同时通过KV缓存技术实现高效推理，并保持原生帧率以实现精确的模态对齐（见图4右侧4）。第三，我们引入一系列系统级、实现级和模型级优化，实现 $38\times$ 推理加速，实现7赫兹的实时控制。我们将在第3.1节详细介绍模型架构，并在第3.2节介绍实时执行。

3.1. 模型架构

问题表述。 DreamZero 联合预测视频 $\mathbf{o}_{l:l+H}$ 和动作 $\mathbf{a}_{l:l+H}$ ，条件为语言指令 $\mathbf{c}$ 、本体感知状态 $\mathbf{q}_l$ 以及视觉观测（包括当前和过去的时序历史 $\mathbf{o}_{0:l}$ ），其中 $H > 0$ 是固定时域， l 是从轨迹中随机采样的索引。注意，视频和动作的联合预测是以下两部分的分解：(1) 自回归视频预测和(2) 基于逆向动力学模型（IDM）的动作预测：

$$\underbrace{\pi_0(\mathbf{o}_{l:l+H}, \mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{DREAMZERO}} = \underbrace{\pi_0(\mathbf{o}_{l:l+H} \mid \mathbf{o}_{0:l}, \mathbf{c}, \mathbf{q}_l)}_{\text{video prediction}} \underbrace{\pi_0(\mathbf{a}_{l:l+H} \mid \mathbf{o}_{0:l+H}, \mathbf{q}_l)}_{\text{IDM}} \quad (1)$$

Instead of using two separate models (video prediction model and inverse dynamics model) to model the decomposed objective (Li et al., 2026; Pai et al., 2025), we train a single model end-to-end with joint prediction objective. We believe that this end-to-end design enables better video-action alignment through a deep integration between the two modalities. Since pretrained video models are already optimized on the video prediction objective on diverse web-scale video data, DREAMZERO only needs to additionally learn to predict videos for the robot embodiment videos and extract corresponding actions from the generated videos. We further hypothesize that this encourages better generalization than the conventional practice of training VLA from VLM, as our approach explicitly learns temporal dynamics from video frames used both as conditioning inputs and prediction targets.

Model Architecture. The model architecture is shown in Figure 4. To retain the generalization capability of video models, we introduce minimal additional parameters: state encoders, action encoders, and decoders. For robot training data that contains multiple views, we concatenate all views into a single frame instead of making architectural changes to the backbone model.

In particular, DREAMZERO is trained to predict video frames and corresponding actions autoregressively. Autoregressive generation possesses the following advantages: (1) it enables faster inference speed by utilizing KV-cache, (2) the policy model can leverage the visual observation history as guidance for the next generation, and (3) it avoids the modality alignment challenges (video, action, and language alignment) inherent to bidirectional models. Concretely, bidirectional diffusion typically requires processing fixed-length sequences, which often necessitates video subsampling that distorts native FPS, potentially harming video-action alignment. On the other hand, autoregressive generation leverages KV caching to support arbitrarily long contexts within a single forward pass. This preserves the native frame rate, ensuring precise alignment between video frames and robot actions. Some details illustration of this difference is provided in Appendix B.

We introduce autoregressive modeling only for the video modality to avoid error propagation coming from closed-loop action prediction. DREAMZERO is trained to predict video frames in a *chunk* manner; each chunk has a fixed number of latent frames K to match the action horizon. Chunk-wise generation enables training on variable length of videos, similar to how LLMs are trained on variable length of language tokens. We provide more details on the QKV attention masking strategy for the different modalities in Appendix C.

Training Objective. Similar to recent video diffusion models and VLAs, we employ flow-matching (Albergo et al., 2023; Lipman et al., 2022; Liu et al., 2022) as the training objective (Ali et al., 2025; Team Wan, 2025; Teng et al., 2025). Unlike recent WAMs (Kim et al., 2026; Li et al., 2025; Liao et al., 2025; Zhu et al., 2025), DREAMZERO shares the denoising timestep between video and action modality for faster convergence at the beginning of training. Also, we apply teacher forcing (Gao et al., 2024; Jin et al., 2024) as a training objective; the model is trained to denoise the noisy current chunk conditioned on the clean previous chunks.

Formally, given a chunk index $k > 0$ and the denoising timestep $t_k \in [0, 1]$, we denote the corresponding noisy video latent vector for original video $\mathbf{o}^k$ as $\mathbf{z}_{t_k}^k$ and noisy normalized actions as $\mathbf{a}_{t_k}^k$. All frames within the same chunk share the same timestep t_k , while different chunks are assigned independent timesteps. Our model denoises $\mathbf{z}_{t_k}^k$ and $\mathbf{a}_{t_k}^k$, defined as linear interpolations between clean vectors and random Gaussian noises:

$$\mathbf{z}_{t_k}^k = t_k \mathbf{z}_1^k + (1 - t_k) \mathbf{z}_0^k, \quad \mathbf{a}_{t_k}^k = t_k \mathbf{a}_1^k + (1 - t_k) \mathbf{a}_0^k, \quad (2)$$

where $\mathbf{z}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{a}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\mathbf{z}_1^k$ and $\mathbf{a}_1^k$ are a clean video latent vector and a normalized action, respectively. Thus, the clean context from the previous chunks can be denoted as $\mathcal{C}_k = \{(\mathbf{z}_1^j, \mathbf{a}_1^j)\}_{j=1}^{k-1}$.

We train the model $\mathbf{u}_\theta$ to predict the joint velocity for both modalities using the following flow-matching objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{z}, \mathbf{a}, \{t_k\}} \left[\frac{1}{K} \sum_{k=1}^K w(t_k) \left\| \mathbf{u}_\theta([\mathbf{z}_{t_k}^k, \mathbf{a}_{t_k}^k]; \mathcal{C}_k, \mathbf{c}, \mathbf{q}_k, t_k) - \mathbf{v}^k \right\|^2 \right], \quad (3)$$

我们没有像传统做法那样使用两个独立模型（视频预测模型和逆向动力学模型）来建模上述分解目标（Li等人, 2026; Pai等人, 2025），而是训练一个单一模型，以端到端的方式实现联合预测目标。我们相信，这种端到端设计能够通过两种模态的深度融合实现更好的视频-动作对齐。由于预训练视频模型已经在多样化的网络规模视频数据上针对视频预测目标进行了优化，DREAMZERO 只需额外学习预测机器人具身视频，并从生成的视频中提取对应动作。我们进一步假设，与从视觉-语言模型训练 VLA 的常规做法相比，这种方法能够促进更好的泛化，因为我们的方法显式地从视频帧中学习时间动力学，而这些视频帧同时作为条件输入和预测目标。

模型架构。 模型架构如图4所示。为保留视频模型的泛化能力，我们引入了少量额外参数：状态编码器、动作编码器和解码器。对于包含多视图的机器人训练数据，我们将所有视图拼接成一个单帧，而不对骨干模型进行架构修改。

具体而言，DREAMZERO 经过训练，能够自回归地预测视频帧和相应动作。自回归生成具有以下优势：（1）通过利用KV缓存技术实现更快的推理速度；（2）策略模型可以利用视觉观测历史作为下一轮生成的指导；（3）它避免了双向模型固有的模态对齐挑战（视频、动作与语言的对齐）。具体来说，双向扩散通常需要处理固定长度序列，这往往需要进行视频下采样，从而扭曲原始帧率，可能损害视频-动作对齐。另一方面，自回归生成利用KV缓存技术支持在单次前向传播中处理任意长的上下文。这保留了原始帧率，确保视频帧与机器人动作之间的精确对齐。关于这一差异的详细说明见附录B。

我们仅对视频模态引入自回归建模，以避免闭环动作预测导致的误差传播。DREAMZERO 被训练为以块的方式预测视频帧；每个块具有固定数量的潜在帧 K 以匹配动作范围。逐块生成使得能够在可变长度视频上进行训练，类似于大语言模型在可变长度语言令牌上的训练方式。我们在附录C中提供了关于不同模态的QKV注意力掩码策略的更多细节。

训练目标。与最近的视频扩散模型和视觉-语言-动作模型类似，我们采用流匹配（Albergo et al., 2023; Lipman et al., 2022; Liu et al., 2022）作为训练目标（Ali et al., 2025; Team Wan, 2025; Teng et al., 2025）。与最近的WAMs（Kim et al., 2026; Li et al., 2025; Liao et al., 2025; Zhu et al., 2025）不同，DreamZero在视频和动作模态之间共享去噪时间步，以便在训练初期实现更快的收敛。此外，我们应用教师强制（Gao et al., 2024; Jin et al., 2024）作为训练目标；模型被训练为在干净先前块的条件下去噪当前块。

形式上，给定块索引 $k > 0$ 和去噪时间步 $t_k \in [0, 1]$ ，我们将原始视频 $\mathbf{o}^k$ 对应的带噪视频潜在向量记为 $\mathbf{z}_{t_k}^k$ ，将带噪归一化动作记为 $\mathbf{a}_{t_k}^k$ 。同一块内的所有帧共享相同的时间步 t_k ，而不同的块被分配独立的时间步。我们的模型对 $\mathbf{z}_{t_k}^k$ 和 $\mathbf{a}_{t_k}^k$ 进行去噪，它们被定义为干净向量与随机高斯噪声之间的线性插值：

$$\mathbf{z}_{t_k}^k = t_k \mathbf{z}_1^k + (1 - t_k) \mathbf{z}_0^k, \quad \mathbf{a}_{t_k}^k = t_k \mathbf{a}_1^k + (1 - t_k) \mathbf{a}_0^k, \quad (2)$$

其中 $\mathbf{z}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{a}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ，以及 $\mathbf{z}_1^k$ 和 $\mathbf{a}_1^k$ 分别是干净视频潜在向量和归一化动作。因此，来自先前块的干净上下文可以表示为 $\mathcal{C}_k = \{(\mathbf{z}_1^j, \mathbf{a}_1^j)\}_{j=1}^{k-1}$ 。

我们训练模型 $\mathbf{u}_\theta$ 来预测两个模态的联合速度，使用以下流匹配目标：

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{z}, \mathbf{a}, \{t_k\}} \left[\frac{1}{K} \sum_{k=1}^K w(t_k) \left\| \mathbf{u}_\theta([\mathbf{z}_{t_k}^k, \mathbf{a}_{t_k}^k]; \mathcal{C}_k, \mathbf{c}, \mathbf{q}_k, t_k) - \mathbf{v}^k \right\|^2 \right], \quad (3)$$

where $w(t_k) > 0$ is a predefined weight function for t_k , $\mathbf{c}$ is the text condition, $\mathbf{q}_k$ is the proprioceptive states of k -th chunk, and the velocity $\mathbf{v}^k := [\mathbf{z}_1^k, \mathbf{a}_1^k] - [\mathbf{z}_0^k, \mathbf{a}_0^k]$. To enable efficient training, we perform trajectory-level updates and apply attention masking (e.g., see Figure 14 for details) so that the current noisy chunk can attend to clean context of previous chunks. We provide the pseudo-code in Algorithm 1.

Model Inference. As shown in Figure 4, during inference, DREAMZERO jointly denoises video and action chunks, leveraging KV caching for efficiency (Huang et al., 2025; Teng et al., 2025; Yin et al., 2025). Unlike pure video generation, our closed-loop setting allows ground-truth observations to replace generated frames in the KV cache after each action execution (see Figure 14). This eliminates the compounding error problem inherent to autoregressive video generation—a key advantage unique to WAMs. Moreover, as a stateful policy, DREAMZERO can leverage visual history for tasks requiring memory.² We provide the pseudo-code of inference in Algorithm 2

3.2. Real-time Execution of DREAMZERO

Diffusion-based WAMs inherit powerful generalization from video foundation models, but their iterative denoising process creates a fundamental tension with reactive robotic control. We address two questions: (1) What prevents WAMs from being reactive policies? (2) How do we resolve this for real-time control?

3.2.1. The Reactivity Gap

Reactive policies must respond to environmental changes within tens of milliseconds. A naive implementation of DREAMZERO on a single GPU requires approximately 5.7 seconds per action chunk due to three bottlenecks: (1) iterative denoising across 16 diffusion steps required for smooth actions, (2) the computational cost of a 14B parameter DiT backbone, and (3) sequential execution that blocks robot motion during inference. This latency makes closed-loop control infeasible.³

3.2.2. Asynchronous Closed-Loop Execution

Our first step towards resolving this is through asynchronous execution that decouples inference from action execution. Rather than waiting for each inference to complete, the motion controller continuously executes the most recent action chunk while inference runs concurrently on the latest observation. This structure transforms the latency constraint from “inference must complete before the robot moves” to “inference must complete before the current action chunk expires.” In our experiments, we deploy policies at an action horizon of 48 steps at 30Hz control frequency (1.6 seconds per chunk) for bimanual manipulation robots. Hence, we target inference latency below approximately 200ms to ensure sufficient overlap for smooth, reactive control.

3.2.3. System-level Optimizations

Given the asynchronous execution structure, we optimize inference throughput through parallelism and caching.

- **CFG Parallelism.** Classifier-free guidance (Ho and Salimans, 2022) requires two forward passes (conditional and unconditional). We distribute these across two GPUs, reducing per-step latency by 47%.
- **DiT Caching.** We exploit the directional consistency of velocity predictions during flow matching. When cosine similarity between successive velocities exceeds a threshold, we reuse cached velocities, reducing effective DiT steps from 16 to 4 with minimal quality loss on action prediction.

3.2.4. Implementation-level Optimizations

We further reduce latency through compiler and kernel enhancements.

²In this work, we do not explicitly evaluate or post-train DREAMZERO on tasks that can only succeed with memory. We leave this for future work.

³One might expect that generating only actions (not video) would accelerate inference, but at 14B scale we empirically found out that the speed gain is minimal—the number of diffusion steps and the number of DiT blocks dominate latency. Moreover, because video and action are jointly trained for strong cross-modal alignment, naively reducing action denoising steps degrades quality. This motivates DREAMZERO-Flash.

其中 $w(t_k) > 0$ 是一个预定义的权重函数，用于 t_k , $\mathbf{c}$ 是文本条件， $\mathbf{q}_k$ 是第 k 块的本体感觉状态，而速度 $\mathbf{v}^k := [\mathbf{z}_1^k, \mathbf{a}_1^k] - [\mathbf{z}_0^k, \mathbf{a}_0^k]$ 。为了实现高效训练，我们执行轨迹级更新并应用注意力掩码（例如，详见图 14 中的细节），以便当前噪声块可以关注先前块的干净上下文。我们提供了伪代码，在算法1。

模型推理。如图4，在推理过程中，DREAMZERO 联合去噪视频和动作块，利用KV缓存技术提升效率（Huang等人，2025；Teng等人，2025；Yin等人，2025）。与纯视频生成不同，我们的闭环设定允许在每次动作执行后，用真实观测替换KV缓存中生成的帧（见图14）。这消除了自回归视频生成固有的累积误差问题——这是WAMs独有的关键优势。此外，作为一种有状态策略，DREAMZERO 可以利用视觉历史来处理需要记忆的任务。²。我们在算法2

3.2. DreamZero的实时执行

基于扩散的世界行动模型继承了视频基础模型的强大泛化能力，但其迭代去噪过程与反应式机器人控制之间存在根本性的矛盾。我们解决两个问题：（1）是什么阻止WAMs成为反应式策略？（2）我们如何解决这一问题以实现实时控制？

3.2.1. 反应性差距

反应式策略必须在几十毫秒内响应环境变化。在单GPU上对DreamZero的朴素实现每个动作块需要大约5.7秒，原因有三个瓶颈：（1）需要16个扩散步数的迭代去噪以实现平滑动作，（2）140亿参数DiT骨干的计算成本，以及（3）顺序执行在推理期间阻塞机器人运动。这种延迟使得闭环控制不可行。³

3.2.2. 异步闭环执行

我们解决这个问题的第一步是通过异步执行，将推理与动作执行解耦。运动控制器不是等待每次推理完成，而是持续执行最新的动作块，同时推理在最新的观测上并发运行。这种结构将延迟约束从“推理必须在机器人移动之前完成”转变为“推理必须在当前动作块过期之前完成”。在我们的实验中，我们对双臂操作机器人在30Hz控制频率下部署了动作范围为48步的策略（每块1.6秒）。因此，我们的目标是推理延迟低于大约200毫秒，以确保足够的重叠以实现平滑、反应式控制。

3.2.3. 系统级优化

鉴于异步执行结构，我们通过并行性和缓存来优化推理吞吐量。

- **CFG并行。**无分类器引导（Ho和Salimans, 2022）需要两次前向传播（有条件和无条件）。我们将这两次传播分配到两个GPU上，从而将每步延迟降低47%。
- **DiT缓存。**我们利用流匹配过程中速度预测的方向一致性。当连续速度之间的余弦相似度超过阈值时，我们复用缓存速度，将有效DiT步数从16步减少到4步，同时动作预测的质量损失极小。

3.2.4. 实现级优化

我们通过编译器和内核增强进一步降低延迟。

²在本工作中，我们未显式评估或对DreamZero进行后训练以处理仅凭记忆才能成功的任务。我们将其留待未来工作。³人们可能会认为仅生成动作（而非视频）能加速推理，但在14B规模下，我们通过经验发现速度提升微乎其微——扩散步数和DiT块的数量主导了延迟。此外，由于视频和动作是联合训练以实现强跨模态对齐的，单纯减少动作去噪步数会降低质量。这促使了DreamZero-Flash的诞生。

- **Torch Compile and CUDA Graphs.** We apply `torch.compile` with CUDA Graphs to eliminate CPU overhead and fuse operators. Static shapes cause recompilations only during the first trajectory.
- **Post-Training Quantization.** On Blackwell architecture, we quantize weights and activations to NVFP4 while keeping sensitive operations (QKV, Softmax) in FP8 and non-linear operations in FP16.
- **Kernel and Scheduler Enhancements.** We use the cuDNN backend for attention and migrate scheduler operations to GPU to eliminate CPU-GPU synchronization stalls.

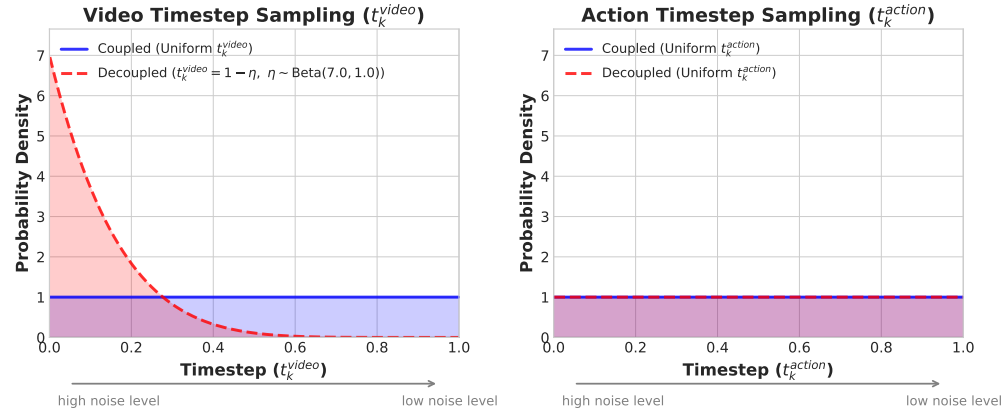


Figure 5: **Decoupled Noise Schedules.** DREAMZERO (blue) uses coupled noise for video and action (both uniform). DREAMZERO-Flash (red) biases video toward high-noise states via a Beta distribution while keeping action noise uniform, training the model to predict clean actions from noisy visual context.

3.2.5. Model-level Optimizations: DREAMZERO-Flash

Even with system optimizations, the number of diffusion steps remains the primary latency bottleneck. However, naively reducing steps degrades action quality because residual visual noise propagates into action predictions.

DREAMZERO-Flash addresses this by decoupling video and action noise schedules during training. The key insight is that, at inference time, actions should denoise to their final values while being conditioned on a still-noisy video representation within the current chunk, since with very few denoising steps (e.g., fewer than 4), the generated video tokens may remain inaccurate and thus provide a noisy conditioning signal. Standard DREAMZERO samples a shared timestep $t_k \sim \mathcal{U}(0, 1)$ for both modalities. This creates a train-test mismatch: during training, the model learns to predict actions when video and action are at the *same* noise level, but few-step or single-step inference requires predicting clean actions while video remains partially noisy.

DREAMZERO-Flash closes this gap by biasing video timesteps toward high-noise states via $t_k^{\text{video}} = 1 - \eta$, where $\eta \sim \text{Beta}(\alpha, \beta)$ with $\alpha > \beta$. In practice, we use Beta(7, 1) as an example configuration, yielding $\mathbb{E}[t_k^{\text{video}}] = 0.125$ (predominantly noisy), while action timesteps remain uniform (Figure 5). During training, this exposes the model to configurations where it must predict clean actions from noisy visual context, directly matching the few-step or single-step inference regime. As a result, we reduce the diffusion steps from four to one, cutting inference from $\sim 350\text{ms}$ to $\sim 150\text{ms}$ with minimal performance loss (Table 3). Moreover, the Flash formulation enables flexible training configurations—such as varying the noise sampling ratios of video and action—to better align training with different few-step or single-step inference regimes. In practice, we mainly apply Flash training as the final stage following the main DREAMZERO model training.

Action Chunk Smoothing. To suppress high-frequency noise in generated actions, we upsample chunks to $2\times$ resolution, apply a Savitzky-Golay filter, and downsample to original resolution.

3.2.6. Summary

Table 1 summarizes cumulative speedups. System and implementation optimizations yield $\sim 9\times$ speedup on H100 and $\sim 16\times$ on GB200; adding DREAMZERO-Flash achieves $38\times$ on GB200, reducing latency from 5.7s to

- **Torch Compile和CUDA Graphs.** 我们使用`torch.compile`结合CUDA Graphs来消除CPU开销并融合算子。静态形状仅在第一条轨迹期间导致重新编译。
- **训练后量化。**在Blackwell架构上，我们将权重和激活量化为NVFP4，同时将敏感操作（QKV、Softmax）保持在FP8，非线性操作保持在FP16。
- **内核和调度器增强。**我们使用cuDNN后端进行注意力计算，并将调度器操作迁移到GPU以消除CPU-GPU同步停滞。

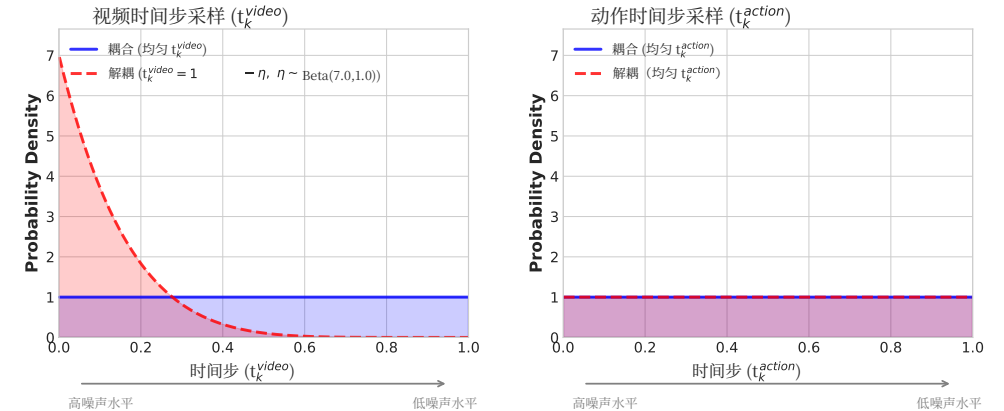


图5: **解耦噪声调度。**DreamZero（蓝色）对视频和动作使用耦合噪声（均匀）。DreamZero-Flash（红色）通过Beta分布使视频偏向高噪声状态，同时保持动作噪声均匀，训练模型从噪声视觉上下文中预测干净动作。

3.2.5. 模型级别优化: DreamZero-Flash

即使经过系统优化，扩散步数仍然是主要的延迟瓶颈。然而，简单地减少步数会降低动作质量，因为残差视觉噪声会传播到动作预测中。

DreamZero-Flash 通过在训练期间解耦视频和动作噪声调度来解决此问题。其关键洞察在于，推理时，动作应在以当前块内仍带噪声的视频表示为条件的情况下，去噪至其最终值；因为当去噪步数非常少（例如少于4步）时，生成的视频令牌可能仍不准确，从而提供有噪声的条件信号。标准 DreamZero 为两种模态采样一个共享时间步 $t_k \sim \mathcal{U}(0, 1)$ 。这造成了训练-测试不匹配：在训练中，模型学习在视频和动作处于相同噪声水平时预测动作，而少步或单步推理则要求在视频仍部分有噪声时预测干净动作。

DREAMZERO-Flash 通过将视频时间步偏向高噪声状态来弥合这一差距，采用 $t_k^{\text{video}} = 1 - \eta$ ，其中 $\eta \sim \text{Beta}(\alpha, \beta)$ 参数为 $\alpha > \beta$ 。实践中，我们使用 Beta(7,1) 作为示例配置，得到 $\mathbb{E}[t_k^{\text{video}}] = 0.125$ （主要处于高噪声状态），而动作时间步保持均匀（图 5）。训练期间，这使模型面对必须从噪声视觉上下文中预测干净动作的配置，直接匹配少步或单步推理模式。因此，我们将扩散步数从四步减少到一步，将推理时间从 $\sim 350\text{毫秒}$ 降至 $\sim 150\text{毫秒}$ ，且性能损失极小（表 3）。此外，Flash formulation 实现了灵活的训练配置——例如改变视频和动作的噪声采样比例——以更好地将训练与不同的少步或单步推理模式对齐。实践中，我们主要将 Flash训练 作为主 DREAMZERO 模型训练之后的最终阶段。

动作块平滑。为抑制生成动作中的高频噪声，我们将块上采样至 $2\times$ 分辨率，应用Savitzky-Golay滤波器，再下采样回原始分辨率。

3.2.6. 总结

桌子 1 总结了累积加速。系统和实现优化产生 $\sim 9\times$ 在H100上的加速和 $\sim 16\times$ 在GB200上；添加 DREAMZERO-Flash achieves $38\times$ 在GB200上，将延迟从5.7秒减少到

150ms. With the exception of DiT caching and quantization, all system and implementation-level optimizations are mathematically equivalent to baseline and show no measurable performance degradation.

Optimization	H100	GB200
Baseline	1×	1.1×
<i>System-level</i>		
+ CFG Parallelism	1.9×	1.8×
+ DiT Caching	5.5×	5.4×
<i>Implementation-level</i>		
+ Torch Compile + CUDA Graphs	8.9×	10.9×
+ Kernel & Scheduler Opts.	9.6×	14.8×
+ Quantization (NVFP4)	—	16.6×
<i>Model-level</i>		
+ DREAMZERO-Flash	—	38×

Table 1: **Cumulative inference speedups.** Each row includes all optimizations above it. Entries marked “—” indicate features not applicable to that hardware.

4. Experimental Setup

We validate our main hypotheses about learning from diverse data on two robot embodiments: the AgiBot G1 mobile bimanual manipulator and the Franka single-arm robot. We pretrain separately for each embodiment, leaving multi-embodiment training for future work. For cross-embodiment experiments, we utilize both the YAM robot and human egocentric data. The experimental setup for AgiBot G1 is illustrated in Figure 7.

We compare against two state-of-the-art Vision-Language-Action models (VLAs): GR00T N1.6 (Bjorck et al., 2025) and $\pi_{0.5}$ (Physical Intelligence, 2025). For each baseline, we evaluate two initialization strategies: (1) *from-scratch*, using pretrained VLM weights without prior robot data training for a fair apple-to-apple comparison with DREAMZERO, and (2) *from-pretrained*, using official checkpoints pretrained on thousands of hours of cross-embodiment robot data. Both variants are then trained on identical data as DREAMZERO: ~ 500 hours of teleoperation data we collected for AgiBot G1, and DROID (Khazatsky et al., 2024) for Franka. We keep the compute budget comparable across all methods by matching total batch size and gradient steps.⁴

4.1. Pretraining

Data. Our data collection philosophy differs from that of existing VLAs. While recent works have shown that VLAs can learn effective policies from moderate-sized datasets, these approaches typically rely on structured, task-focused demonstrations to ensure consistent behavior. We hypothesize that learning to only predict actions without encoding the knowledge about future world states makes it challenging to leverage highly heterogeneous, non-repetitive data effectively, as the model must implicitly infer dynamics from noisy state-action pairs. In contrast, we hypothesize DREAMZERO’s world modeling objective enables effective learning from diverse demonstrations, allowing us to prioritize breadth and utility over repetition during data collection.

Using AgiBot G1, we collect approximately 500 hours of teleoperation data across 22 unique environments (see Figure 15), including homes, restaurants, supermarkets, coffee shops, and offices—prioritizing task diversity and real-world utility over task-specific repetition. As shown in Figure 6, each episode averages around 4.4 minutes and encompasses approximately 42 subtasks—significantly longer-horizon than typical robotic manipulation datasets (Khazatsky et al., 2024; Walke et al., 2023). The skill distribution reflects real-world deployment requirements: navigation enables movement between workspaces, while torso adjustments allow

⁴For from-pretrained baselines, this constitutes continual training on top of the official weights.

150毫秒。除了DiT缓存和量化之外，所有系统和实现级别优化在数学上等同于基线，并且没有显示出可衡量的性能下降。

优化	H100	GB200
基线	1×	1.1×
<i>系统级</i>		
+ CFG并行	1.9×	1.8×
+ DiT缓存	5.5×	5.4×
<i>实现级别</i>		
+ Torch Compile + CUDA Graphs	8.9×	10.9×
+ 内核与调度器优化	9.6×	14.8×
+ 量化 (NVFP4)	—	16.6×
<i>模型级别</i>		
+ DreamZero-Flash	—	38×

表1: **累积推理加速。** 每一行包含其上方所有优化。标记为 “—” 的条目表示该特性不适用于该硬件。

4. 实验设置

我们在两种机器人实体上验证了关于从多样化数据学习的主要假设：AgiBot G1 移动双臂机械臂和 Franka 单臂机器人。我们分别针对每种机器人实体进行预训练，将多具身化训练留待未来工作。在跨具身化实验中，我们同时使用了 YAM 机器人和人类第一人称数据。AgiBot G1 的实验设置如图7所示。

我们与两种最先进的视觉-语言-动作模型 (VLA) 进行比较：GR00T N1.6 (Bjorck等人,2025) 和 $\pi_{0.5}$ (Physical Intelligence, 2025)。对于每种基线，我们评估两种初始化策略：（1）从头训练，使用预训练 VLM 权重，无需事先的机器人数据训练，以便与 DREAMZERO进行公平的同类比较；（2）基于预训练，使用在数千小时跨具身化机器人数据上预训练过的官方检查点。随后，两种变体在与 DREAMZERO相同的数据上进行训练： ~ 500 小时我们为 AgiBot G1 收集的遥操作数据，以及用于 Franka 的 DROID 数据 (Khazatsky等人,2024)。我们通过匹配总批次大小和梯度步数，使所有方法的计算预算保持可比。⁴

4.1. 预训练

数据。 我们的数据收集理念与现有视觉-语言-动作模型 (VLAs) 有所不同。虽然近期研究已表明，视觉-语言-动作模型 (VLAs) 可以从中等规模的数据集中学习有效的策略，但这些方法通常依赖于结构化的、以任务为中心的演示来确保行为的一致性。我们假设，仅学习预测动作而不编码关于未来世界状态的知识，会使得有效利用高度异质、非重复的数据变得困难，因为模型必须从噪声状态-动作对中隐式推断动力学。相比之下，我们假设 DreamZero 的世界建模目标能够从多样的演示中实现有效学习，使我们能够在数据收集过程中优先考虑广度和实用性，而非重复性。

使用 AgiBot G1，我们在 22 个独特的环境（见图 15）中收集了约 500 小时的遥操作数据，包括家庭、餐厅、超市、咖啡店和办公室——优先考虑任务多样性和真实世界实用性，而非重复特定任务。如图 6所示，每个片段平均时长约 4.4 分钟，包含约 42 个子任务——比典型的机器人操作数据集 (Khazatsky 等人, 2024; Walke 等人, 2023) 的视野更长。技能分布反映了真实世界的部署需求：导航支持在工作空间之间移动，而躯干调整允许

⁴对于基于预训练的基线模型，这构成了在官方权重基础上进行的持续训练。

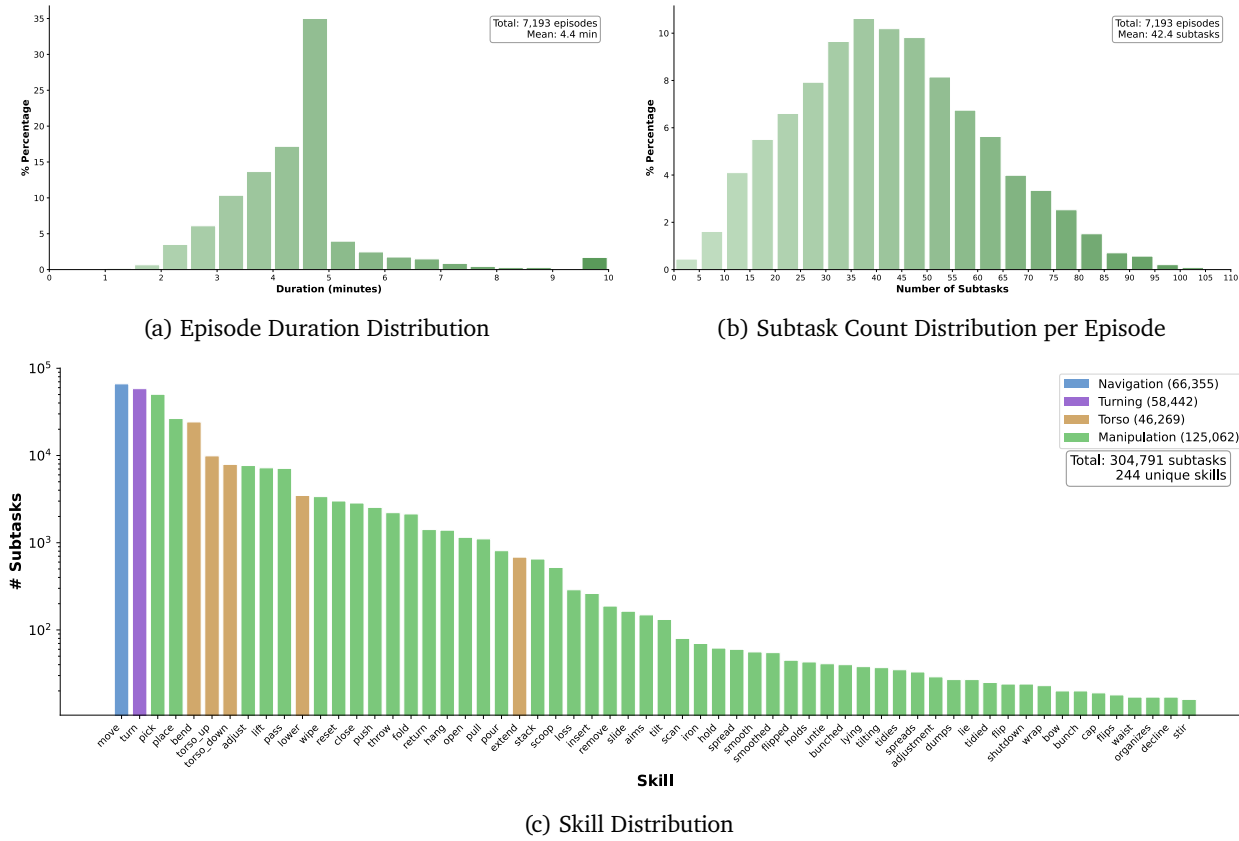


Figure 6: **Distribution statistics** for the AgiBot pretraining corpus: episode durations, subtask density, and skill coverage across 7.2K episodes (~500 hours).

interaction with objects at varying heights (shelves, cabinets). Additional details on the data collection pipeline are provided in Appendix E.⁵

We also validate DREAMZERO on the Franka single-arm robot using DROID (Khazatsky et al., 2024), one of the most heterogeneous publicly available robotic datasets to demonstrate the effectiveness of WAMs on diverse, open-source data and enables reproducibility prior to the release of our in-house AgiBot dataset. We open-source the checkpoint and inference code to run some DROID-sim evals in PolaRiS (Jain et al., 2025).⁶

Training. We use Wan2.1-I2V-14B-480P (Team Wan, 2025), a 14B image-to-video diffusion model, as the backbone for DREAMZERO. We train for 100K steps with a global batch size of 128 for AgiBot and 100K steps with a global batch size of 128 for DROID datasets. We update all DiT blocks, the state encoder, action encoder, and action decoder, while freezing the text encoder, image encoder, and VAE.⁷ For both datasets, we filter out idle actions and use relative joint positions as the default action representation. We also conduct some ablations (Section 5.2) where we initialize from Wan2.1-I2V-5B-480P to see the effect of model size (5B vs. 14B).

Evaluation Protocol. We evaluate models out of the box after pretraining. Our default evaluation setting is *unseen environments, unseen objects*—because our pretraining and post-training data were collected in a different geographic location from our evaluation sites, every benchmark inherently tests out-of-distribution generalization rather than interpolation within the training distribution. We evaluate on two categories: *seen* and *unseen* tasks. We define the granularity of a task as a combination of the motion required for the task and

⁵We plan to open-source this dataset in upcoming releases. Some samples can be found at https://dreamzero0.github.io/training_data_gallery

⁶Available at <https://github.com/dreamzero0/dreamzero>.

⁷We experimented with LoRA (Hu et al., 2022) but found it led to suboptimal results.

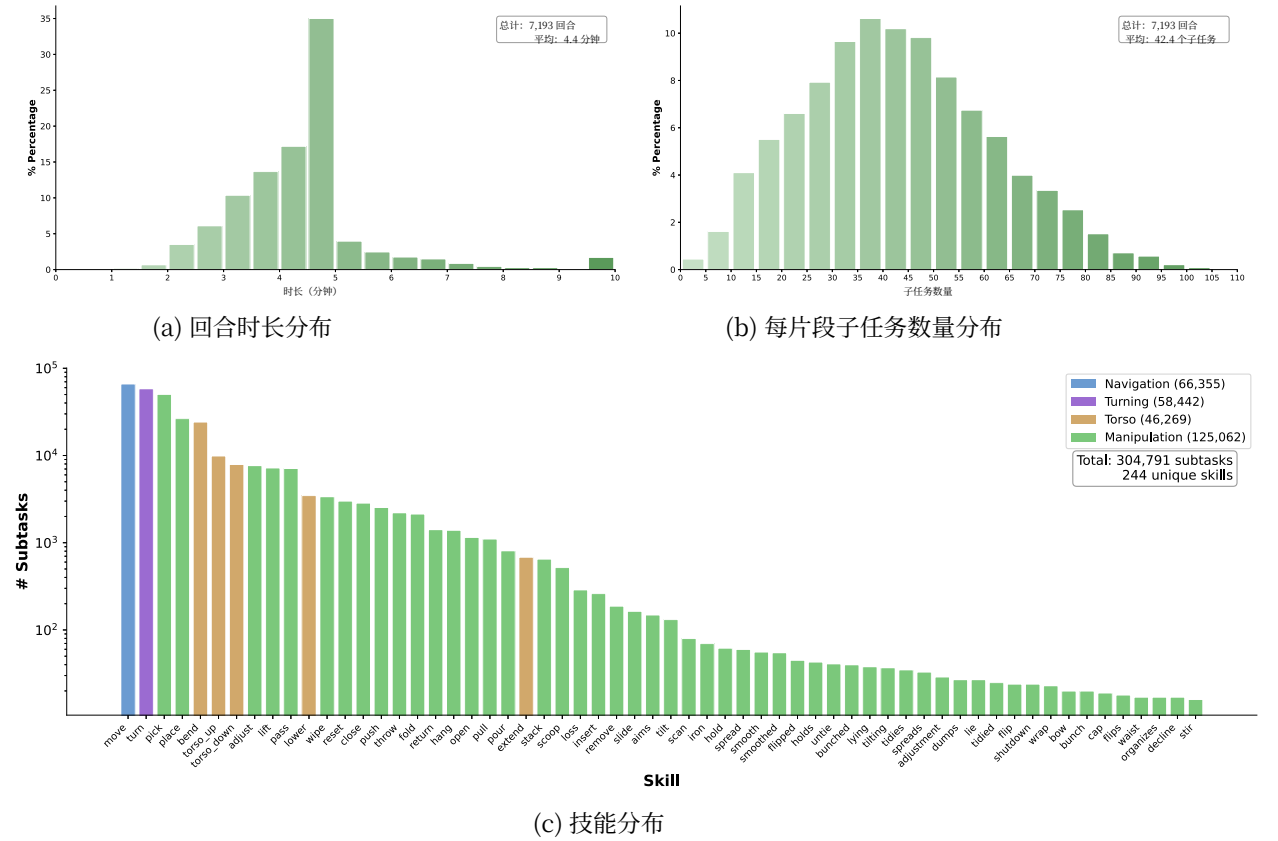


图6: 分布统计针对AgiBot预训练语料库: 回合时长、子任务密度以及技能覆盖率 across 7.2K 个回合 (~500 小时)。

与不同高度（架子、柜子）的物体进行交互。关于数据采集流程的更多细节见附录E。⁵

我们还验证了 DREAMZERO 在 Franka 单臂机器人上使用 DROID (Khazatsky 等人, 2024)，这是最具异质性的公开机器人数据集之一，以证明 WAMs 在多样化、开源数据上的有效性，并在我们内部 AgiBot 数据集发布之前实现可复现性。我们开源了检查点和推理代码，以便在 PolaRiS (Jain 等人, 2025) 中运行一些 DROID 模拟评估。⁶

训练。 我们使用 Wan2.1-I2V-14B-480P (TeamWan, 2025)，一个 14B 图像到视频扩散模型，作为 DREAMZERO 的骨干网络。我们在 AgiBot 数据集上训练了 100K 步，全局批次大小为 128；在 DROID 数据集上也训练了 100K 步，全局批次大小为 128。我们更新了所有 DiT 块、状态编码器、动作编码器和动作解码器，同时冻结了文本编码器、图像编码器和 VAE。⁷ 对于两个数据集，我们过滤掉空闲动作，并使用相对关节位置作为默认动作表示。我们还进行了一些消融实验（第 5.2 节），其中我们从 Wan2.1-I2V-5B-480P 初始化，以观察模型规模（5B vs. 14B）的影响。

评估协议。 我们在预训练后直接对模型进行开箱评估。我们的默认评估设置为未见环境、未见物体——因为我们的预训练和后训练数据是在与评估地点不同的地理位置收集的，每个基准测试都天然地测试分布外泛化，而不是训练分布内的插值。我们在两个类别上进行评估：见过和未见任务。我们将任务粒度定义为任务所需动作与

⁵我们计划在未来的版本中开源此数据集。部分样本可在以下网址找到：https://dreamzero0.github.io/training_数据_画廊⁶可在以下网址获取<https://github.com/dreamzero0/dreamzero>。⁷我们尝试了LoRA (Hu等人, 2022) 但发现它导致了次优结果。

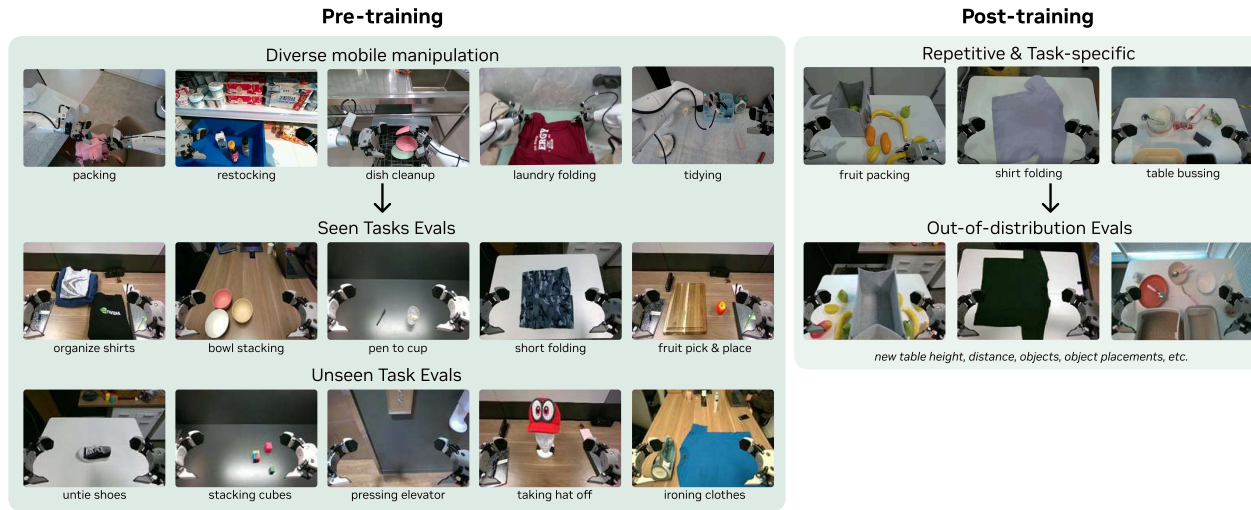


Figure 7: **AgiBot Evaluation Set-up**. We are first-citizens of generalization evals, where the default setting is *unseen* environment and *unseen* objects.

the object type. For example, if the training data contains folding a red-colored shirt and evaluate the model to fold a black-colored shirt with a different size, it is considered as a *seen* task. On the other hand, if we evaluate the model to fold socks, it is considered as an unseen task because the motion required to fold socks is different from folding a shirt (See samples in Figure 7).

AgiBot Evaluation Protocol. For *seen* tasks, we select 10 tasks from the pretraining distribution, including pick-and-place variants, stacking, wiping, and folding; we run 8 rollouts per task across 4 robots, each in different environments and different objects (80 rollouts total per checkpoint). We divide 10 seen tasks into three categories: PnP-Easy (Pick and place fruit, Wipe the mess, Take out fruit from bag), PnP-Hard (Pick and place fork/spoon, put the pen in pen holder, put the cup on the coaster, stack bowls/cups in a row), and Contact-Rich Manipulation (fold shirts, fold shorts, stack clothes). For *unseen* tasks, we evaluate 10 tasks absent from training—such as ironing, painting, pulling carts, cube stacking, removing a hat from a mannequin, and untying shoe laces—with 8 rollouts per task across 4 robots (80 rollouts total per checkpoint). The full list of each evaluation rollout initial frame and prompt is provided in Appendix F, and some evaluation rollouts can be found here.⁸

DROID Evaluation Protocol. We evaluate on 20 *seen* tasks and 20 *unseen* tasks (verbs absent from DROID), performing 2 rollouts per task, for a total of 80 evaluation rollouts across 40 tasks for each checkpoint. We compare DREAMZERO against the publicly released $\pi_{0.5}$ -DROID and an internally trained GR00T N1.6-DROID checkpoint. Object positions are fixed across checkpoints to ensure fairness. Each rollout is scored from 0 to 1.0 based on partial task completion; full details are provided in Appendix G.

4.2. Post-training

Beyond pre-training, we evaluate whether WAMs improve fine-tuning performance on task-specific data using the AgiBot robot.

Data. We collect post-training data on three downstream tasks:

- *Shirt folding* (33 hrs): Fold a flattened t-shirt through 5 sequential stages. We randomize initial shirt position across 2 shirt types.

⁸<https://dreamzero0.github.io> for main evaluation rollouts and https://dreamzero0.github.io/evals_gallery for accumulation of unique evaluation rollouts.



对象类型的组合。例如，如果训练数据包含折叠一件红色衬衫，而评估模型折叠一件不同尺寸的黑色衬衫，则这被视为一个见过任务。另一方面，如果我们评估模型折叠袜子，则其被视为未见任务，因为折叠袜子所需的动作与折叠衬衫不同（示例见图7）。

AgiBot评估协议。对于已见任务，我们从预训练分布中选择10个任务，包括拾取与放置变体、堆叠、擦拭和折叠；我们在4台机器人上每个任务运行8次，每台机器人处于不同的环境和不同的物体中（每个检查点共80次运行）。我们将10个已见任务分为三个类别：PnP-Easy（拾取并放置水果，擦拭污渍，从袋子中取出水果）、PnP-Hard（拾取并放置叉子/勺子，将笔放入笔筒，将杯子放在杯垫上，将碗/杯子排成一排堆叠）和Contact-Rich Manipulation（折叠衬衫，折叠短裤，堆叠衣物）。对于未见过的任务，我们评估10个训练中未出现的任务——如熨烫、绘画、拉车、立方体堆叠、从人体模型上取下帽子、解开鞋带——每个任务在4台机器人上运行8次（每个检查点共80次运行）。每次评估运行的初始帧和提示词的完整列表见附录F，部分评估运行可在此处找到。⁸

DROID评估协议。我们在20个已见任务和20个未见任务（DROID中缺失的动词）上进行评估，每个任务执行2次运行，每个检查点共40个任务80次评估运行。我们将DREAMZERO与公开发布的 $\pi_{0.5}$ -DROID以及内部训练的GR00T N1.6-DROID检查点进行比较。物体位置在不同检查点间固定以确保公平。每次运行根据部分任务完成情况评分0到1.0；完整细节见附录G。

4.2. 后训练

除了预训练，我们还评估WAMs在使用AgiBot机器人的任务特定数据上是否能提升微调性能。

数据。我们在三个下游任务上收集后训练数据：

- 衬衫折叠（33小时）：通过5个连续阶段折叠一件平铺的T恤。我们在2种衬衫类型中随机化初始衬衫位置。

⁸独特评估的模型 dreamzero0.github.io 用于主要评估运行和 https://dreamzero0.github.io/evals_ 图库用于累积评估运行。

- *Fruit packing* (12 hrs): Pack 10 fruits from a table into a bag. We randomize fruit combinations and positions of fruits and bag.
- *Table bussing* (40 hrs): Clear 5 pieces of trash into a trash bin and 5 pieces of dishware (dish, bowl, fork, and spoon) into a dish bin. We randomize object types, combinations, and positions.

Training. We post-train for 50K steps per task. As in pretraining, we update all parameters except the text encoder, image encoder, and VAE.

Evaluation Protocol. We measure average task progress across 10 rollouts per task. Task progress is defined as: (1) folding stages completed out of 5 for shirt folding, (2) fruits successfully packed out of 10 for fruit packing and (3) items cleared for table bussing. Following Barreiros et al. (2025), we apply an image overlay to the initial scene to reduce variance.

5. Experimental Results

5.1. Main Results

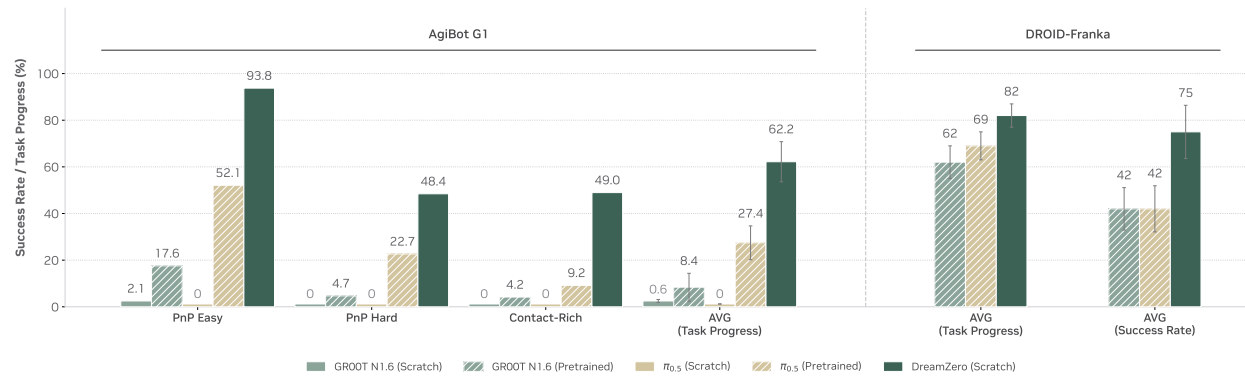


Figure 8: **Seen Task Evaluation.** DREAMZERO effectively learns from diverse data and generalizes to new environments, outperforming VLAs across all task categories. VLAs trained *from scratch* achieve near-zero success, while *pretrained* VLAs show modest performance—likely benefiting from embodiment-specific knowledge acquired through repetitive demonstrations during pretraining.

We evaluate the zero-shot generalization performance of DREAMZERO compared to baseline models and investigate the following research questions:

Q1. Do WAMs learn better from diverse, non-repetitive data?

We evaluate pretrained models out-of-the-box on tasks present in the pretraining data, but in zero-shot environments with unseen objects. Results are shown in Figure 8.

On AgiBot G1, *from-scratch* VLAs achieve near-zero task progress score across all categories. Even on simple pick-and-place tasks (PnP Easy), VLAs occasionally reach toward the correct object but fail to interact accurately with unseen objects in novel environments. In contrast, DREAMZERO successfully learns from heterogeneous data, achieving 62.2% average task progress—over $2\times$ higher than the best *pretrained* VLA baseline (27.4%), despite those baselines being pretrained on thousands of hours of cross-embodiment robot data before continued training on our data mix. On DROID-Franka, we show a similar result as well; DREAMZERO which is only trained on the DROID dataset outperforms pre-trained baseline models trained on multiple robot embodiment data.

We attribute this gap to the joint video-action formulation: while VLAs require massive robot data to learn direct observation-to-action mappings, WAMs leverage video generation as a strong prior for action prediction, enabling effective learning of diverse data and generalization to unseen environments. Notably, we observe tight

- 水果包装（12小时）：将10个水果从桌子上装入一个袋子。我们随机化水果组合以及水果和袋子的位置。
- 餐桌清理（40小时）：将5块垃圾清理到垃圾桶，将5件餐具（盘子、碗、叉子和勺子）清理到餐具盒。我们随机化物体类型、组合和位置。

训练。我们每个任务进行5万步的后训练。与预训练一样，我们更新除文本编码器、图像编码器和VAE之外的所有参数。

评估协议。我们衡量每个任务10次运行的平均任务进度。任务进度定义为：(1) 衬衫折叠中完成的折叠阶段数（共5个），(2) 水果包装中成功包装的水果数（共10个），(3) 餐桌清理中清理的物品数。遵循 Barreiros等人（2025），我们对初始场景应用图像叠加以减少方差。

5. 实验结果

5.1. 主要结果

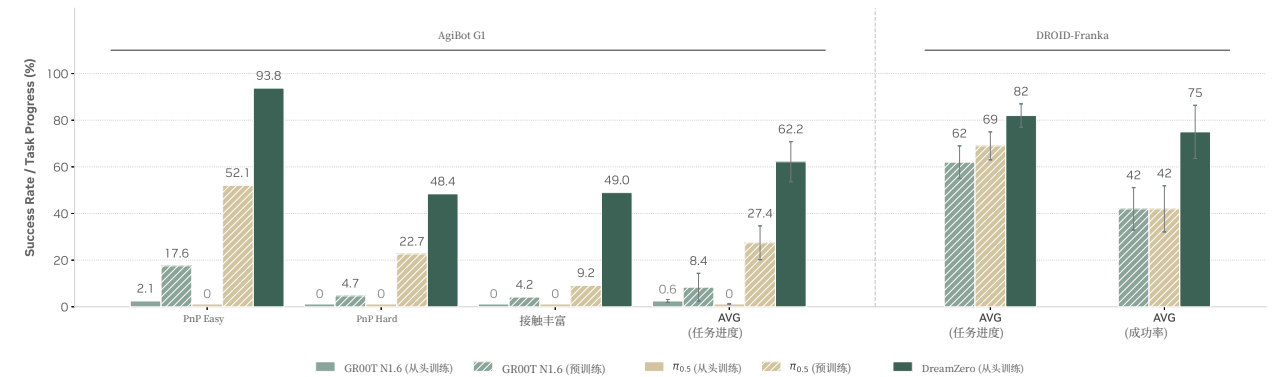


图8：已见任务评估。DreamZero有效地从多样化数据中学习并泛化到新环境，在所有任务类别上均优于视觉-语言-动作模型。从头训练的视觉-语言-动作模型实现近乎零成功率，而预训练的视觉-语言-动作模型表现一般——可能得益于在预训练期间通过重复演示获得的具身化特定知识。

我们评估了DreamZero与基线模型相比的零样本泛化性能，并探讨以下研究问题：

Q1. WAMs是否从多样、非重复的数据中学习得更好？

我们在预训练数据中存在的任务上评估开箱即用的预训练模型，但在零样本环境中使用未见物体。结果如图8所示。

在AgiBot G1上，从头训练的视觉-语言-动作模型在所有类别上获得近乎零任务进度得分。即使在简单的抓取放置任务（PnP Easy）中，视觉-语言-动作模型偶尔会伸向正确的物体，但无法在新环境中与未见物体准确交互。相比之下，DreamZero成功从异构数据中学习，实现了62.2%的平均任务进度——超过 $2\times$ ，高于最好的预训练的VLA基线（27.4%），尽管这些基线在继续在我们的数据混合上训练之前已经在数千小时的跨具身机器人数据上进行了预训练。在DROID-Franka上，我们同样展示了相似的结果；仅使用DROID数据集训练的DreamZero优于在多个机器人具身数据上训练的预训练基线模型。

我们将这一差距归因于联合视频动作公式化：虽然视觉-语言-动作模型需要大量的机器人数据来学习从观察到动作的直接映射，但WAMs利用视频生成作为动作预测的强先验，从而能够有效地学习多样化数据并泛化到未见过的环境。值得注意的是，我们观察到紧密

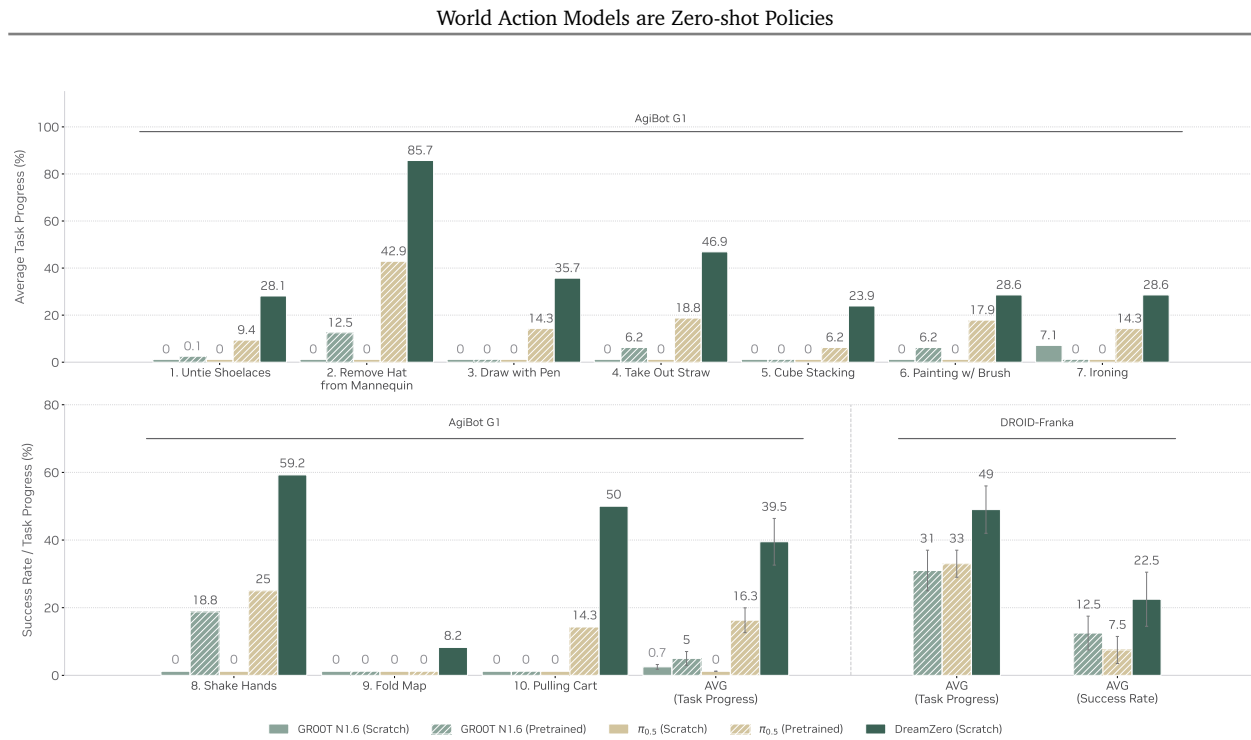


Figure 9: **Zero-shot Generalization to Unseen Tasks.** DREAMZERO achieves non-trivial task progress on 10 tasks absent from training, while VLAs struggle across both embodiments.

alignment between generated videos and real-world execution, even for suboptimal behaviors (Figure 16). Most DREAMZERO failures stem from video generation errors rather than action prediction—the policy faithfully executes whatever trajectory the video predicts. This suggests that improvements to the video backbone would directly translate to better WAM performance.

Q2. Do WAMs generalize to unseen tasks?

Figure 9 evaluates generalization to 10 tasks entirely absent from the pretraining distribution, including untying shoelaces, ironing, painting with a brush, and shaking hands.

On AgiBot G1, *from-scratch* VLAs achieve near-zero task progress ($< 1\%$), while DREAMZERO reaches 39.5% on average—with strong performance on tasks like “Remove Hat from Mannequin” (85.7%) and “Shake Hands” (59.2%). DREAMZERO also significantly outperforms pretrained VLA baselines (39.5% vs. 16.3%), even though those baselines may have encountered some of these tasks during cross-embodiment pretraining. Also on the DROID-Franka setup, DREAMZERO significantly outperforms (49% task progress, 22.5% success rate) other pretrained baselines (31% task progress, 12.5% success rate for GR00T N1.6 and 33% task progress, 7.5% success rate for $\pi_{0.5}$).

Qualitatively, we observe that pretrained VLAs often reach toward objects and attempt grasping regardless of the instruction, suggesting they overfit to dominant training behaviors (e.g., pick-and-place) rather than understanding novel task semantics, accounting for their partial task progress despite failing to complete the intended tasks. In contrast, DREAMZERO performs visual planning for unseen tasks and executes them successfully, with strong alignment between generated videos and real-world actions.

Beyond structured evaluation, we conduct free-form testing on over 100+ additional tasks, including “Pop the balloon” and “Press elevator button”, by doing free-form prompting with verbal instructions.⁹

⁹Rollouts of these tasks are provided in https://dreamzero0.github.io/evals_gallery/.

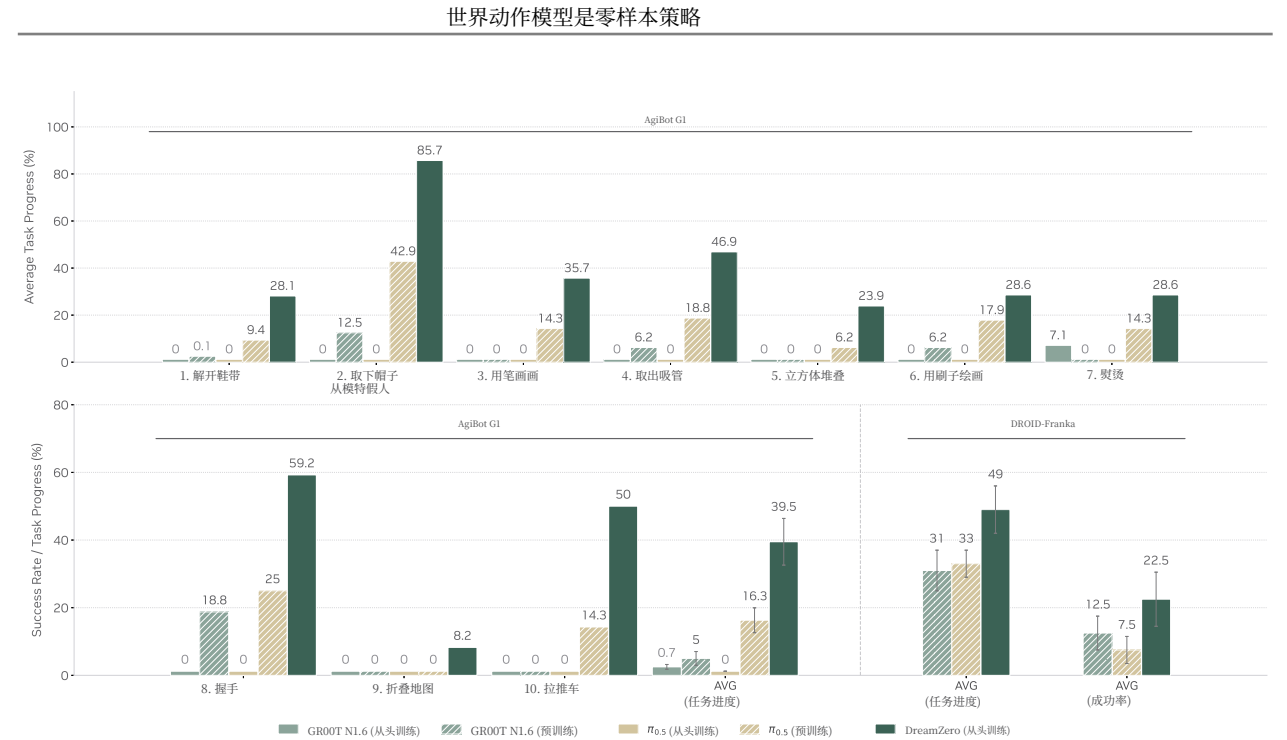


图9: **对未见任务的零样本泛化。** DreamZero在训练中未涉及的10个任务上取得了显著的任务进度，而VLAs在两个具身上都表现不佳。

对齐生成视频与实际执行之间的紧密关系，即使对于次优行为也是如此（图16）。大多数DREAMZERO失败的根源在于视频生成错误而非动作预测——策略忠实地执行视频预测的任何轨迹。这表明，对视频骨干网络的改进将直接转化为更好的WAM性能。

问题2：WAMs能否泛化到未见任务？

图9评估了在预训练分布中完全未出现的10个任务上的泛化能力，包括解鞋带、熨烫、用画笔绘画以及握手。

在AgiBot G1上，从头训练的VLA模型实现了近乎为零的任务进度（ $< 1\%$ ），而DreamZero平均达到39.5%——在“从模特假人取下帽子”（85.7%）和“握手”（59.2%）等任务上表现强劲。DreamZero也显著优于预训练VLA基线（39.5%对16.3%），尽管这些基线可能在跨具身预训练期间遇到过其中一些任务。同样，在DROID-Franka设置中，DreamZero（任务进度49%，成功率22.5%）显著优于其他预训练基线（GR00T N1.6任务进度31%，成功率12.5%；以及 $\pi_{0.5}$ 任务进度33%，成功率7.5%）。

从定性角度观察，预训练视觉-语言-动作模型常常伸手抓向物体并尝试抓住，而不考虑指令内容，这表明它们过度拟合了主导训练行为（例如拾取与放置），而非理解新任务语义，这也解释了为何它们尽管未能完成预期任务，却仍取得了部分任务进度。相比之下，DreamZero能对未见任务进行视觉规划并成功执行，生成的视频与真实世界动作之间具有强一致性。

除了结构化评估，我们还通过自由形式提示结合口头指令，在 100+ 个额外任务上进行了自由形式测试，包括“戳气球”和“按电梯按钮”。⁹

⁹这些任务的运行结果位于https://dreamzero0.github.io/evals_gallery/。

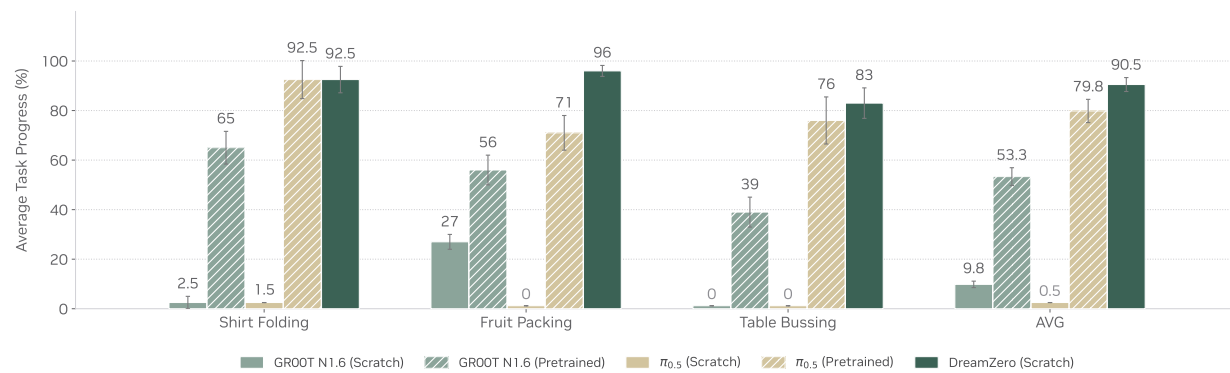


Figure 10: **Posttraining Results.** WAMs enable stronger post-training results across three tasks, indicating that environment generalization of DREAMZERO is retained after post-training.

Q3. Do WAMs improve post-training performance?

We investigate whether WAMs retain their generalization even after fine-tuning on task-specific data. Figure 10 shows results on three tasks with varying distribution diversity.

DREAMZERO matches or outperforms VLA baselines across all tasks: comparable performance on shirt folding and table bussing while significantly outperforming on fruit packing. Similar to the findings from Figure 8 and Figure 9, *from-scratch* baselines fail to learn accurate motions to grasp the target objects; this means that *from-scratch* VLAs tend to overfit to the training data and fail to generalize to scenarios where we vary the table height, table distance, objects, and object placements, mostly due to the evaluation site being in a different geographic location (see Figure 7 for samples). Although pretraining on multiple robot embodiments with repetitive data largely boosts the post-training generalization performance for pretrained baselines, DREAMZERO still matches or outperforms pretrained VLA baselines without cross embodiment pretraining. Since we still evaluate on unseen environments for post-training, this implies that the environment generalization of DREAMZERO is retained after post-training.

Q4. Do WAMs enable strong cross-embodiment transfer to unseen tasks?

Having shown that WAMs generalize to unseen tasks (Figure 9), we now investigate whether this generalization can be further improved by leveraging video data from different embodiments performing the same tasks. Crucially, we use only the video prediction objective for the cross-embodiment data (no actions), while maintaining the joint video-action objective for the AgiBot pretraining data; the cross-embodiment data thus serves as additional visual experience to strengthen the world model’s understanding of task dynamics and expected behavior.

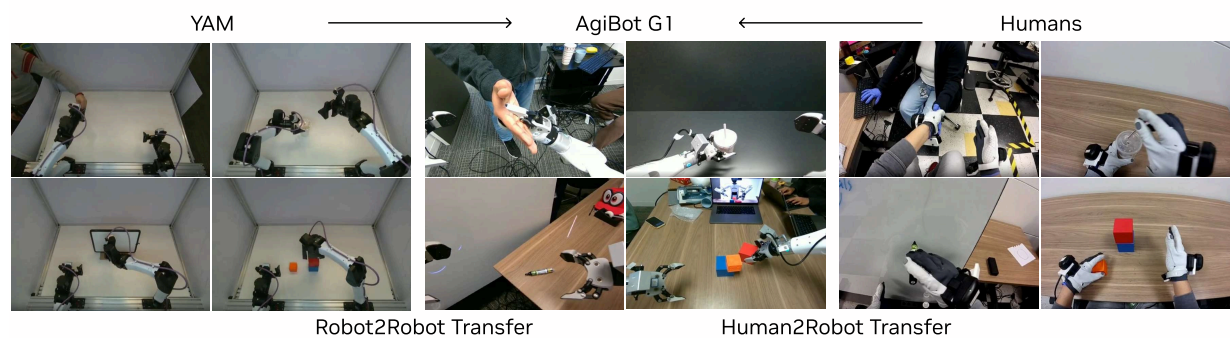


Figure 11: **Cross-Embodiment Transfer.** We explore robot-to-robot (YAM → AgiBot) and human-to-robot embodiment transfer to unseen tasks.

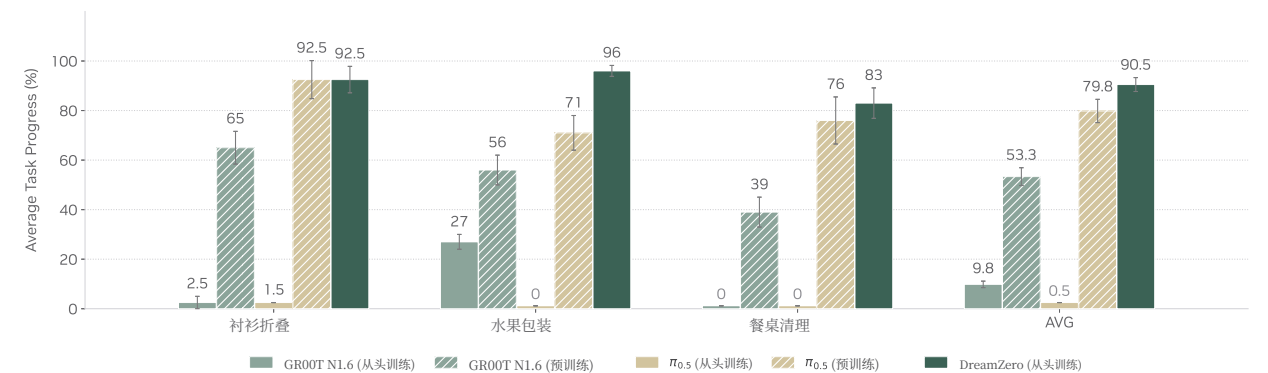


图10: **后训练结果。** WAMs在三个任务上实现了更强的后训练结果，表明DreamZero的环境泛化能力在后训练后得以保留。

问题3：WAMs能否提升后训练性能？

我们研究WAMs是否在任务特定数据上微调后仍能保持其泛化能力。图10显示了三个具有不同分布多样性的任务的结果。

DREAMZERO 在所有任务上与VLA基线相当或更优：在衬衫折叠和餐桌清理上性能相当，在水果包装上显著领先。与图8和图9的发现类似，从头训练的基线无法学习抓取目标物体的精确动作；这意味着从头训练的VLA往往过度拟合训练数据，无法泛化到桌子高度、桌子距离、物体和物体放置变化的情景，这主要是由于评估地点位于不同的地理位置（参见图7中的示例）。尽管在多个机器人实体上使用重复数据进行预训练大幅提升了预训练基线在后训练中的泛化性能，DREAMZERO 在没有跨实体预训练的情况下仍然与预训练VLA基线持平或更优。由于我们在后训练中仍然在未见环境中进行评估，这意味着DREAMZERO 的环境泛化能力在后训练后得以保留。

Q4. WAMs能否实现对未见任务的强大跨具身迁移？

我们已经证明WAMs能够泛化到未见任务（图9），现在我们研究是否可以通过利用不同具身执行相同任务的视频数据来进一步提升这种泛化。关键的是，我们对跨实体数据仅使用视频预测目标（无动作），同时保持对AgiBot预训练数据使用联合视频-动作目标；跨实体数据因此作为额外的视觉经验来强化世界模型对任务动态和预期行为的理解。

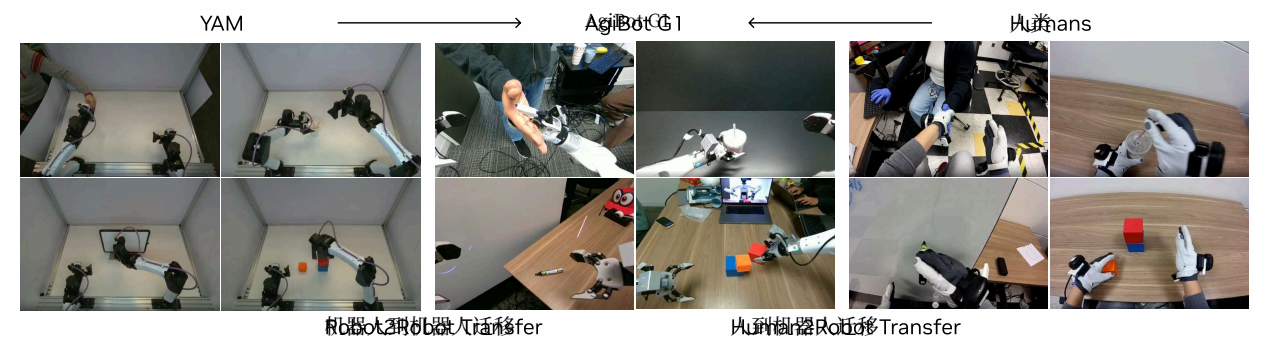


图11: **跨具身迁移。** 我们探索了机器人到机器人（YAM → AgiBot）和人到机器人对未见任务的具身迁移。

We explore two settings (Figure 11): (1) *Robot-to-robot transfer* using the bimanual YAM robot, and (2) *Human-to-robot transfer* using egocentric human demonstrations. For each setting, we collect 72 multi-view trajectories of the 9 unseen tasks (8 demonstrations per task, 20 minutes for YAM, 12 minutes for human).¹⁰ We then co-train from the DREAMZERO-AgiBot checkpoint on a 1:1 mix with pretraining data for 10K steps.

Results on the 9 unseen tasks (Table 2) show that both transfer settings improve performance over the baseline DREAMZERO. Robot-to-robot transfer yields the largest gain (38.3% $\rightarrow$ 55.4%), likely due to the narrower embodiment gap; both YAM and AgiBot are bimanual parallel grippers. Human-to-robot transfer also improves performance (38.3% $\rightarrow$ 54.3%), despite the larger morphological gap and dynamic egocentric viewpoints.

Method	Task Progress
DREAMZERO	38.3% $\pm$ 7.6%
DREAMZERO + Human2Robot Transfer	54.3% $\pm$ 10.4%
DREAMZERO + Robot2Robot Transfer	55.4% $\pm$ 9.5%

Table 2: **Cross-Embodiment Transfer Results.** Average task progress on unseen tasks ($\pm$ standard error). Both transfer settings improve over baseline (result from Table 9) using only 10–20 minutes of video-only demonstration data.

These results point to a promising property of WAMs: unlike recent VLA approaches to embodiment transfer (Kareer et al., 2025; Team, 2025), our method relies solely on visual information without action labels. While current success rates remain moderate, the consistent improvement from just 10–20 minutes of video-only data provides an early signal that cross-embodiment visual experience transfers meaningfully. This opens a potential scaling pathway: abundant human video data—orders of magnitude larger than robot datasets—could enable WAMs to acquire diverse skills without action annotation, pending further research into strengthening the transfer mechanism.

Q5. Do WAMs enable few-shot new embodiment adaptation?

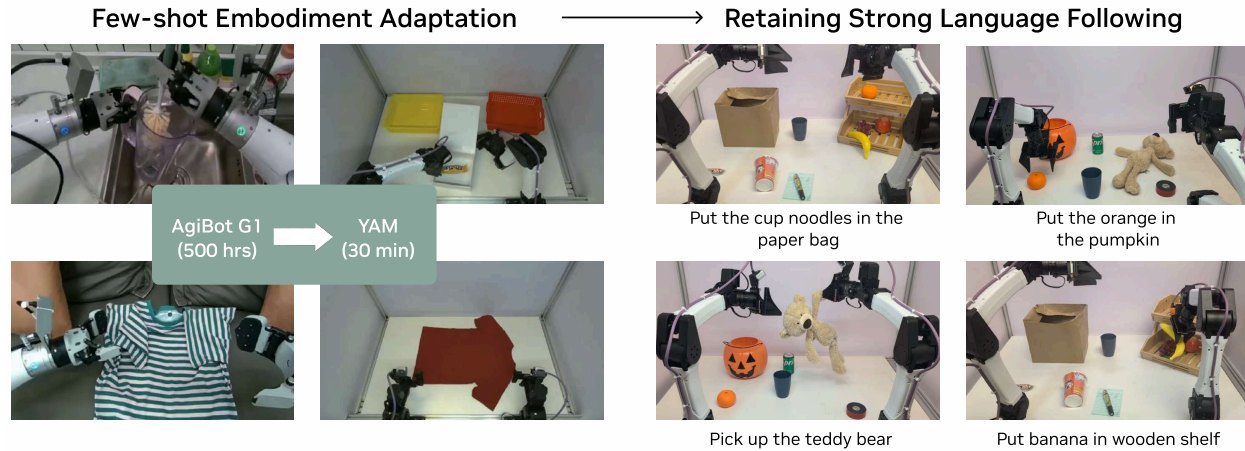


Figure 12: **Few-shot Embodiment Adaptation.** We explore few-shot embodiment adaptation by post-training on 30 minutes of new embodiment *play* data and evaluating on pick-and-place variants requiring strong language following.

We post-trained the DREAMZERO-AgiBot checkpoint on a new bimanual manipulator (YAM robot) using only 55 trajectories across 11 unique tasks ($\sim$ 30 minutes of data).¹¹ As illustrated in Figure 12, despite limited data and diversity, the post-trained policy retains strong language following ability, even generalizing to novel

¹⁰We exclude *Pulling Cart* task since data collection through teleoperation was infeasible with our bimanual YAM robot setup.

¹¹We visualize the entire 30 minutes of play data https://dreamzero0.github.io/yam_gallery/

我们探索两种设置（图11）：（1）机器人到机器人迁移使用YAM双机械臂机器人，以及（2）人类到机器人迁移使用第一人称人类演示。对于每种设置，我们收集72个多视角轨迹，涵盖9个未见任务（每个任务8次演示，YAM用时20分钟，人类用时12分钟）。¹⁰然后我们从DREAMZERO-AgiBot检查点开始联合训练，以1:1的比例混合预训练数据，训练10K步。

在9个未见任务上的结果（表2）显示，两种迁移设置均提升了相对于基线的性能DREAMZERO。机器人到机器人迁移获得了最大提升（38.3% $\rightarrow$ 55.4%），这可能是由于较窄的形态差距；YAM和AgiBot都是双手平行夹爪。人类到机器人迁移也提升了性能（38.3% $\rightarrow$ 54.3%），尽管存在较大的形态差距和动态第一人称视角。

方法	任务进度
DreamZero	38.3% $\pm$ 7.6%
DreamZero +人到机器人迁移	54.3% $\pm$ 10.4%
DreamZero +机器人到机器人迁移	55.4% $\pm$ 9.5%

表2: **跨具身迁移结果。** 未见任务的平均任务进度（ $\pm$ 标准误差）。两种迁移设置均优于基线（结果来自表9），且仅使用10–20分钟的纯视频演示数据。

这些结果指出了WAMs的一个有前景特性：与近期面向具身迁移的VLA方法（Kareer et al., 2025; Team, 2025）不同，我们的方法仅依靠视觉信息，无需动作标签。虽然当前成功率仍属中等，但仅凭10–20分钟的仅视频数据便实现了一致改进，这为一个早期信号提供了证据：跨具身视觉经验能够进行有意义的迁移。这开辟了一条潜在的扩展路径：丰富的人类视频数据——其规模比机器人数据集大数个数量级——可让WAMs在不需动作标注的情况下获取多样技能，但仍需进一步研究以强化迁移机制。

Q5. WAMs 是否支持小样本新具身适应？

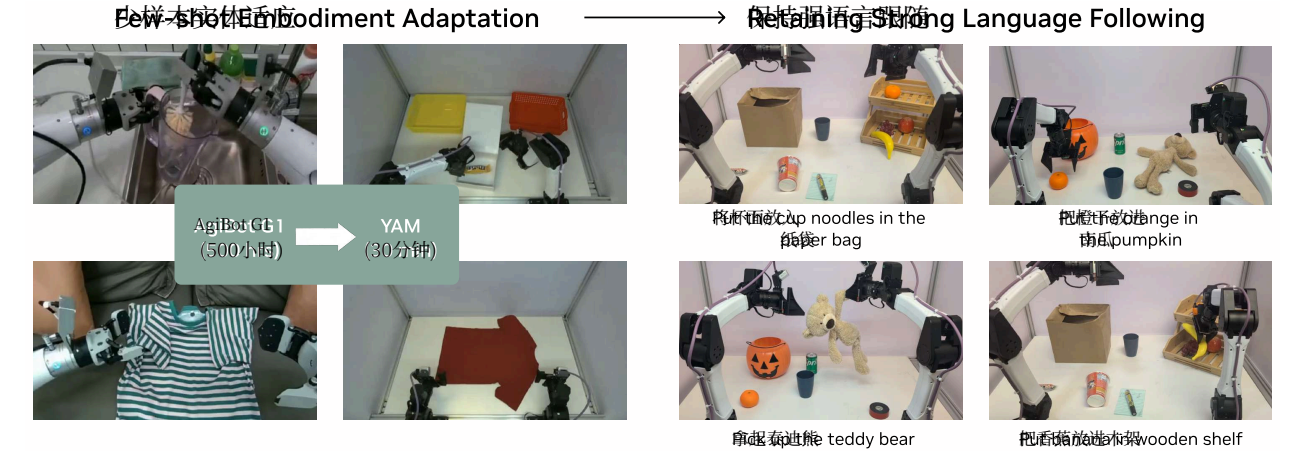


图12: **小样本具身适应。** 我们在新具身的30分钟游戏数据上进行后训练，并在需要强语言跟随的拾取与放置变体上进行评估，来探索小样本具身适应。

我们对DREAMZERO-AgiBot 检查点进行了后训练，用于新型双臂机械臂（YAM 机器人），仅使用了 11 个独特任务中的 55 条轨迹（ $\sim$ 30 分钟的数据）。¹¹ 如图12所示，尽管数据有限且多样性不足，后训练策略仍保留了强大的语言跟随能力，甚至泛化到了训练中未见的新

¹⁰我们排除了拉推车任务，因为在我们的YAM双机械臂机器人设置中通过遥操作收集数据是不可行的。¹¹我们可视化了全部30分钟的游戏数据https://dreamzero0.github.io/yam_gallery/

objects unseen during training, including pumpkins, teddy bears, pens, cup noodles, and paper bags. Even with minimal data, we observe tight video-action alignment, demonstrating very efficient cross-embodiment transfer.

We hypothesize that two factors enable this efficiency: (1) the visual similarity of AgiBot G1 and YAM embodiment (both equipped with bi-manual parallel grippers), and (2) more fundamentally, learning an implicit IDM from predicted videos may be inherently more sample-efficient than direct policy learning—the model only needs to learn the mapping from visual futures to actions, while leveraging the pretrained video model’s existing understanding of physical dynamics. Consistent with our AgiBot findings, failures primarily stem from video prediction errors rather than action extraction, suggesting that increasing task diversity during post-training could further improve performance.¹²

Q6. Does DREAMZERO-Flash maintain performance with fewer denoising steps?

We evaluate whether DREAMZERO-Flash can maintain task performance under aggressive single-step denoising. As shown in Table 3, reducing DREAMZERO from 4 denoising steps to 1 step drops task progress substantially (83% $\rightarrow$ 52%) on the table bussing task. In contrast, DREAMZERO-Flash achieves a higher average success rate (74%) at single-step inference, sitting only 9% below the 4-step baseline while being $\sim 2\times$ faster. This suggests that decoupled noise scheduling offers a more effective speed–accuracy trade-off for real-time deployment.

Method	Denoising steps	Task Progress	Inference speed	$\times$ Speed up
DREAMZERO	4	83% $\pm$ 6.1%	350ms	1
DREAMZERO	1	52% $\pm$ 10.2%	150ms	2.33 $\times$
DREAMZERO-Flash	1	74% $\pm$ 10.1%	150ms	2.33 $\times$

Table 3: **DREAMZERO-Flash Evaluation.** Task progress on table bussing with varying denoising steps ($\pm$ standard error). DREAMZERO-Flash recovers most of the 4-step performance using only 1 denoising step.

5.2. Model and Data Ablations

We conduct ablations to isolate the contributions of data diversity, model scale, and architecture. Due to computational constraints, all ablation models are trained with 50K steps and batch size 32, and evaluated on *PnP Easy* tasks for consistent comparison.

Architecture	Model Size	Data	Task Progress
<i>Q1. Data Diversity</i>			
DREAMZERO (AR)	14B	Repetitive	33% $\pm$ 4.2%
DREAMZERO (AR)	14B	Diverse	50% $\pm$ 6.3%
<i>Q2. Model Scale</i>			
DREAMZERO (AR)	5B	Diverse	21% $\pm$ 4.2%
DREAMZERO (AR)	14B	Diverse	50% $\pm$ 6.3%
VLA	5B	Diverse	0% $\pm$ 0.0%
VLA	14B	Diverse	0% $\pm$ 0.0%
<i>Q3. Architecture (Bidirectional vs. AR)</i>			
DREAMZERO (BD)	14B	Diverse	50% $\pm$ 14.4%
DREAMZERO (AR)	14B	Diverse	50% $\pm$ 6.3%

Table 4: **Model and Data Ablations.** Task progress on PnP Easy tasks ($\pm$ standard error). AR = autoregressive, BD = bidirectional. All models trained with 50K steps and batch size 32.

¹²In this specific post-training experiment, we only utilized 11 short, global language annotations unique for each task. We hypothesize diversifying the language can also enable stronger tranfer, but leave further investigation to future work.

物体，包括南瓜、泰迪熊、笔、杯面和纸袋。即使数据极少，我们也观察到了紧密的视频-动作对齐，展示了非常高效的跨具身迁移。

我们推测，这种高效性得益于两个因素：（1）AgiBot G1与YAM具身形态的视觉相似性（两者均配备双机械臂平行夹爪），以及（2）更根本的一点——从预测视频中学习隐式IDM可能天生比直接学习策略更具样本效率——模型只需学习从视觉未来到动作的映射，同时利用预训练视频模型对物理动力学的已有理解。与我们在AgiBot上的发现一致，失败主要源于视频预测误差而非动作提取，这表明在后训练期间增加任务多样性可以进一步提升性能。¹²

Q6. DreamZero-Flash在减少去噪步数时能否保持性能?

我们评估DREAMZERO-Flash是否能在激进的单步去噪下保持任务性能。如表3所示，将DREAMZERO 从4步去噪减少到1步，会使餐桌清理任务的任务进度大幅下降（83% $\rightarrow$ 52%）。相比之下，DREAMZERO-Flash在单步推理时实现了更高的平均成功率（74%），仅比4步基线低9%，同时速度提升 $\sim 2\times$ 倍。这表明，解耦噪声调度为实时部署提供了更有效的速度-精度权衡。

方法	去噪步数	任务进度	推理速度	$\times$ 加速
DreamZero	4	83% $\pm$ 6.1%	350ms	1
DreamZero	1	52% $\pm$ 10.2%	150ms	2.33 $\times$
DreamZero-Flash	1	74% $\pm$ 10.1%	150ms	2.33 $\times$

表3: **DREAMZERO-Flash评估。** 使用不同去噪步数的餐桌清理任务进度（ $\pm$ 标准误差）。DreamZero-Flash仅使用1步去噪就能恢复大部分4步性能。

5.2. 模型与数据消融

我们进行消融实验，以分离数据多样性、模型规模和架构的贡献。由于计算资源限制，所有消融模型均以50K步数和批次大小32进行训练，并在*PnP Easy* 任务上进行评估，以进行一致比较。

架构	模型规模	Data	任务进度
<i>Q1. 数据多样性</i>			
DreamZero (AR)	14B	重复	33% $\pm$ 4.2%
DreamZero (AR)	14B	多样	50% $\pm$ 6.3%
<i>Q2. 模型规模</i>			
DreamZero (AR)	5B	多样	21% $\pm$ 4.2%
DreamZero (AR)	14B	多样	50% $\pm$ 6.3%
VLA	5B	多样	0% $\pm$ 0.0%
VLA	14B	多样	0% $\pm$ 0.0%
<i>Q3. 架构（双向 vs. 自回归）</i>			
DreamZero (BD)	14B	多样	50% $\pm$ 14.4%
DreamZero (自回归)	14B	多样	50% $\pm$ 6.3%

表4: **模型与数据消融。** 任务进展 PnP Easytasks ($\pm$ 标准误差). 自回归 = 自回归, 双向 = 双向. 所有模型使用50K步数和批次大小32进行训练.

¹²在这个特定的后训练实验中，我们仅使用了每个任务独有的11个简短、全局的语言注释。我们假设多样化的语言也能实现更强的迁移，但将进一步的调查留给未来的工作。

Q1. Does data diversity improve generalization?

We compare DREAMZERO trained on 500 hours of *diverse* data versus 500 hours of *repetitive* data, where the latter contains 70 tasks with many repeated demonstrations per task using similar object positions and configurations. As shown in Table 4, diverse data substantially improves generalization (33% $\rightarrow$ 50%), even on simple pick-and-place tasks. We hypothesize this reflects WAMs’ learning dynamics: since video prediction is largely inherited from pretraining, the key challenge is learning inverse dynamics. A robust IDM requires diverse state-action correspondences across varied contexts, which repetitive data inherently lacks.

Q2. Does WAM performance scale with model size?

For VLAs, scaling model size improves semantic reasoning but not necessarily action prediction. We find that WAMs exhibit clearer scaling behavior: the 14B model significantly outperforms the 5B model (50% vs. 21%), with the smaller model prone to visual hallucinations that propagate to erroneous actions.

To ensure fair comparison, we also scale VLA baselines to match DREAMZERO’s size by initializing from 8B and 32B pretrained VLMs (Yang et al., 2025), truncating to the first half of the transformer blocks, and attaching DiT-based action modules following Bjorck et al. (2025). As shown in Table 4, larger VLAs still fail to learn from diverse data (0% task progress), often hovering near objects without making contact. This suggests that scaling model capacity alone does not address VLAs’ difficulty with diverse data distributions.

Q3. Does autoregressive architecture outperform bidirectional?

We compare DREAMZERO’s autoregressive (AR) architecture against a bidirectional (BD) variant. While task progress is similar (Table 4), the AR model produces substantially smoother motions—backpropagating through entire action sequences enables better temporal consistency. Additionally, AR inference is 3–4 $\times$ faster due to KV caching.

6. Discussion and Future Work

Scaling Laws of WAMs. We have identified that leveraging larger video backbone model and training on diverse data boosts downstream performance in Table 4. However, we still have lacking evidence for scaling laws for robot foundation models, specifically for WAMs. Similar to scaling laws for language models (Kaplan et al., 2020), scaling laws for WAMs depending on the model size, dataset size, and the training compute need to be explored to determine the optimal configuration to extract the maximal capability of WAMs. We expect that the tendency of scaling WAMs to be different from VLAs, showing a more direct scaling law for actions. We leave deep investigation on scaling laws for WAMs as future work.

Learning from In-the-wild Human Data. Although we have investigated leveraging egocentric human data to boost the performance on unseen tasks (Section 5), our experiments are still constrained to small scale in-lab data (only 12 minutes). Recently, a large amount of human video data has been released that has more diverse distribution compared to robot data (Chen et al., 2026; Grauman et al., 2022; Hoque et al., 2025). Since WAMs are pretrained on diverse internet video data, we hypothesize that leveraging large-scale egocentric human video data that are related to robot manipulation tasks would lead to stronger transfer to downstream robot tasks compared to current VLAs. We leave this direction as future work.

Faster Inference. Through model and system optimizations, we enable DREAMZERO to run at 7Hz using 2 GB200s. However, compared to current VLAs which runs up to over 20Hz on consumer GPUs, DREAMZERO is still computationally expensive due the large parameter size and the iterative denoising nature of video models. In the future, if smaller video backbone models also have strong generalization capability, WAMs could potentially be utilized as a real-time System 1 model on a lightweight edge device.

Long-horizon Reasoning. Current DREAMZERO architecture functions primarily as a System 1 model. Although DREAMZERO has a concept of visual memory, it is currently short-horizon (6 seconds). Robust long-horizon execution will require either a System 2 planner or WAMs with significantly extended context

Q1. 数据多样性是否提升了泛化能力?

我们比较 DREAMZERO 在500小时的 多样化的 数据与500小时的 重复的 数据，其中后者包含70个任务，每个任务有多次重复演示，使用相似的物体位置和配置。如表 4，多样化数据显著提升泛化（33% $\rightarrow$ 50），即使在简单的抓取放置任务上也是如此。我们假设这反映了WAMs的学习动态：由于视频预测在很大程度上继承自预训练，关键挑战在于学习逆动力学。一个稳健的IDM需要跨不同上下文的多样化状态-动作对应关系，而重复数据本质上缺乏这一点。

Q2. WAM性能是否随模型规模扩展?

对于视觉-语言-动作模型，扩大模型规模能提升语义推理，但未必提升动作预测。我们发现WAMs展现出更清晰的扩展行为：14B模型显著优于5B模型（50% vs. 21%），而较小的模型容易产生视觉幻觉，这些幻觉会传播到错误动作中。

为了确保公平比较，我们还扩大了VLA基线以匹配DREAMZERO的规模，通过从8B和32B预训练VLM初始化（Yang等人，2025），截取Transformer块的前半部分，并附加基于DiT的动作模块，按照Bjorck等人（2025）。如表4，更大VLAs仍然无法从多样化数据中学习（0%任务进度），经常在物体附近徘徊而不进行接触。这表明仅扩大模型容量并不能解决VLAs在多样化数据分布上的困难。

Q3. 自回归架构是否优于双向?

我们比较 DREAMZERO的自回归（AR）架构与双向（BD）变体。虽然任务进度相似（桌子 4），但AR模型产生了明显更平滑的动作——通过整个动作序列的反向传播实现了更好的时间一致性。此外，AR推理由于KV缓存技术而快了3–4 $\times$ 倍。

6. 讨论与未来工作

WAMs的扩展定律。我们已经发现，利用更大的视频骨干模型并在多样化数据上进行训练可以提升桌子 4中的下游性能。然而，我们仍然缺乏机器人基础模型扩展定律的证据，特别是针对WAMs。类似于语言模型的扩展定律（Kaplan等人，2020），WAMs的扩展定律取决于模型规模、数据集规模和训练计算，需要探索以确定最佳配置，从而发挥WAMs的最大能力。我们预计WAMs的扩展趋势将不同于视觉-语言-动作模型，展现出更直接的动作扩展定律。我们将对WAMs扩展定律的深入研究留作未来工作。

从野外人类数据中学习。尽管我们研究了利用自我中心人类数据来提升未见任务的性能（第5节），但我们的实验仍然局限于小规模实验室数据（仅12分钟）。近期，大量人类视频数据被发布，其分布比机器人数据更加多样（Chen等人,2026; Grauman等人,2022; Hoque等人, 2025）。由于WAMs是在多样化的互联网视频数据上预训练的，我们假设，利用与机器人操作任务相关的大规模自我中心人类视频数据，将比当前VLAs更有效地迁移到下游机器人任务。我们将此方向留作未来工作。

更快的推理。通过模型与系统优化，我们使DreamZero能够在2块GB200上以7赫兹运行。然而，与当前在消费级GPU上可运行超过20赫兹的VLAs相比，DreamZero由于参数规模大以及视频模型的迭代去噪特性，计算成本仍然很高。未来，如果更小的视频骨干模型也具备强大的泛化能力，WAMs或许能够在轻量级边缘设备上作为实时的系统1模型使用。

长程推理。当前的DreamZero架构主要作为系统1模型运行。尽管DreamZero具有视觉记忆的概念，但目前仍是短程的（6秒）。稳健的长程执行需要系统2规划器或上下文显著扩展的WAMs

windows. For the former, both modular dual-system architectures (Shi et al., 2025) and unified approaches (Deng et al., 2025) offer promising directions. For the latter, techniques from video-based world models that maintain coherent generation over extended horizons (Ball et al., 2025; HunyuanWorld, 2025) could be adapted to expand WAM context length.

High-Precision Tasks. While DREAMZERO generalizes broadly across tasks and environments, it inherits limitations common to behavior cloning on tasks requiring sub-centimeter precision, such as key insertion or fine assembly. Our diverse pretraining strategy prioritizes breadth, which may underrepresent the dense demonstrations needed for these high-precision manipulation. That said, recent work (Kim et al., 2026) showed promising results that WAMs may actually hold an advantage for high-precision manipulation tasks with millimeter tolerance, an encouraging signal that the trade-off between broad generalization and fine-grained dexterity may be reconcilable with further investigation.

Embodiment Design for WAMs. We hypothesize two key factors will shape the best robot embodiments for future WAM development: (1) *Degrees of freedom*: Higher-DOF robots will require more play data to learn an accurate implicit IDM, as the mapping from visual futures to motor commands grows combinatorially with kinematic complexity. Quantifying the accuracy of implicit IDMs remains a challenge. (2) *Human similarity*: Embodiments that more closely resemble humans—particularly humanoids with dexterous manipulation capabilities—may transfer more efficiently despite higher DOF, as they can leverage both the motion priors from video pretraining and the massive scale of human egocentric videos. These factors pull in opposite directions—yet human-like embodiments may win out by trading mechanical simplicity for access to web-scale human data—the fuel for next-generation robot foundation models.

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窗口。对于前者，模块化双系统架构（Shi et al., 2025）和统一方法（Deng et al., 2025）都提供了有前途的方向。对于后者，来自基于视频的世界模型的技术（这些模型在扩展的时间范围内保持连贯生成，Ball et al., 2025; HunyuanWorld,2025）可以进行调整以扩展WAM上下文长度。

高精度操作任务。虽然DREAMZERO 在任务和环境上具有广泛的泛化能力，但它继承了行为克隆在需要亚厘米精度的任务（如钥匙插入或精密装配）中的常见局限性。我们的多样化预训练策略优先考虑广度，这可能导致此类高精度操作所需的密集演示数据不足。尽管如此，近期研究（Kim等人, 2026）显示出令人鼓舞的结果：WAMs实际上可能在毫米公差的高精度操作任务中具备优势，这是一个积极的信号，表明广泛泛化与细粒度灵巧性之间的权衡可能通过进一步研究得以调和。

WAMs的具身设计。我们假设两个关键因素将塑造最适合未来WAM开发的机器人具身化：（1）自由度：高自由度机器人需要更多的游戏数据来学习准确的隐式IDM，因为从视觉未来到电机指令的映射随着运动学复杂度呈组合式增长。量化隐式IDM的准确性仍是一个挑战。（2）人类相似性：与人类更相似的具身——尤其是具有灵巧操作能力的人形机器人——尽管自由度更高，但可能更高效地实现迁移，因为它们能够同时利用视频预训练的运动先验和大量人类第一人称视频。这两个因素相互矛盾——然而，类人具身可能通过用机械简单性换取网络规模人类数据（下一代机器人基础模型的燃料）而胜出。

7. 致谢

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A. Comparison with Alternative World Model Architectures

The term “world model” encompasses a broad family of approaches beyond video-based prediction. Here we discuss how WAMs differ from these alternatives.

Latent-Space World Models. Joint Embedding Predictive Architectures (JEPAs) (Assran et al., 2023, 2025; Bardes et al., 2024; LeCun, 2022) predict future states in abstract latent spaces rather than pixel space, offering computational efficiency and the ability to discard unpredictable details. V-JEPA 2 (Assran et al., 2025) demonstrates this approach on robotics, achieving zero-shot planning on manipulation tasks after post-training on 62 hours of robot data. Similarly, the Dreamer series (Hafner et al., 2019, 2020, 2023, 2025) learns compact latent dynamics models for model-based reinforcement learning. However, these approaches model dynamics as $p(s_{t+1}|s_t, a_t)$ —predicting next state conditioned on current state and action—and thus require goal-conditioned planning or search at test time to yield trajectories.

3D Point Cloud World Models. PointWorld (Huang et al., 2025) unifies state and action in a shared 3D spatial domain, predicting scene dynamics as 3D point flows conditioned on robot actions. This formulation enables embodiment-agnostic learning and real-time integration with model-predictive control (MPC). However, like latent world models, it requires explicit optimization (e.g., MPPI sampling) at inference to generate action trajectories.

Key Distinction. These alternative approaches share a common characteristic: they model forward dynamics and require separate inverse dynamics models or explicit planning/search procedures at deployment. In contrast, WAMs jointly model $p(o_{t:t+H}, a_{t:t+H} | o_{0:t}, c)$, directly producing action trajectories aligned with predicted visual futures without test-time optimization. This enables real-time closed-loop control at 7Hz—a frequency that would be challenging for search-based methods—while inheriting rich spatiotemporal priors from video pretraining.

B. Bidirectional vs. Autoregressive WAMs

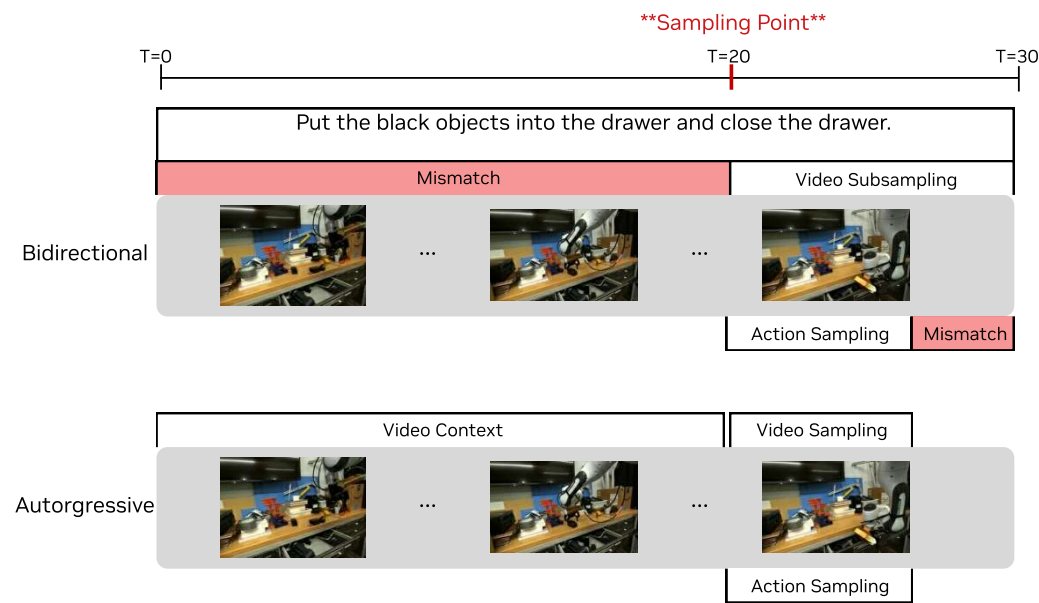


Figure 13: **Bidirectional vs. Autoregressive WAMs.** When the sampling point falls mid-task ($T=20$), bidirectional WAMs must subsample video to align with the language caption, distorting native FPS and degrading video-action alignment. Autoregressive WAMs avoid this trade-off by conditioning on video context, preserving both language-video correspondence and native frame rate.

A. 与其他世界模型架构的比较

“世界模型”一词涵盖了基于视频预测之外的广泛方法系列。此处我们讨论 WAMs 与这些替代方法的不同之处。

潜在空间世界模型。 联合嵌入预测架构 (JEPAs) (Assran等人, 2023, 2025; Bardes等人, 2024; LeCun, 2022) 预测未来状态在抽象潜空间而非像素空间中, 提供了计算效率并能够丢弃不可预测的细节。V-JEPA 2 (Assran等人, 2025) 演示了这种方法在机器人领域的应用, 经过对62小时机器人数据后训练后, 在操作任务上实现了零样本规划。类似地, Dreamer系列 (Hafner等人, 2019, 2020, 2023, 2025) 学习紧凑的潜在动力学模型用于基于模型的强化学习。然而, 这些方法将动力学建模为 $p(s_{t+1}|s_t, a_t)$ ——根据当前状态和动作预测下一个状态——因此需要在测试时进行目标条件规划或搜索以生成轨迹。

3D点云世界模型。 Pointworld (Huang等人, 2025) 将状态和动作统一在共享的3D空间域中, 根据机器人动作将场景动力学预测为3D点流。这种表述实现了具身无关学习, 并与模型预测控制 (MPC) 实时集成。然而, 与潜在世界模型类似, 它需要在推理时通过显式优化 (例如MPPI采样) 来生成动作轨迹。

关键区别。 这些替代方法有一个共同特征: 它们对正向动力学进行建模, 并且在部署时需要独立的逆向动力学模型或显式的规划/搜索过程。相比之下, WAMs 联合建模 $p(o_{t:t+H}, a_{t:t+H} | o_{0:t}, c)$, 直接生成与预测的视觉未来对齐的动作轨迹, 无需测试时优化。这使得我们能够以 7赫兹 的频率进行实时闭环控制——这一频率对于基于搜索的方法来说将具有挑战性——同时继承了来自视频预训练的丰富时空先验。

B. 双向与自回归WAMs



图13: **双向与自回归WAMs.** 当采样点落在任务中期 ($T=20$) 时, 双向WAMs必须对视频进行子采样以对齐语言描述, 这会扭曲原始帧率并降低视频-动作对齐质量。自回归WAMs通过以视频上下文为条件来避免这一权衡, 同时保持语言-视频对应关系和原始帧率。

Figure 13 illustrates a key challenge for bidirectional WAMs when training closed-loop systems. Given a language annotation in a given long-horizon demonstration, the model must learn that the instruction corresponds to a specific video interval. In bidirectional architectures, without subsampling, the model receives a language instruction (e.g., ‘put the black objects into the drawer’) but usually generates video covering only a fraction of the task interval, causing the language to describe actions not yet visible in the predicted frames. This mismatch leads to significant language-following degradation, as measured via an internal video prediction benchmark.

A natural solution is to subsample the video to match the task caption interval. However, this creates a new problem for closed-loop training: we need the model to receive observations from arbitrary points within a task—not just the beginning. When the sampling point falls mid-task (e.g., $T=20$ in Figure 13), subsampling distorts the native FPS, making video-action alignment substantially harder to learn.

Autoregressive WAMs can sidestep this dilemma. By conditioning on video context rather than subsampling, they preserve both the language-video correspondence and the native frame rate, enabling tight alignment across all three modalities: language, video, and action.

C. Model and Training details

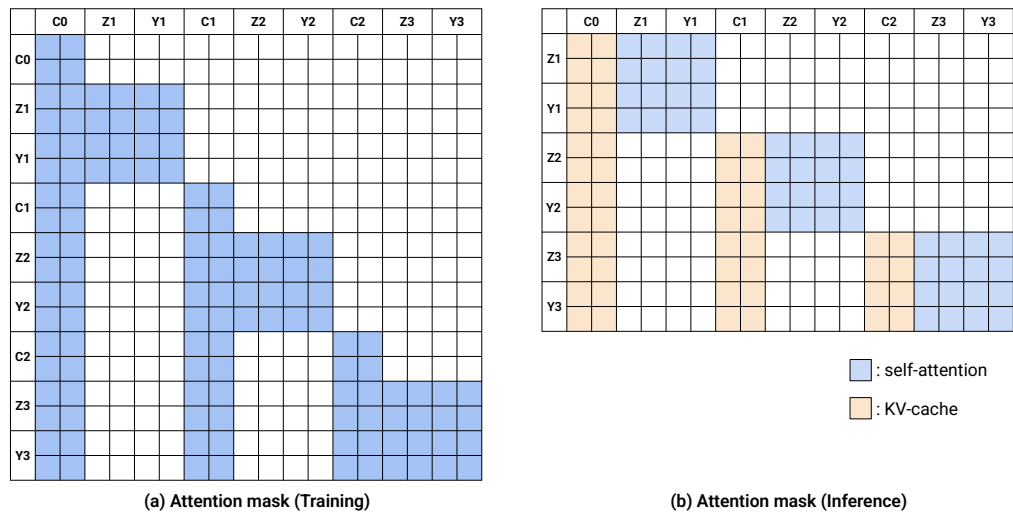


Figure 14: **Attention strategy of DREAMZERO.** (a) QKV Self-Attention mask for DreamZero training. Y-axis shows the Query (Q) and X-axis shows the Key/Value (KV). Given conditioning frames (C0, C1, C2), we train the model to predict the velocities of next frames (Z1, Z2, Z3) and actions (Y1, Y2, Y3). (b) During inference, we compute the KV-cache of conditional frames and concatenate them to predict the action and frames. For example, Y3 (action) is able to attend to C0, C1, and C2, taking into account previous visual observations as history to predict the current actions during both training and inference. Note that the C0, C1, and C2 during inference is replaced with the GT observations.

We visualize the attention mask for training and inference in Figure 14. For DREAMZERO, we set each chunk as $K = 2$ latent frames. From preliminary results, we have observed that $K = 2$ outperforms $K = 1$ empirically. We set the number of chunks $M = 4$ by default. If the trajectory length is shorter than $M = 4$, M can be smaller than 4. For Agibot training data, the video is sampled at 5FPS ratio and action is sampled at 30Hz. We use the action horizon of $H = 48$. Therefore, the video and action span 1.6 seconds per chunk. For DROID training, the video is sampled at 5FPS ratio and action is sampled at 15Hz. We use the action horizon of $H = 24$. Therefore, same as Agibot, the video and action span 1.6 seconds per chunk. The maximum context length is 8 latent frames (4×2), which is equivalent to 33 raw frames, spanning 6.6 seconds. We leave increasing the visual context for WAMs as future work.

图13展示了训练闭环系统时双向WAMs面临的一个关键挑战。给定一个特定长视界演示中的语言标注，模型必须学会指令对应一个特定的视频区间。在双向架构中，如果没有下采样，模型会接收一条语言指令（例如“将黑色物体放入抽屉”），但通常生成的视频只覆盖任务区间的一小部分，导致语言描述的动作在预测帧中尚未可见。这种不匹配会导致显著的语言跟随退化，如通过内部视频预测基准所衡量的那样。

一种自然的解决方案是对视频进行下采样以匹配任务描述间隔。然而，这给闭环训练带来了一个新问题：我们需要模型接收任务中任意点的观测——而不仅仅是开始。当采样点落在任务中期（例如， $T=20$ 在图 13），下采样会扭曲原始帧率，使得视频-动作对齐变得更加难以学习。

自回归WAMs可以规避这一困境。通过以视频上下文为条件而非下采样，它们既保留了语言-视频对应关系，又保留了原始帧率，从而在语言、视频和动作这三种模态之间实现紧密对齐。

C. 模型与训练细节

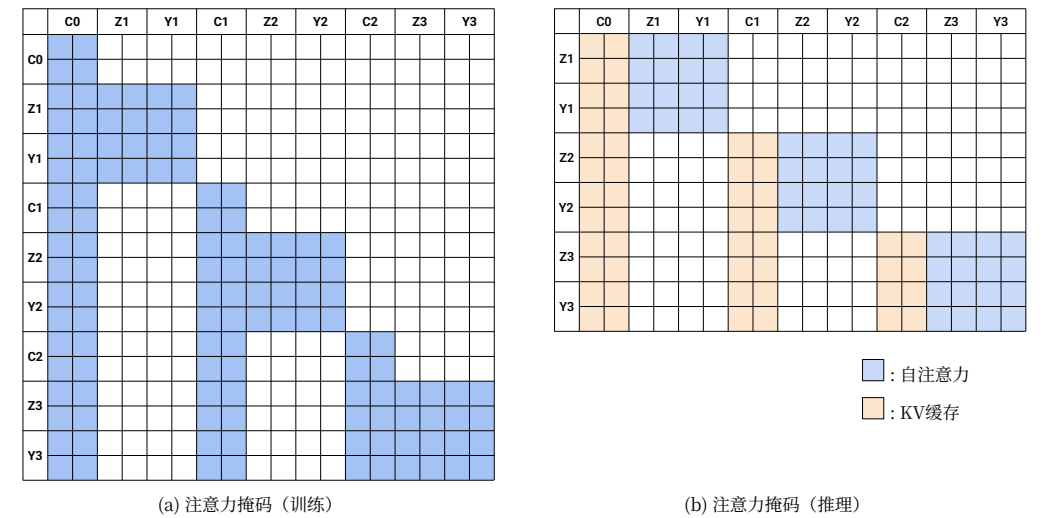


图14: **注意力策略的 DREAMZERO.** (a) DreamZero训练的QKV自注意力掩码。Y轴表示查询(Q)，X轴表示键/值(KV)。给定条件帧(C0, C1, C2)，我们训练模型预测下一帧的速度(Z1, Z2, Z3)和动作(Y1, Y2, Y3)。 (b) 在推理期间，我们计算条件帧的KV缓存并拼接它们以预测动作和帧。例如，Y3（动作）能够关注C0、C1和C2，将先前的视觉观测作为历史信息考虑，以在训练和推理过程中预测当前动作。注意，推理期间的C0、C1和C2被替换为真实观测。

我们在图14中展示了训练和推理的注意力掩码。对于DREAMZERO，我们将每个块设置为 $K = 2$ 潜在帧。从初步结果中，我们观察到 $K = 2$ 在经验上优于 $K = 1$ 。我们将块的数量设置为 $M = 4$ 作为默认值。如果轨迹长度短于 $M = 4$ ， M 可以小于4。对于Agibot训练数据，视频以5帧率比率采样，动作以30Hz采样。我们使用的动作范围为 $H = 48$ 。因此，视频和动作每块跨越1.6秒。对于DROID训练，视频以5帧率比率采样，动作以15Hz采样。我们使用的动作范围为 $H = 24$ 。因此，与Agibot相同，视频和动作每块跨越1.6秒。最大上下文长度是8个潜在帧（ 4×2 ），相当于33个原始帧，跨越6.6秒。我们将来会为WAMs增加视觉上下文作为未来工作。

Algorithm 1 DREAMZERO Training (Flow Matching)

```

1: Input: Dataset  $\mathcal{D}$ , Text condition  $\mathbf{c}$ 
2: Hyperparams: Number of chunks  $M$ 
3: Model:  $\mathbf{u}_\theta$  (Joint Video-Action DiT)
4: while not converged do
5:   Sample trajectory  $\tau \sim \mathcal{D}$ 
6:   Encode video to clean latents  $\mathbf{z}_1^{1:M}$ , normalize actions  $\mathbf{a}_1^{1:M}$ 
7:   Split  $\tau$  into  $M$  chunks
8:   for  $k = 1, \dots, M$  do  $\triangleright$  Chunk-wise Training
9:     Define clean context  $\mathcal{C}_k \leftarrow \{(\mathbf{z}_1^j, \mathbf{a}_1^j)\}_{j=1}^{k-1} \triangleright$  TF History
10:    Sample timestep  $t_k \sim \mathcal{U}(0, 1)$ 
11:    // Optional: DreamZero-Flash Decoupling
12:    if Flash Mode then
13:       $t_{vid} \sim \text{Beta}(7, 1)$ ,  $t_{act} \sim \mathcal{U}(0, 1)$ 
14:    else
15:       $t_{vid} \leftarrow t_k$ ,  $t_{act} \leftarrow t_k$ 
16:    end if
17:    Sample noise  $\mathbf{z}_0^k, \mathbf{a}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
18:    Interpolate (Eq. 2):
19:       $\mathbf{z}_t^k \leftarrow t_{vid}\mathbf{z}_1^k + (1 - t_{vid})\mathbf{z}_0^k$ 
20:       $\mathbf{a}_t^k \leftarrow t_{act}\mathbf{a}_1^k + (1 - t_{act})\mathbf{a}_0^k$ 
21:    Predict velocity:
22:       $\mathbf{v}_{pred} \leftarrow \mathbf{u}_\theta([\mathbf{z}_t^k, \mathbf{a}_t^k]; \mathcal{C}_k, \mathbf{c}, \mathbf{q}_k, t_k)$ 
23:    Target vel.  $\mathbf{v}^k := [\mathbf{z}_1^k, \mathbf{a}_1^k] - [\mathbf{z}_0^k, \mathbf{a}_0^k]$ 
24:    Loss  $\mathcal{L} \leftarrow \|\mathbf{v}_{pred} - \mathbf{v}^k\|^2 \triangleright$  Eq. 3
25:    Update  $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$ 
26:  end for
27: end while

```

Algorithm 2 DREAMZERO Inference (Closed-Loop Control)

```

1: Input: Instruction  $\mathbf{c}$ , Initial Image  $\mathbf{o}_{init}$ , State  $\mathbf{q}_{init}$ 
2: Hyperparams: Steps  $N$ , Cache Thresh  $\epsilon$ 
3: Init:  $\mathcal{KV} \leftarrow \emptyset$ ,  $\mathbf{v}_{prev} \leftarrow \emptyset$ ,  $\mathbf{q}_{curr} \leftarrow \mathbf{q}_{init}$ 
4: // 1. Prefill Cache (Context Phase,  $t = 0$ )
5:  $\mathbf{z}_{init} \leftarrow \text{VAE}(\mathbf{o}_{init})$ 
6: // Pass clean video, no action/state
7:  $(\cdot, \cdot; \mathcal{KV}) \leftarrow \mathbf{u}_\theta([\mathbf{z}_{init}, \emptyset]; \mathcal{KV}, \mathbf{c}, \emptyset, t = 0, \text{update}=\text{True})$ 
8: // 2. Autoregressive Loop
9: while task not done do
10:   Sample  $\mathbf{x}_0 = [\mathbf{z}_0, \mathbf{a}_0] \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \triangleright$  Noise at  $t = 0$ 
11:   // Joint Denoising (Flow Matching  $t : 0 \rightarrow 1$ )
12:   for  $i = 0 \dots N - 1$  do
13:      $t_i, t_{i+1} \leftarrow \text{Scheduler}(i, N)$ 
14:     // Optimization: DiT Caching
15:     if  $\mathbf{v}_{prev} \neq \emptyset$  and  $\text{CosSim}(\mathbf{v}_{prev}, \mathbf{v}_{last}) > \epsilon$  then
16:        $\mathbf{v}_i \leftarrow \mathbf{v}_{prev}$ 
17:     else
18:        $(\mathbf{v}_i^{\text{vid}}, \mathbf{v}_i^{\text{act}}, \cdot) \leftarrow \mathbf{u}_\theta(\mathbf{x}_{t_i}; \mathcal{KV}, \mathbf{c}, \mathbf{q}_{curr}, t_i, \text{update}=\text{False})$ 
19:        $\mathbf{v}_i \leftarrow [\mathbf{v}_i^{\text{vid}}, \mathbf{v}_i^{\text{act}}]$ 
20:        $\mathbf{v}_{prev} \leftarrow \mathbf{v}_i$ 
21:     end if
22:     // Solver Step
23:      $\mathbf{x}_{t_{i+1}} \leftarrow \mathbf{x}_{t_i} + \text{Step}(\mathbf{v}_i, t_i, t_{i+1})$ 
24:   end for
25:   // 3. Execution & Cache Update
26:    $\hat{\mathbf{a}} \leftarrow \text{Filter}(\mathbf{x}_1^{\text{action}}) \triangleright$  Clean Action  $t = 1$ 
27:   Async Execute  $\hat{\mathbf{a}}$  on Robot
28:   // Critical: Inject Ground Truth ( $t = 0$ )
29:   Real observation  $\mathbf{o}_{real}, \mathbf{q}_{real}$ 
30:    $\mathbf{q}_{curr} \leftarrow \mathbf{q}_{real}$ ,  $\mathbf{z}_{real} \leftarrow \text{VAE}(\mathbf{o}_{real})$ 
31:    $(\cdot, \cdot; \mathcal{KV}) \leftarrow \mathbf{u}_\theta([\mathbf{z}_{real}, \emptyset]; \mathcal{KV}, \mathbf{c}, \emptyset, t = 0, \text{update}=\text{True})$ 
32:   *Discard predicted video latent from  $\mathbf{x}_1$ 
33: end while

```

D. Real-time Execution Details

This appendix provides additional details on the system-, implementation-, and model-level optimizations introduced in Section 3.2.

D.1. System-level Optimizations

CFG Parallelism. To address the computation bottleneck inherent to DiT, we employed Classifier-Free Guidance (CFG) parallelism. Standard CFG requires two distinct model evaluations—the conditional and unconditional forward passes—which are typically executed sequentially. We parallelized these operations by distributing the conditioned and null-conditioned score estimations across two independent GPUs. This effectively reduced the latency per diffusion step by nearly half, with no impact on overall model quality.

算法1 DreamZero训练 (流匹配)

```

1: 输入: 数据  $\mathcal{D}$ , 文本条件  $\mathbf{c}$ 
2: 超参数: 块数  $M$ 
3: 模型:  $\mathbf{u}_\theta$  (联合视频动作DiT)
4: 当 未收敛执行
5:   样本轨迹  $\tau \sim \mathcal{D}$ 
6:   将视频编码为干净的潜在变量  $\mathbf{z}_1^{1:M}$ , 然后归一化动作  $\mathbf{a}_1^{1:M}$ 
7:   将  $\tau$  分割成  $M$  块
8:   对于  $k = 1, \dots, M$  执行 分块训练
9:     定义干净上下文  $\mathcal{C}_k \leftarrow \{(\mathbf{z}_1^j, \mathbf{a}_1^j)\}_{j=1}^{k-1} \triangleright$  TF历史
10:    采样时间步  $t_k \sim \mathcal{U}(0, 1)$ 
11:    // 可选: DreamZero-Flash解耦
12:    如果 Flash模式则
13:       $t_{vid} \sim \text{贝塔}(7, 1)$   $t_{act} \sim \mathcal{U}(0, 1)$ 
14:    else
15:       $t_{vid} \leftarrow t_k$   $t_{act} \leftarrow t_k$ 
16:    结束如果
17:    采样噪声  $\mathbf{z}_0^k, \mathbf{a}_0^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
18:    插值 (等式 2):
19:       $\mathbf{z}_t^k \leftarrow t_{vid}\mathbf{z}_1^k + (1 - t_{vid})\mathbf{z}_0^k$ 
20:       $\mathbf{a}_t^k \leftarrow t_{act}\mathbf{a}_1^k + (1 - t_{act})\mathbf{a}_0^k$ 
21:    预测速度:
22:       $\mathbf{v}_{pred} \leftarrow \mathbf{u}_\theta([\mathbf{z}_t^k, \mathbf{a}_t^k]; \mathcal{C}_k, \mathbf{c}, \mathbf{q}_k, t_k)$ 
23:    目标速度  $\mathbf{v}^k := [\mathbf{z}_1^k, \mathbf{a}_1^k] - [\mathbf{z}_0^k, \mathbf{a}_0^k]$ 
24:    损失  $\mathcal{L} \leftarrow \|\mathbf{v}_{pred} - \mathbf{v}^k\|^2 \triangleright$  方程 3
25:    更新  $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$ 
26:  结束 for
27: 结束 while

```

算法2 DreamZero Inference (闭环控制控制)

```

1: 输入: 指令  $\mathbf{c}$ , 初始图像  $\mathbf{o}_{init}$ , 状态  $\mathbf{q}_{init}$ 
2: 超参数: 步数  $N$ , 缓存阈值  $\epsilon$ 
3: 初始化:  $\mathcal{KV} \leftarrow \emptyset$ ,  $\mathbf{v}_{prev} \leftarrow \emptyset$ ,  $\mathbf{q}_{curr} \leftarrow \mathbf{q}_{init}$ 
4: // 1. 预填充缓存 (上下文阶段,  $t = 0$ )
5:  $\mathbf{z}_{init} \leftarrow \text{VAE}(\mathbf{o}_{init})$ 
6: // 传递干净视频, 无动作/状态
7:  $(\cdot, \cdot; \mathcal{KV}) \leftarrow \mathbf{u}_\theta([\mathbf{z}_{init}, \emptyset]; \mathcal{KV}, \mathbf{c}, \emptyset, t = 0, \text{更新}=\text{True})$ 
8: // 2. 自回归循环
9: 当 任务未完成执行
10:   采样  $\mathbf{x}_0 = [\mathbf{z}_0, \mathbf{a}_0] \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \triangleright$  噪声 at  $t = 0$ 
11:   // 联合去噪 (流匹配  $t : 0 \rightarrow 1$ )
12:   for  $i = 0 \dots N - 1$  do
13:      $t_i, t_{i+1} \leftarrow \text{调度器}(i, N)$ 
14:     // 优化: DiT缓存
15:     if  $\mathbf{v}_{prev} \neq \emptyset$  和 余弦相似度( $\mathbf{v}_{prev}, \mathbf{v}_{last}$ )  $> \epsilon$  then
16:        $\mathbf{v}_i \leftarrow \mathbf{v}_{prev}$ 
17:     else
18:        $(\mathbf{v}_i^{\text{vid}}, \mathbf{v}_i^{\text{act}}, \cdot) \leftarrow \mathbf{u}_\theta(\mathbf{x}_{t_i}; \mathcal{KV}, \mathbf{c}, \mathbf{q}_{curr}, t_i, \text{更新}=\text{False})$ 
19:        $\mathbf{v}_i \leftarrow [\mathbf{v}_i^{\text{vid}}, \mathbf{v}_i^{\text{act}}]$ 
20:        $\mathbf{v}_{prev} \leftarrow \mathbf{v}_i$ 
21:     end if
22:     // 求解器步骤
23:      $\mathbf{x}_{t_{i+1}} \leftarrow \mathbf{x}_{t_i} + \text{步骤}(\mathbf{v}_i, t_i, t_{i+1})$ 
24:   结束循环
25:   // 3. 执行与缓存更新
26:    $\hat{\mathbf{a}} \leftarrow \text{过滤器}(\mathbf{x}_1^{\text{action}}) \triangleright$  清理动作  $t = 1$ 
27:   异步执行  $\hat{\mathbf{a}}$  在机器人上
28:   // 关键: 注入真实值 ( $t = 0$ )
29:   真实观测  $\mathbf{o}_{real}, \mathbf{q}_{real}$ 
30:    $\mathbf{q}_{curr} \leftarrow \mathbf{q}_{real}$ ,  $\mathbf{z}_{real} \leftarrow \text{VAE}(\mathbf{o}_{real})$ 
31:    $(\cdot, \cdot; \mathcal{KV}) \leftarrow \mathbf{u}_\theta([\mathbf{z}_{real}, \emptyset]; \mathcal{KV}, \mathbf{c}, \emptyset, t = 0, \text{update}=\text{True})$ 
32:   *丢弃从  $\mathbf{x}_1$  中预测的视频潜在在表示
33: 结束 while 循环

```

D. 实时执行细节

本附录提供了关于系统级、实现级和模型级优化的更多细节，这些优化在第3.2节。

D.1. 系统级优化

CFG并行。 为了解决DiT固有的计算瓶颈，我们采用了无分类器引导（CFG）并行。标准CFG需要两个不同的模型评估——条件前向传递和无条件前向传递——通常顺序执行。我们通过将条件分数估计和空条件分数估计分布到两个独立的GPU上来并行化这些操作。这有效地将每个扩散步数的延迟减少了近一半，而对整体模型质量没有影响。

DiT Caching. The iterative nature of DiT inference imposes a significant computational bottleneck for real-time applications. Previous efforts have focused on training-free caching techniques that exploit temporal redundancies across diffusion steps. TeaCache (Liu et al., 2024) leverages the relative L_1 difference between timestep-embedding modulated inputs as a heuristic to estimate output variance and skip redundant computations, while TaylorSeer (Liu et al., 2025) employs higher-order Taylor expansions to extrapolate future latent states from historical derivatives. In DREAMZERO, we implement a caching mechanism that exploits the directional consistency of velocity vectors learned during flow matching.

During inference, the model tracks cosine similarity between successive velocity predictions. When this metric exceeds a predefined threshold (τ), the model bypasses the DiT forward pass for a window of a few steps by reusing the cached velocity vector. This adaptive scheduling concentrates computational resources on critical trajectory updates, reducing the average number of DiT steps from 16 to 4 with minimal degradation to predicted video and action fidelity.

Asynchronous Execution. Sequential execution of action blocks introduces stalls while waiting for upstream models to produce the next chunk, making the robot more open-loop and less reactive to real-time state changes, especially with large chunk sizes. We use an asynchronous execution mechanism that decouples model inference from action execution, allowing both stages to run concurrently. The motion controller always executes the most recent action scheduled for the current timestamp, while the inference module always uses the latest observation.

D.2. Implementation-level Optimizations

Torch Compile and CUDA Graphs. Inference is predominantly CPU-bound due to kernel launch overheads and Python execution. We employ torch.compile with CUDA Graphs (mode="reduce-overhead") and enforce full graph capture (fullgraph=True) to eliminate graph breaks. In addition to reducing CPU overhead, torch.compile decreases memory bandwidth requirements through operator fusion. We apply compilation to five model components: the diffusion transformer, scheduler, text encoder, image encoder, and VAE. We enforce static shapes (dynamic=False), which results in multiple recompilations during the first inference trajectory due to the evolving KV cache shape. From the second trajectory onward, inference proceeds without recompilation. To enable error-free compilation, we refactor the model to follow a functional programming paradigm: the KV cache is passed explicitly as input and returned as output of the compiled function.

Post-Training Quantization. We implement a mixed-precision strategy using the NVIDIA Model Optimizer (NVIDIA Corporation, 2024) on Blackwell (SM100) architecture. We quantize model weights and activations to NVFP4 (E2M1) while maintaining sensitive QKV projections and Softmax operations in FP8 (E4M3). To preserve numerical stability, we employ FP16 accumulation for non-linear operations including LayerNorm and RoPE. This configuration improves latency with negligible impact on generated video and action quality.

Kernel-Level Enhancements. We use the cuDNN backend for dot-product attention via PyTorch Scaled Dot-Product Attention (SDPA), requiring PyTorch version ≥ 2.9 . Earlier versions of the Transformer Engine library may also access these efficient cuDNN kernels.

Scheduler Optimizations. The initial Flow UniPC scheduler (Zhao et al., 2023) implementation required CPU execution for several operations, causing frequent CPU–GPU synchronization and GPU stalls. We migrated these operations to GPU, eliminating unnecessary CPU overhead.

D.3. Model-level Optimizations — DREAMZERO-Flash

In the standard DREAMZERO formulation, video and action modalities share the same denoising timestep t_k :

$$t_k^{\text{video}} = t_k^{\text{action}} = t_k, \quad t_k \sim \mathcal{U}(0, 1) \quad (4)$$

DiT缓存。 DiT推理的迭代性质对实时应用造成了显著的计算瓶颈。先前的工作集中于利用扩散步数间时间冗余的免训练缓存技术。TeaCache (Liu et al., 2024) 利用了相对 L_1 时间步嵌入调制输入之间的差异，作为估计输出方差和跳过冗余计算的启发式方法，而TaylorSeer (Liu et al., 2025) 采用高阶泰勒展开从历史导数中推断未来潜在状态。在DREAMZERO中，我们实现了一种利用流匹配过程中学习到的速度向量的方向一致性的缓存机制。

在推理过程中，模型会追踪连续速度预测之间的余弦相似度。当该指标超过预设阈值 (τ) 时，模型会在几个步长的窗口内跳过DiT前向传播，直接复用缓存的速度向量。这种自适应调度将计算资源集中于关键轨迹更新，使平均DiT步数从16降至4，同时对预测视频与动作保真度的影响极小。

异步执行。 顺序执行动作块时，会因等待上游模型生成下一个块而产生停滞，使机器人更偏向开环运行，对实时状态变化的响应性降低，尤其是在块尺寸较大时。我们采用异步执行机制，将模型推理与动作执行解耦，使两个阶段能够并发运行。运动控制器始终执行当前时间戳下最新调度的动作，而推理模块则始终使用最新的观测数据。

D.2. 实现级优化

Torch Compile 与 CUDA Graphs. 推理过程主要受限于CPU开销，原因在于内核启动开销和Python执行。我们采用torch.compile结合CUDA Graphs（模式="reduce-overhead"）并强制全图捕获（fullgraph=True），以消除图断裂。除了降低CPU开销外，torch.compile还通过算子融合减少了内存带宽需求。我们对五个模型组件应用了编译：扩散Transformer、调度器、文本编码器、图像编码器和VAE。我们强制使用静态形状（dynamic=False），这会导致在首次推理轨迹期间因KV缓存形状变化而发生多次重新编译。从第二条轨迹开始，推理过程无需重新编译。为实现无错误编译，我们按照函数式编程范式重构了模型：KV缓存作为编译函数的显式输入传入，并作为输出返回。

训练后量化。 我们使用英伟达公司的NVIDIA Model Optimizer (NVIDIA Corporation, 2024) 在Blackwell (SM100)架构上实现了混合精度策略。我们将模型权重和激活值量化至NVFP4 (E2M1)，同时将敏感的QKV投影和Softmax运算保留为FP8 (E4M3)。为保持数值稳定性，我们对包括LayerNorm和RoPE在内的非线性运算使用FP16累加。此配置降低了延迟，对生成视频与动作质量的影响可忽略。

内核级增强。 我们通过PyTorch缩放点积注意力（SDPA）使用cuDNN后端进行点积注意力，要求PyTorch版本 ≥ 2.9 。更早版本的Transformer Engine库也可能访问这些高效的cuDNN内核。

调度器优化。 初始的Flow UniPC调度器 (Zhao 等人, 2023) 实现中，多个操作需在CPU上执行，导致频繁的CPU-GPU同步和GPU停滞。我们将这些操作迁移至GPU，消除了不必要的CPU开销。

D.3. 模型级优化——DreamZero-Flash

在标准 DreamZero 公式化中，视频和动作模态共享相同的去噪时间步 t_k :

$$t_k^{\text{video}} = t_k^{\text{action}} = t_k, \quad t_k \sim \mathcal{U}(0, 1) \quad (4)$$

DREAMZERO-Flash decouples these schedules by biasing video timesteps toward lower values (higher noise) while keeping action timesteps uniform:

$$t_k^{\text{video}} = 1 - \eta, \quad \eta \sim \text{Beta}(\alpha, \beta), \quad t_k^{\text{action}} \sim \mathcal{U}(0, 1) \quad (5)$$

where $\alpha > \beta$ (e.g., $\alpha = 7, \beta = 1$). Since $\text{Beta}(\alpha, \beta)$ with $\alpha > \beta$ concentrates mass near $\eta \approx 1$, the transformed variable $t_k^{\text{video}} = 1 - \eta$ is biased toward 0, corresponding to high-noise video states. The noisy samples become:

$$\mathbf{z}_{t_k^{\text{video}}}^k = t_k^{\text{video}} \mathbf{z}_1^k + (1 - t_k^{\text{video}}) \mathbf{z}_0^k, \quad \mathbf{a}_{t_k^{\text{action}}}^k = t_k^{\text{action}} \mathbf{a}_1^k + (1 - t_k^{\text{action}}) \mathbf{a}_0^k \quad (6)$$

For $\text{Beta}(7, 1)$, $\mathbb{E}[\eta] = 0.875$, yielding $\mathbb{E}[t_k^{\text{video}}] = 0.125$ compared to 0.5 in the coupled setting. This exposes the model during training to configurations where actions must be predicted from predominantly noisy visual context (Figure 5), aligning training with rapid-action-denoising inference where actions denoise from noise level $1 \rightarrow 0$ in one step while video remains partially noisy.

Action Chunk Smoothing. Generated action chunks may contain high-frequency noise from denoising. We apply filtering to ensure stable real-world behavior: first upsampling the action chunk to $2\times$ resolution via cubic interpolation, then applying a Savitzky-Golay filter (window size 21, polynomial order 3) to suppress noise while preserving trajectory shape, and finally downsampling to original resolution.

E. AgiBot Diverse Data Collection Strategy

Our data collection philosophy prioritizes *diversity over repetition*. Unlike conventional approaches that collect hundreds of demonstrations per task in controlled lab settings, we collect data across 22 real-world environments spanning homes, restaurants, supermarkets, coffee shops, offices, warehouses, laboratories, and hotels (Figure 15).

E.1. Daily Collection Workflow

Each day, teleoperators receive a printed task sheet listing available tasks for their assigned area (e.g., kitchen area, checkout counter). For each episode, they select three *tasks* (usually very coarse-grained like “tidy up”) from this sheet and execute them consecutively. Each task typically requires 1–2 minutes, resulting in approximately 5-minute episodes.

At the end of each day, teleoperators log the frequency count for each task. Once a task was collected in 50 episodes, it is deprecated and removed from the task sheet. Teleoperators are incentivized to propose new tasks, which they inevitably must do as existing tasks become deprecated. This mechanism continuously expands the task distribution throughout data collection, yielding a long-tail of diverse behaviors.

Because we prioritize utility over repetition, our tasks are naturally more coarse-grained than typical robot learning datasets. Examples include organizing items, ground garbage cleaning, shopping basket return, toy box tidying, table tidying, and clothes hanging—but the full set grows organically as the collection progresses.

E.2. Multi-Task Episode Structure

The three-task episode structure serves two purposes: it maximizes diversity within each episode and encourages the model to learn smooth task transitions. For example, a single episode might involve (1) clearing dishes from a table, (2) wiping the table surface, and (3) organizing condiments. This design yields an average of 42 subtasks per episode (Figure 6), significantly more than typical single-task datasets.

This combination of environment diversity, task deprecation with forced expansion, and multi-task episodes produces a heterogeneous dataset that differs substantially from conventional robot learning corpora. Rather

DreamZero-Flash 通过将视频时间步偏向较低值（更高噪声），同时保持动作时间步均匀，来解耦这些调度：

$$t_k^{\text{video}} = 1 - \eta, \quad \eta \sim \text{Beta}(\alpha, \beta), \quad t_k^{\text{action}} \sim \mathcal{U}(0, 1) \quad (5)$$

其中 $\alpha > \beta$ （例如， $\alpha = 7, \beta = 1$ ）。由于 $\text{Beta}(\alpha, \beta)$ 与 $\alpha > \beta$ 将质量集中在 $\eta \approx 1$ 附近，变换后的变量 $t_k^{\text{video}} = 1 - \eta$ 偏向0，对应高噪声视频状态。噪声样本变为：

$$\mathbf{z}_{t_k^{\text{video}}}^k = t_k^{\text{video}} \mathbf{z}_1^k + (1 - t_k^{\text{video}}) \mathbf{z}_0^k, \quad \mathbf{a}_{t_k^{\text{action}}}^k = t_k^{\text{action}} \mathbf{a}_1^k + (1 - t_k^{\text{action}}) \mathbf{a}_0^k \quad (6)$$

对于 $\text{Beta}(7, 1)$ ， $\mathbb{E}[\eta] = 0.875$ ，得出 $\mathbb{E}[t_k^{\text{video}}] = 0.125$ 与 0.5 在耦合设置中相比。这使得模型在训练期间暴露于必须从主要包含噪声的视觉上下文（图5）中预测动作的配置，从而使训练与快速动作去噪推理保持一致，其中动作从噪声水平 $1 \rightarrow 0$ 一步去噪，而视频保持部分噪声。

动作块平滑。 生成的动作块可能包含来自去噪的高频噪声。我们应用过滤以确保稳定的真实世界行为：首先通过三次插值将动作块上采样到 $2\times$ 分辨率，然后应用Savitzky-Golay滤波器（窗口大小21，多项式阶数3）以在保持轨迹形状的同时抑制噪声，最后下采样到原始分辨率。

E. AgiBot 多样化数据收集策略

我们的数据收集理念优先考虑多样而非重复。与在受控实验室环境中为每个任务收集数百次演示的传统方法不同，我们收集数据跨越22个真实世界环境，涵盖家庭、餐厅、超市、咖啡店、办公室、仓库、实验室和酒店（图15）。

E.1. 日常收集工作流程

每天，远程操作员会收到一份打印的任务表，列出其负责区域（例如厨房区域、收银台）的可执行任务。对于每个片段，他们从该表中选择三个任务（通常是像“整理”这样非常粗粒度的任务）并连续执行。每个任务通常需要1–2分钟，从而产生大约5分钟的回合。

每天结束时，远程操作员会记录每个任务的频率计数。一旦某个任务在50个回合中被收集，它就会被废弃并从任务表中移除。远程操作员被激励提出新任务，随着现有任务被废弃，他们不可避免地必须这样做。这种机制在整个数据收集过程中不断扩展任务分布，产生多样化行为的长尾。

因为我们优先考虑实用性而非重复性，我们的任务自然比典型的机器人学习数据集更加粗粒度。示例包括整理物品、地面垃圾清理、购物篮归位、玩具盒整理、桌子整理和挂衣服——但随着数据收集的进行，完整集合会自然增长。

E.2. 多任务片段结构

三任务片段结构有两个目的：最大化每个片段内的多样性，并鼓励模型学习平滑的任务切换。例如，一个片段可能涉及 (1) 从桌子上清理碗碟，(2) 擦拭桌子表面，以及 (3) 整理调味品。这种设计使得每个片段平均包含42个子任务（图6），远超典型的单任务数据集。

这种环境多样性、任务淘汰与强制扩展以及多任务片段的组合，产生了一个与常规机器人学习语料库截然不同的异构数据集。

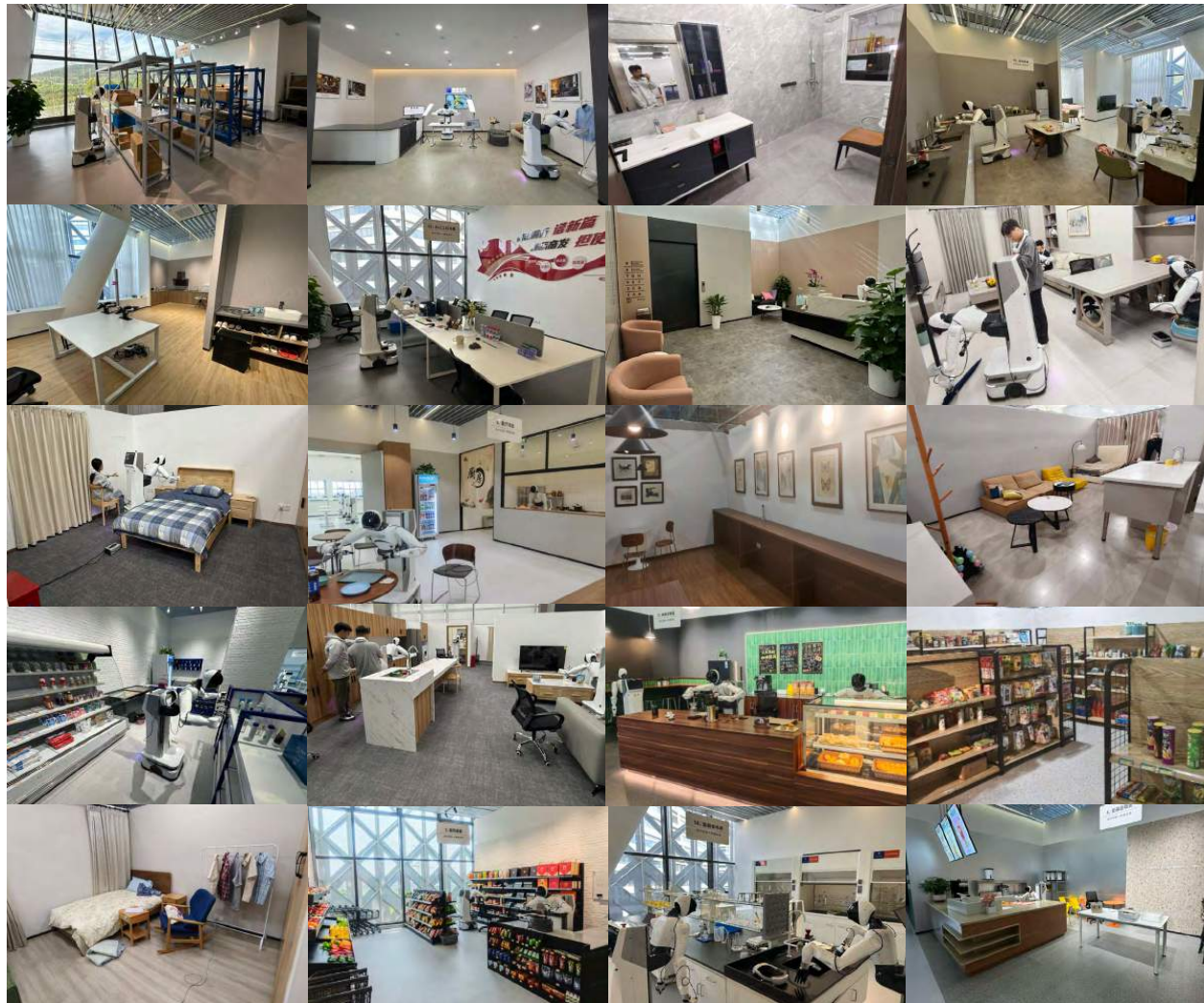


Figure 15: **Data Collection Environments.** We collect teleoperation data across 22 diverse real-world environments, including offices, laboratories, restaurants, supermarkets, coffee shops, warehouses, homes, hotels, and retail stores. This diversity enables DREAMZERO to generalize to unseen environments without task-specific fine-tuning.

than learning narrow task-specific policies, DREAMZERO learns generalizable skills that transfer across environments and tasks. See https://dreamzero0.github.io/training_data_gallery/ for examples of these coarse-grained tasks (the videos shown in the website concatenated each coarse-grained tasks)

F. AgiBot Evaluation Details

We provide the evaluation setup for both seen tasks in Tables 5 and unseen tasks in Table 6 for Agibot. Each row shows the initial frame and instruction for 4 robots. We conduct 2 rollouts per robot for each task by varying the objects, locations and the robot arm to use for the task (The current table mostly shows the evaluation using the left arm).

G. DROID Evaluation Details

We provide the evaluation setup for both seen and unseen tasks in Tables 7a and 7b for DROID. We conduct 2 rollouts per task by varying the location of the objects.

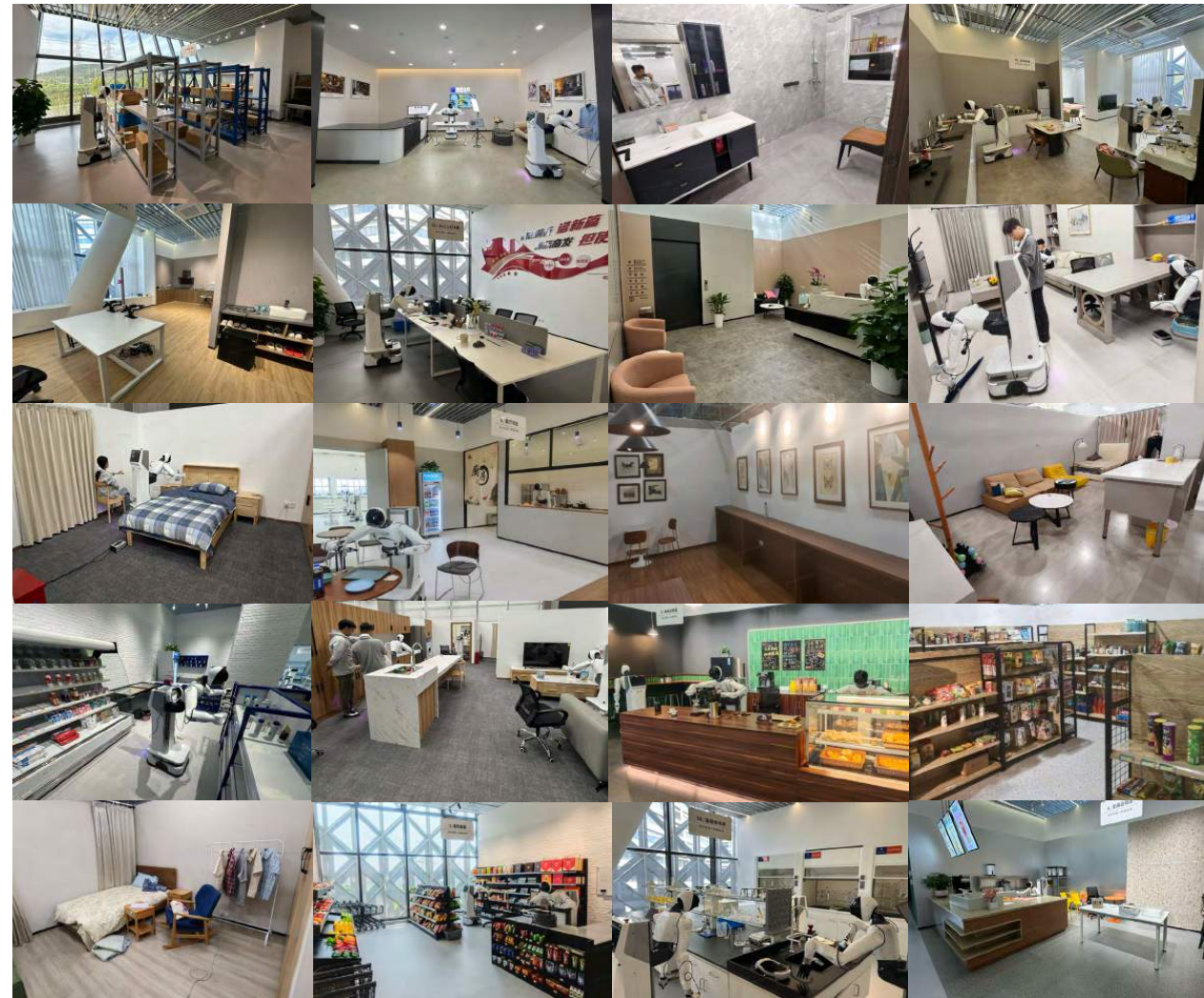


图15: **数据收集环境。**我们在22个多样的真实世界环境中收集遥操作数据，包括办公室、实验室、餐厅、超市、咖啡馆、仓库、住宅、酒店和零售店。这种多样性使得DreamZero能够在无需任务特定微调的情况下泛化到未见环境。

而不是学习狭窄的任务特定策略，DREAMZERO 学习可泛化的技能，这些技能能够在不同环境和任务之间迁移。参见https://dreamzero0.github.io/training_数据_图库/以获取这些粗粒度任务的示例（网站中展示的视频已拼接了每个粗粒度任务）

F. AgiBot 评估详情

我们提供了已见任务的评估设置，见于表5和未见任务的评估设置，见表6用于AgiBot。每一行显示4台机器人的初始帧和指令。对于每个任务，每台机器人执行2次运行，通过改变物体、位置以及用于任务的手臂（当前表格主要展示使用左臂进行评估）。

G. DROID 评估详情

我们提供了已见和未见任务的评估设置，见于表7a和7b用于DROID。每个任务执行2次运行，通过改变物体的位置。

Table 5: Seen Tasks Evaluation Setup for AgiBot G1

Category	#	Task	Image	Instruction	Image	Instruction	Image	Instruction	Image	Instruction
PnP Easy	1	PnP Fruit		The left arm picks up the Banana on the table and places it in the Blue Plate.		The left arm picks up the lime on the table and places it on the light green plate.		The left arm picks up the peach on the table and places it on the blue checked bowl.		The left arm picks up the green pear on the table and places it on the blue checked bowl.
	2	Taking out Fruit		The left arm picks up the Mango from the plastic bag and places it into the Wooden Basket		The left arm picks up the yellow pear from the plastic bag and places it onto the blue tray.		The left arm picks up the purple grapes from the plastic bag and places it into the brown basket.		The left arm picks up the watermelon from the plastic bag and places it onto the Blue Plate.
	3	Wipe the Mess		The left arm uses a sponge to wipe the coffee spill off the table.		The left arm used a black cloth to wipe the powder off the table.		The left arm uses a napkin to wipe the creamer spill off the table.		The left arm used a paper towel to wipe the water off the table.
PnP Hard	4	PnP Fork/Spoon		The left arm picks up the Pink Fork from the table and places it onto the blue plate.		The left arm picks up the orange fork from the table and places it into the glass cup.		The left arm picks up the light blue fork from the table and places it onto the orange plate.		The left arm picks up the green fork from the table and places it into the blue plate.
	5	Put Pen in Holder		The left arm picks up the Red Marker pen from the table and placed it into the pen holder.		The left arm picked up the black marker from the table and placed it into the pen holder.		The left arm picked up the white marker pen from the table and placed it into the pen holder.		The left arm picked up the mechanical pencil from the table and placed it into the pen holder.
	6	Put Cup on Coaster		The left arm picks up the clear cup from the table and places it on the grey coaster.		The left arm picks up the plastic cup from the table and places it on the blue coaster.		The left arm picks up the plastic cup from the table and places it on the white coaster.		The left arm picks up the pink cup from the table and places it on the gray coaster.
	7	Stack Bowls/Cups		The robot reaches to grip the green bowl, moves it to the middle wooden bowl, and releases it to stack. It then reaches to grip the white bowl, moves it to the same location, and releases it onto the stack.		The robot reaches its left arm to grip the blue plate, moves it to the middle light green plate, and releases it to stack. It then reaches its right arm to grip the red plate, moves it to the same location, and releases it to finish the task.		The robot reaches its left arm to grip the paper bowl, moves it over the middle white bowl, and releases it to stack. It then reaches its left arm to grip the blue checked bowl, moves it over the stack, and releases it to finish the task.		The robot reaches its left arm to grip the pink bowl, moves it over the middle beige bowl, and releases it to stack. It then reaches its left arm to grip the white bowl, moves it over the stack, and releases it onto the stack.
	8	Folding Shirts		Both arms grip the bottom of the light grey short sleeve and fold it toward the middle. They then pull the short sleeve across the table to the edge. Next, both arms grasp the top of the shirt and fold it down to the middle. Finally, the right arm grips the collar and folds it down to complete the task.		Both arms fold the bottom of the light grey short sleeve to the middle. They then pull the shirt toward the edge of the table. Next, both arms fold the top of the shirt down to the middle. Finally, the right arm grips the collar and folds it down to complete the task.		Both arms fold the bottom of the logo short sleeve to the middle. They then pull the shirt toward the edge of the table. Next, both arms fold the top of the shirt down to the middle. Finally, the left arm grasps the collar and folds it down to complete the task.		Both arms fold the bottom of the gray short sleeve to the middle. They then pull the shirt toward the edge of the table. Next, both arms fold the top of the shirt down to the middle. Finally, the left arm grasps the collar and folds it down to complete the task.
Contact Rich	9	Folding Shorts		Both arms fold the bottom of the tan shorts toward the middle. Then, the right arm folds the shorts in half from right to left to complete the fold.		Both arms fold the bottom of the grey shorts toward the middle. Then, the right arm folds the shorts in half from right to left to complete the fold.		Both arms fold the bottom of the logo short sleeve to the middle. They then pull the shirt toward the edge of the table. Next, both arms fold the top of the shirt down to the middle. Finally, the left arm grasps the collar and folds it down to complete the task.		Both arms fold the bottom of the green shorts toward the middle. Then, the right arm folds the shorts in half from right to left to complete the task.
	10	Stacking Clothes		Both arms pick up the black shirt and place it on the stack of clothes.		Both arms pick up the white shirt and place it on the gray towel.		Both arms pick up the black hoodie and place it on the stack of clothes.		Both arms pick up the dark gray shirt and place it on the stack of clothes.

表5: AgiBot G1的已见任务评估设置

类别	#	Task	图像	指令	图像	指令	图像	指令	图像	指令
PnP Easy	1	PnP水果		左臂拿起桌子上的香蕉并将其放入蓝色盘子。		左臂拿起柠檬在桌子上，然后将其放在浅绿色盘子上。		左臂拿起桌子上的桃子并放置它放在烤盘上。		左臂拿起桌子上的绿梨并把它放在蓝色的棋盘格碗。
	2	取出水果		左臂拿起芒果从塑料袋中并放入木制的篮子		左臂拿起黄色梨从塑料袋子并把它放到蓝色托盘。		左臂拿起紫葡萄从塑料袋子并把它放入棕色篮子。		左臂拿起西瓜从塑料袋子并把它放到蓝色盘子。
	3	擦拭 Mess		左臂用海绵将咖啡污渍从桌子上。		左臂用了黑色布巾擦拭白色粉末离开桌子。		左臂用一张餐巾巾擦拭奶精渍从桌子上。		左臂用了一张纸巾巾擦拭水渍从桌子上。
PnP Hard	4	PnP 叉/勺		左臂拿起粉红从桌子上拿起叉子并把它放到蓝色盘子上。		左臂拿起橙色叉子从桌子上并把它放入玻璃杯中		左臂拿起浅蓝色叉子从桌子上并把它放入玻璃杯中		左臂拿起绿色叉子从桌子上，并把它放入蓝色盘子中。
	5	放入笔 支架		左臂拿起红色马克笔从桌子上并把它放入笔筒中。		左臂拿起了黑色马克笔从桌子上并把它放入笔筒中。		左臂拿起了白色马克笔从桌子上并把它放入笔筒中。		左臂拿起了自动铅笔从桌子上并把它放进笔筒。
	6	把杯子放在杯垫		左臂拿起透明杯子从桌子上并放置它放在灰色杯垫上。		左臂拿起P塑料杯从桌子上并把它放在蓝色杯垫上。		左臂拿起P塑料杯从桌子上并把它放在白色杯垫上。		左臂拿起粉色杯子从桌子上拿起并放置它放在灰色杯垫上。
	7	堆叠碗/杯子		机器人伸手去抓绿色碗，将其移动到中间木碗，并释放它到堆叠上。然后伸手去抓白色碗，并将其移动到同一位置，并释放它到堆叠上。		机器人伸出其左臂以抓取蓝色盘子，移动它到中间浅绿色盘子，并释放它到堆叠上。然后伸出其右臂以抓取红色盘子，并将其移动到中间蓝色盘子，并释放它以完成任务。		机器人伸出左臂以抓取纸碗，移动它到中间白色碗上方，并释放它到堆叠上。然后伸出其左臂去抓取白色碗，移动它到堆叠上，并释放它以完成任务。		机器人伸出其左臂以抓取粉色碗，移动它到中间白色碗的上方，并释放它到堆叠上。然后伸出其左臂去抓取白色碗，移动它到堆叠上，并释放它以完成任务。
	8	折叠 衬衫		双臂抓住下摆浅灰色短袖衬衫并将其向中间折叠。然后他们拉动短袖向桌子的边缘。接下来，双臂抓住短袖的顶部并将其向下折叠至中间。最后，右手臂抓住领子并折叠它向下以完成任务。		双臂将下摆的绿色短袖向中间。然后他们拉动短袖向桌子的边缘。接下来，双臂抓住短袖的顶部并将其向下折叠至中间。最后，右手臂抓住领子并折叠它向下以完成任务。		双臂将logo短袖的下摆折叠至中间。然后他们拉动短袖向桌子的边缘。接下来，双臂抓住短袖的顶部并将其向下折叠至中间。最后，右手臂抓住领子并折叠它向下以完成任务。		双臂将logo短袖的下摆折叠至中间。然后他们拉动短袖向桌子的边缘。接下来，双臂抓住短袖的顶部并将其向下折叠至中间。最后，右手臂抓住领子并折叠它向下以完成任务。
	9	折叠 短裤		双臂将下摆的灰色短袖向中间。然后，右臂将短袖从右向左完成折叠。		双臂将灰色短裤的下摆向中间。然后，右臂将短裤从右向左完成折叠。		双臂将logo短裤的下摆向中间。然后，右臂将短裤从右向左完成折叠。		双臂将logo短裤的下摆向中间。然后，右臂将短裤从右向左完成折叠。
接触 Rich	10	堆叠 衣服		双臂拿起黑色的衬衫并将其放在堆叠上的衣服。		双臂拿起白色衬衫并将其放在灰色毛巾上。		双臂拿起黑色连帽衫并将其放在衣物堆。		双臂拿起深色的灰色衬衫并将其放在衣物堆。

Table 6: Unseen Tasks Evaluation Setup for AgiBot G1

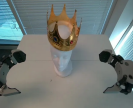
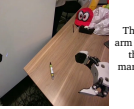
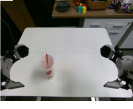
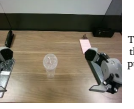
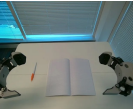
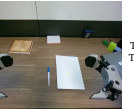
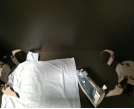
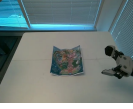
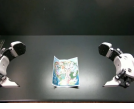
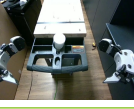
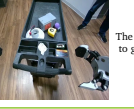
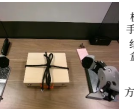
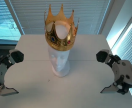
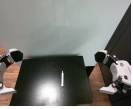
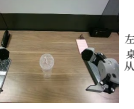
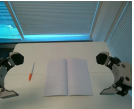
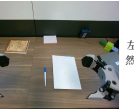
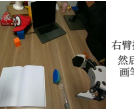
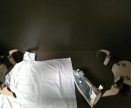
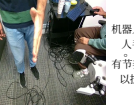
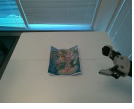
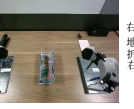
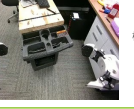
#	Task	Image	Instruction	Image	Instruction	Image	Instruction	Image	Instruction
1	Untie Shoelaces		The robot coordinates both arms to simultaneously grasp the two loops of the shoelace. It then moves both arms outward in synchronized, opposing directions until the shoelace is fully untied.		The robot reaches its left arm to grasp the blue box and hold it steady. It then reaches its right arm toward the blue ribbon and moves it to untie the knot.		The robot coordinates both arms to simultaneously grasp the two loops of the knot of the box. It then moves both arms outward in synchronized, opposing directions until the knot of the box is fully untied.		The robot coordinates both arms to simultaneously grasp the two loops of the knot of the package. It then moves both arms outward in synchronized, opposing directions until the knot of the package is fully untied.
2	Remove /Put Hat		The robot reaches its right arm to grasp the crown and lifts it to remove it from the mannequin's head.		The robot reaches its left arm to grasp the hat and lifts it to remove it from the mannequin's head.		The robot reaches its left arm to grasp the hat and lifts it to remove it from the mannequin's head.		The robot reaches its left arm to grasp the hat and lifts it to remove it from the mannequin's head.
3	Draw Circle		The robot reaches its left arm to pick up the red marker and the left arm moves the marker to draw a circle on the book.		The robot reaches its right arm to pick up the black marker and the right arm draws a circle on the whiteboard with the marker.		The robot reaches its left arm to pick up the black marker and the left arm moves the marker to draw a circle on the paper.		The robot reaches its right arm to pick up the black marker and the right arm moves the marker to draw a line on the whiteboard.
4	Take out Straw		The left arm holds the cup on the table. Then the right arm pulls the straw out of the cup.		The left arm holds the cup on the table. Then the right arm pulls the straw out of the cup.		The left arm holds the cup on the table. Then the right arm pulls the straw out of the cup.		The right arm holds the cup on the table. Then the left arm pulls the straw out of the cup.
5	Cube Stacking		The robot reaches its right arm to pick up the green cube, moves it over the red cube, and releases it to stack. It then reaches its right arm to pick up the yellow cube, moves it over the stack, and releases it to finish the tower.		The robot reaches its right arm to pick up the red cube, moves it over the colorful cube, and releases it to stack. It then reaches its right arm to pick up the green cube, moves it over the red cube, and releases it to finish the tower.		The robot reaches its right arm to pick up the white cube, moves it over the green cube, and releases it to begin the stack. It then reaches its left arm to pick up the blue cube, moves it over the stack, and releases it onto the white cube to finish the task.		The robot reaches its right arm to pick up the red cube, moves it over the blue cube, and releases it to begin the stack. It then reaches its left arm to pick up the orange cube, moves it over the stack, and releases it onto the red cube to complete the three-tier structure.
6	Painting		The left arm grabs the brush. Then left arm paints with the brush on the notebook.		The right arm grabs the brush. Then right arm paints with the brush on the paper.		The left arm grabs the brush. Then left arm paints with the brush on the notebook.		The right arm grabs the brush. Then right arm paints with the brush on the notebook.
7	Ironing		The robot reaches its left arm to grasp the iron and moves it across the shorts to iron it.		The robot reaches its right arm to grasp the iron and moves it across the shirt to iron it.		The robot reaches its left arm to grasp the iron and moves it across the shirt to iron it.		The robot reaches its left arm to grasp the iron and moves it across the shirt to iron it.
8	Shake Hands		The right arm of the robot grasp the human hand to shake hands. It then initiates a rhythmic up-and-down motion to perform the handshake.		The left arm shakes the hand of the human up and down.		The right arm shakes the hand of the human up and down.		The left arm of the robot grasp the human hand to shake hands. It then initiates a rhythmic up-and-down motion to perform the handshake.
9	Folding (Map)		The left arm grabs the left side of the map. The right arm folds the right side of the map. The left arm folds the left side of the map.		The left arm grabs the left side of the map. The right arm folds the right side of the map. The left arm folds the left side of the map.		The right arm grabs the right side of the map. The left arm folds the left side of the map. The right arm folds the right side of the map.		The right arm grabs the right side of the map. The left arm folds the left side of the map. The right arm folds the right side of the map.
10	Pulling Cart		The robot reaches its right arm to grasp the cart and pulls it.		The robot reaches its left arm to grasp the cart and pulls it.		The robot reaches its right arm to grasp the cart and pulls it.		The robot reaches its left arm to grasp the cart and pulls it forward.

表 6:AgiBot G1 的未见任务评估设置

#	Task	图片	指令	图片	指令	图像	指令	图像	指令
1	Untie 鞋带		机器人协调双臂同时抓住鞋带的两个环，然后它移动双臂向外，相反方向直到鞋带完全untied.		机器人伸出其左臂以抓取蓝色盒子并保持它稳定。随后伸出其右臂向外在同步、相反方向，直到蓝色盒子完全解开。		机器人协调双臂同时抓住鞋带的两个环，然后它移动双臂向外在同步、相反的方向，直到蓝色盒子完全解开。		机器人协调双臂同时抓住鞋带的两个环，然后它移动双臂向外在同步、相反的方向，直到蓝色盒子完全解开。
2	移除/放置帽子		机器人伸出它的右臂抓住帽子并将其抬起以从人体模型头部移除它。		机器人伸出它的左臂抓住帽子并将其抬起至将其从人体模型头部取下。		机器人伸出左臂抓住帽子并将其抬起至将其从人体模型头部。		机器人伸出其左臂抓住帽子并将其提起至将其从人体模型头部。
3	Draw 画		机器人伸出左臂去拿起红色记号笔，然后左臂移动记号笔在书上画一个圆。		机器人伸出右臂去拿起红色记号笔，然后右臂画一个圆在白板上用记号笔。		机器人伸出其左臂去拿起红色记号笔，然后左臂移动记号笔在...上画一个圆在纸上。		机器人伸出它的右臂拿起红色记号笔，然后右臂移动记号笔在...上画一条线在白板上。
4	取出 吸管		左臂握住杯子在桌子上。然后右臂将吸管从杯子中取出。		左臂将杯子放在桌子上。然后右臂从杯子中拔出吸管。		左臂将杯子放在桌子上。然后右臂从杯子中拔出吸管。		右臂握住杯子放在桌子上。然后左臂将吸管从杯子中拔出。
5	Cube 堆叠		机器人伸出右臂去拿起红色立方体，将其移至红色立方体上方，并释放它进行堆叠。接着，它伸出右臂去拿起黄色立方体，将其移至堆叠上方，然后释放它来完成堆叠。		机器人伸出它的右臂去拿起红色立方体，并将其移动到蓝色立方体上方，然后松开使其堆叠。然后它伸出它的右臂去拿起绿色立方体，移动它越过堆叠，并释放它到白色立方体上以完成任务。		机器人伸出其右臂去拿起红色立方体，并将其移动到蓝色立方体上方，并释放它进行堆叠。它然后伸出它的左臂去拿起蓝色立方体，移动它越过堆叠，并释放它到白色立方体上以完成任务。		机器人伸出其右臂以拿起红色立方体，将其移动到蓝色立方体上方，并释放它进行堆叠。然后它伸出其左臂以拿起蓝色立方体，并将其移动到堆叠上方，然后释放它到白色立方体上，以完成三层结构。
6	绘画		左臂抓住画笔。然后左臂用画笔在笔记本上。		右臂抓住画笔。然后右臂用画笔在纸上。		左臂抓住画笔。然后左臂用画笔在笔记本上。		右臂抓住画笔。然后右臂用画笔在笔记本上。
7	熨烫		机器人伸出其左臂抓住熨斗并将其移动到短裤上来熨烫。		机器人伸出其右臂抓住熨斗并在衬衫上熨烫。		机器人伸出它的左臂抓住熨斗并移动它穿过衬衫以熨烫它。		机器人伸出它的左臂抓住熨斗并移动它在衬衫上来熨烫。
8	握手		机器人的右臂抓住人手以握手。然后它启动一个有节奏的上下运动以执行握手。		左臂摆动那只手上下摆动人类的手。		右臂摆动那只手上下摆动人类的手。		机器人的左臂抓住人手以握手。然后它开始一个有节奏的上下运动以执行握手。
9	折叠 (地图)		左臂抓住左侧地图的一侧。右臂折叠地图的右侧。左臂折叠左侧地图。		左臂抓住左侧地图的一侧。右臂折叠地图的右侧。左臂折叠左侧地图。		右臂抓住右侧地图的侧面。左臂折叠地图的左侧。右臂折叠右侧地图的侧面。		右臂抓住右侧地图的侧面。左臂折叠地图的左侧。右臂折叠右侧地图的侧面。
10	拉动 Cart		机器人伸出其右臂抓住手推车并拉它。		机器人伸出它的左臂去抓住手推车并拉它。		机器人伸出它的右臂去抓住手推车并拉它。		机器人伸出左臂抓住手推车并拉动它向前。

(a) Seen tasks on DROID

#	Image	Instruction	#	Image	Instruction
1		Move the cup forward then put the marker inside the cup	11		Move the bowl on the left to the right side of the table.
2		Put the marker in the blue box	12		Pick up the pencil and put it on the bowl
3		Remove the pair of gloves from the open drawer and put it on the table	13		Pick the marker up from the table and put it in the bowl
4		Put the marker on table	14		Place the bowl next to the marker
5		Pick up the apple and put it in the basket	15		Remove a lemon from the bowl
6		Put the towel on the white cup	16		Move the grapes to the left
7		Put the towel in the pan	17		Move the green grapes backwards
8		Put the hat on the table	18		Put the bread inside the toaster
9		Put the pair of scissors into the drawer	19		Push the lever on the bread toaster downwards
10		Put the towel in the basket	20		Pick up the cup from the bowl and put it in the other cups

(b) Tasks with unseen verbs on DROID

#	Image	Instruction	#	Image	Instruction
1		<u>Orient</u> the mug so the handle is to the right	11		<u>Hook</u> the hat onto the tripod
2		<u>Fan</u> the burger	12		<u>Pinch</u> the binder clip to release the papers
3		<u>Slice</u> the bread with the knife	13		<u>Withdraw</u> the bread from the toaster and place on the plate
4		<u>Type</u> 'hi' on the keyboard	14		<u>Cinch</u> the drawstring of the bag
5		<u>Extricate</u> the straw from the cup	15		<u>Dispense</u> the mustard onto the bread
6		<u>Reveal</u> the object under the cup	16		<u>Bake</u> the croissant in the oven
7		<u>Match</u> the objects to their corresponding bowl	17		<u>Fry</u> the vegetables in the pan with the spatula
8		<u>Maneuver</u> the blocks through the matching hole	18		<u>Depress</u> the lever on the toaster
9		<u>Affix</u> the magnet to the tray	19		<u>Elevate</u> the yellow block to the highest platform
10		<u>Combine</u> the nuts and batteries	20		<u>Weave</u> the wire through the holes of the box

H. Failure Case Analysis

In Figure 16, we illustrate the generated video by DREAMZERO and execution rollout for both AgiBot and DROID. Overall, the robot execution follows the visual plan generated on the video modality side. For AgiBot video generated by DREAMZERO, the robot picks up the marker with left arm and passes the marker to the right arm. Consistent with the generated video, for the execution rollout, the robot picks up the top part of the marker, but instead of drawing a line on the whiteboard, the left arm passes the marker to the right arm. For DROID video generated by DREAMZERO, the robot picks up the bread instead of opening the oven first. Aligned with the generated video, for the execution rollout, the robot picks up the bread first instead of opening the oven, leading to the rollout being stuck after the robot has reached to the oven with bread held. This implies that improving the language following and visual planning capability of WAMs could potentially lead to better action execution.

(a) DROID上的已见任务

#	图像	指令	#	图像	指令
1		将杯子向前移动，然后放将记号笔放入杯子	11		将左侧的碗移到桌子的右侧。
2		将记号笔放入蓝色盒子	12		拾起铅笔并将其放在碗
3		将手套从打开的抽屉中取出并放在桌子	13		从桌子上拿起记号笔并将其放入碗中
4		把记号笔放在桌子上	14		把碗放在记号笔旁边
5		拿起苹果，把它放进篮子	15		从碗里取出一个柠檬
6		把毛巾放在白色杯子上	16		将葡萄向左移动
7		把毛巾放入平底锅中	17		将青葡萄向后移动
8		把帽子放在桌子上	18		把面包放进烤面包机里
9		把剪刀放入抽屉	19		向下按压面包烤面包机上的手柄
10		把毛巾放进篮子里	20		从碗里拿起杯子并把它放到其他杯子中

(b) 在DROID上包含未见动词的任务

#	图片	指令	#	图片	指令
1		定向马克杯以使把手向右	11		钩住帽子到三脚架上
2		扇汉堡	12		捏住活页夹以释放纸张
3		切面包用刀	13		取出面包从烤面包机并放在盘子上
4		输入'嗨'在键盘上	14		拉紧袋子的拉绳
5		取出吸管从杯子	15		挤出芥末到面包
6		揭示杯子下方的物体	16		烘焙烤箱里的可颂
7		匹配物体与它们的对应的碗	17		煎平底锅中的蔬菜与锅铲
8		操纵积木穿过匹配孔	18		按下烤面包机上的手柄
9		固定磁铁到托盘	19		提升黄色积木到最高平台
10		组合螺母和电池	20		编织电线穿过孔盒子的

H. 失败案例分析

在图16，我们展示由DREAMZERO 以及AgiBot和DROID的执行展开。总的来说，机器人执行遵循在视频模态一侧生成的视觉计划。对于AgiBot视频由DREAMZERO，机器人用左臂拿起记号笔并将记号笔传递给右臂。与生成的视频一致，在执行展开中，机器人拿起记号笔的顶部，但左臂没有在白板上画线，而是将记号笔传递给右臂。对于DROID视频由DREAMZERO，机器人拿起面包而不是先打开烤箱。与生成的视频一致，在执行展开中，机器人先拿起面包而不是先打开烤箱，导致机器人拿着面包到达烤箱后仿真轨迹卡住。这表明提高WAMs的语言跟随和视觉规划能力可能带来更好的动作执行。

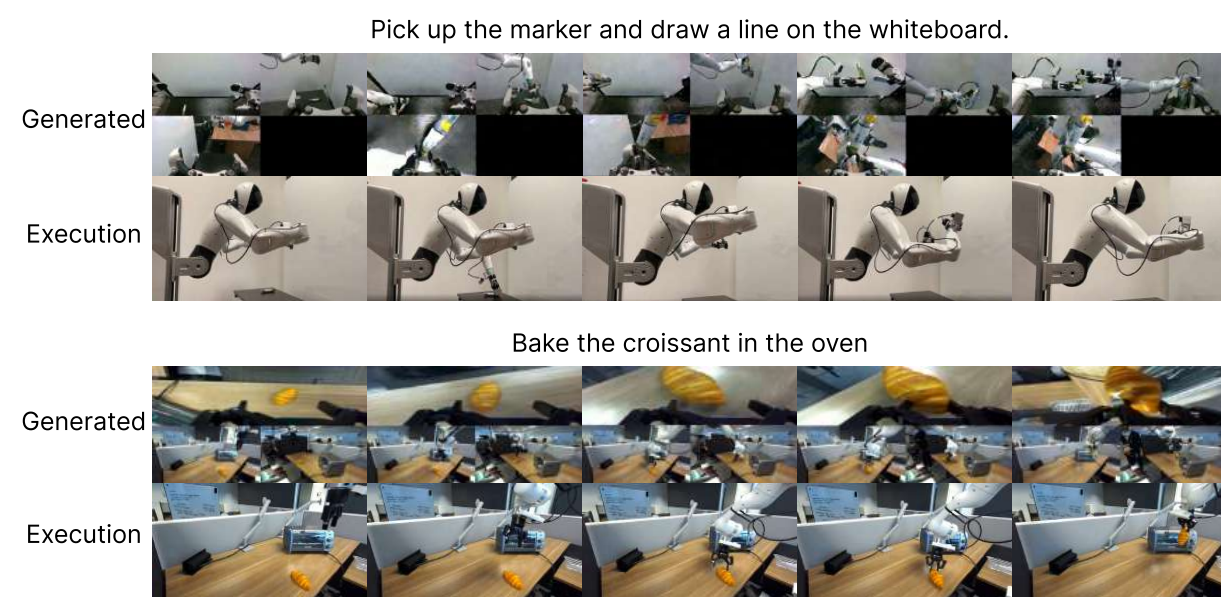


Figure 16: **Illustration of generated and executed pair.** We illustrate the generated video and action execution pair. These two examples show scenarios where the video prediction *failed* and the robot followed the failed plan. DREAMZERO failed to generate video of the Agibot G1 robot drawing in the whiteboard and failed to generate video of the Franka robot opening the oven first.

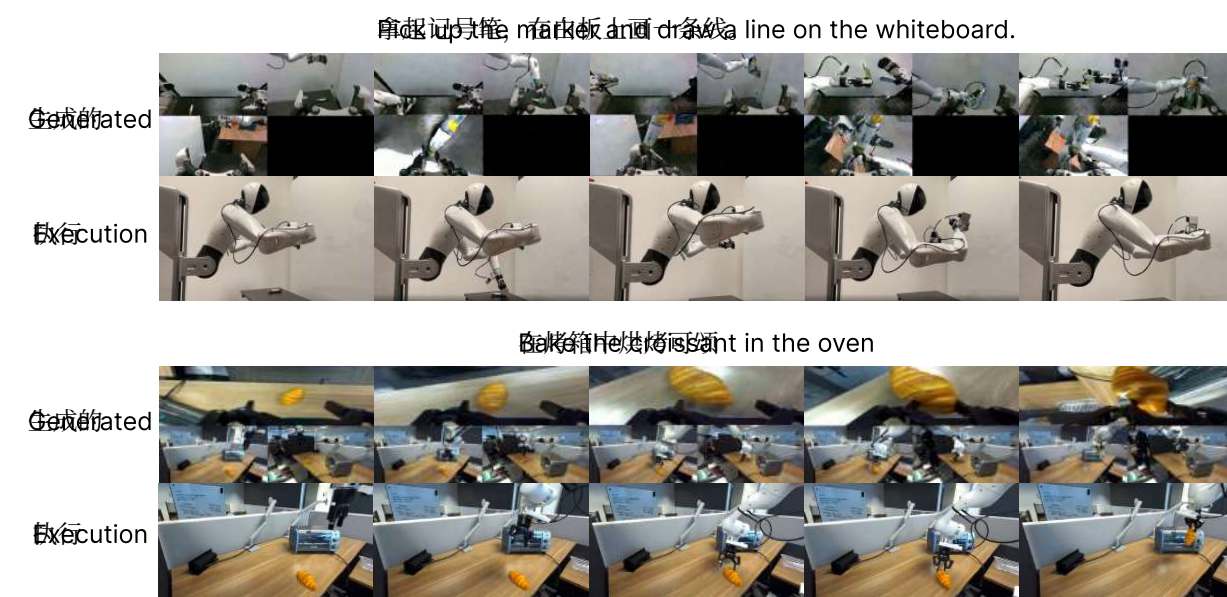


图16: **生成的与执行的配对示例。** 我们展示了生成的视频与动作执行配对。这两个示例展示了视频预测失败的场景，机器人遵循了该失败的规划。DreamZero未能生成AgiBot G1机器人在白板上绘图的视频，也未能生成Franka机器人首先打开烤箱的视频。

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